M.Sc C.S. – II SEM - II
Journal

Roll No.	072
Name	PARAB SAHIL SUBHASHCHANDRA
Subject	Cloud Computing - II

Thakur Educational Trust's (Regd.)
Thakur College of Science & Commerce
UGC Recognised • Affiliated to University Of Mumbai
(NAAC Accredited with Grade "A" [3rd Cycle] & ISO 9001:2015 Certified)

CERTIFICATE

This is here to certify that Mr. PARAB SAHIL SUBHASHCHANDRA , Seat Number 072 of M.Sc. II Computer Science, has satisfactorily completed the required number of experiments prescribed by the UNIVERSITY OF MUMBAI during the academic year 2022 – 2023.

Date:22-09-2022

Place: Mumbai

Teacher In-Charge Head of Department

External Examiner

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Sr no	Experiment	Date	Remarks
1	Execute & check the performance of existing algorithms using CloudSim.	09th June 2022	
2	Install a Cloud Analyst and Integrate with Eclipse/Netbeans. Monitor the performance of an Existing Algorithms.	16th June 2022	
3	Build an application on private cloud.	14th July 2022	
4	Demonstrate any Cloud Monitoring tool.	21st July 2022	
5	Evaluate a Private IAAS Cloud .	04th Aug 2022	
6	Implement FOSS-Cloud Functionality - VDI (Virtual Desktop Infrastructure)	11th Aug 2022	
7	Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure) Infrastructure as a Service (IaaS)	18th Aug 2022	
8	Implement FOSS-Cloud Functionality - VSI Platform as a Service (PaaS)	25th Aug 2022	
9	Implement FOSS-Cloud Functionality - VSI Software as a Service (SaaS)	01st Sep 2022	
10	Explore GCP / AWS / Azure	15th Sep 2022	

Practical No : **1**

Aim : Execute & check the performance of existing algorithms using CloudSim.

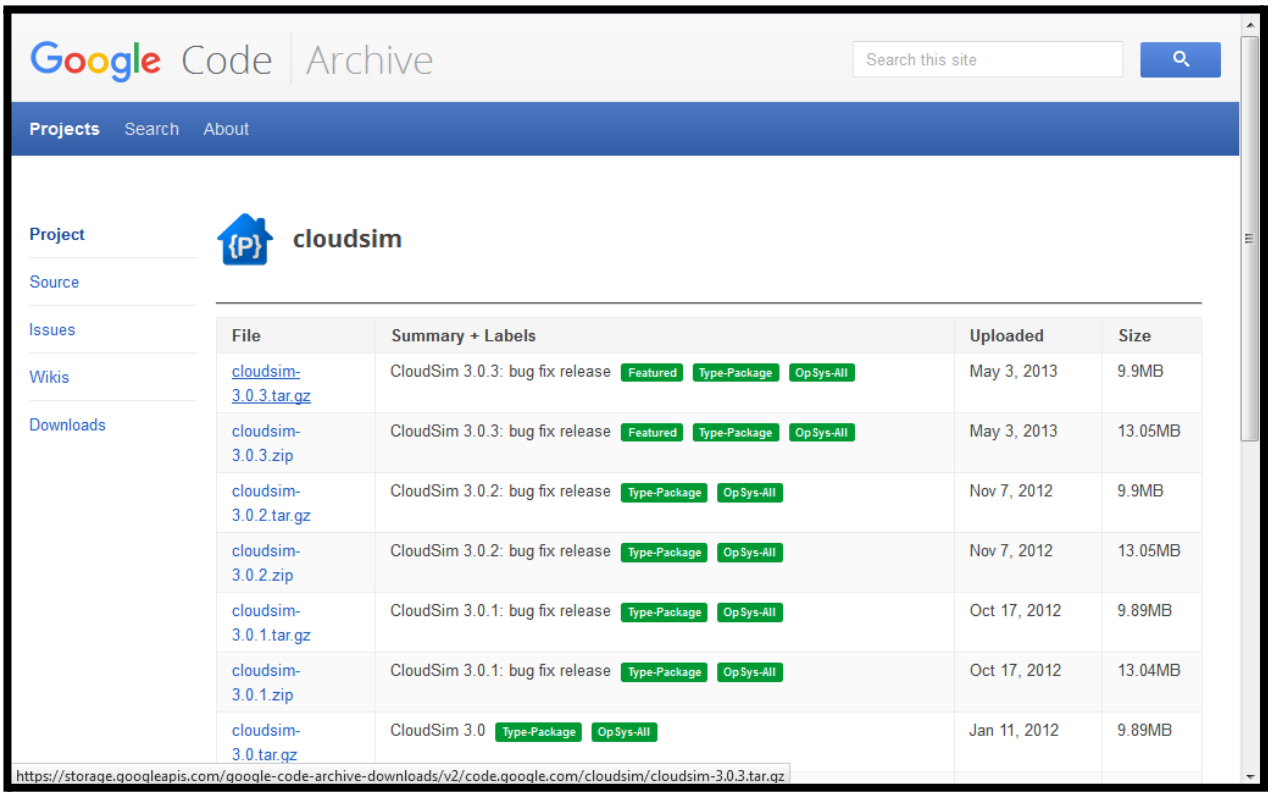
Theory:

CloudSim is an open-source framework, which is used to simulate cloud computing infrastructure and services. It is developed by the CLOUDS Lab organization and is written entirely in Java. It is used for modelling and simulating a cloud computing environment as a means for evaluating a hypothesis prior to software development in order to reproduce tests and results.

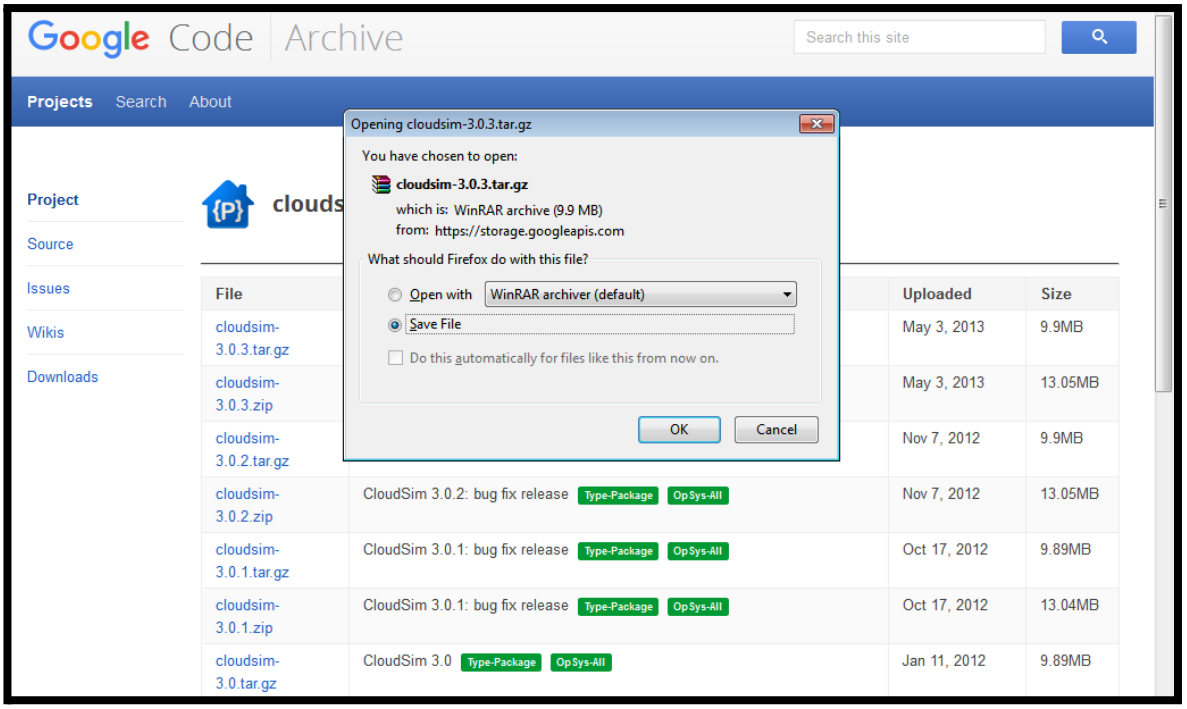
For example, if you were to deploy an application or a website on the cloud and wanted to test the services and load that your product can handle and also tune its performance to overcome bottlenecks before risking deployment, then such evaluations could be performed by simply coding a simulation of that environment with the help of various flexible and scalable classes provided by the CloudSim package, free of cost.

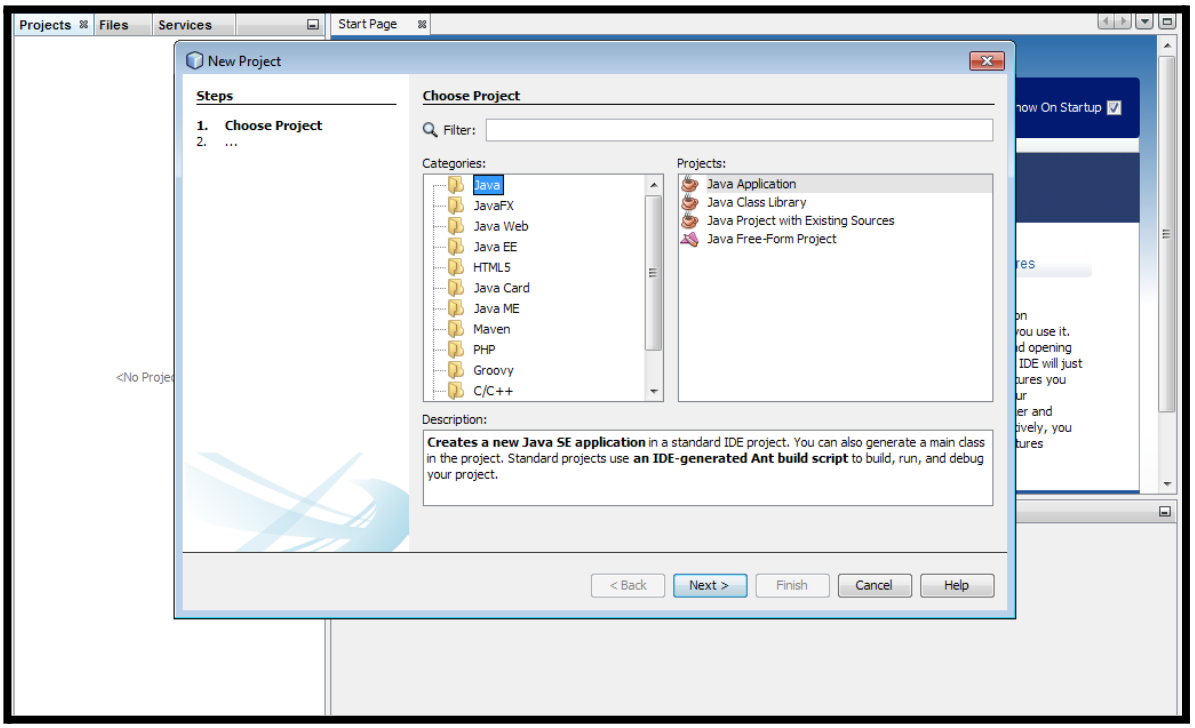
Download CloudSim:

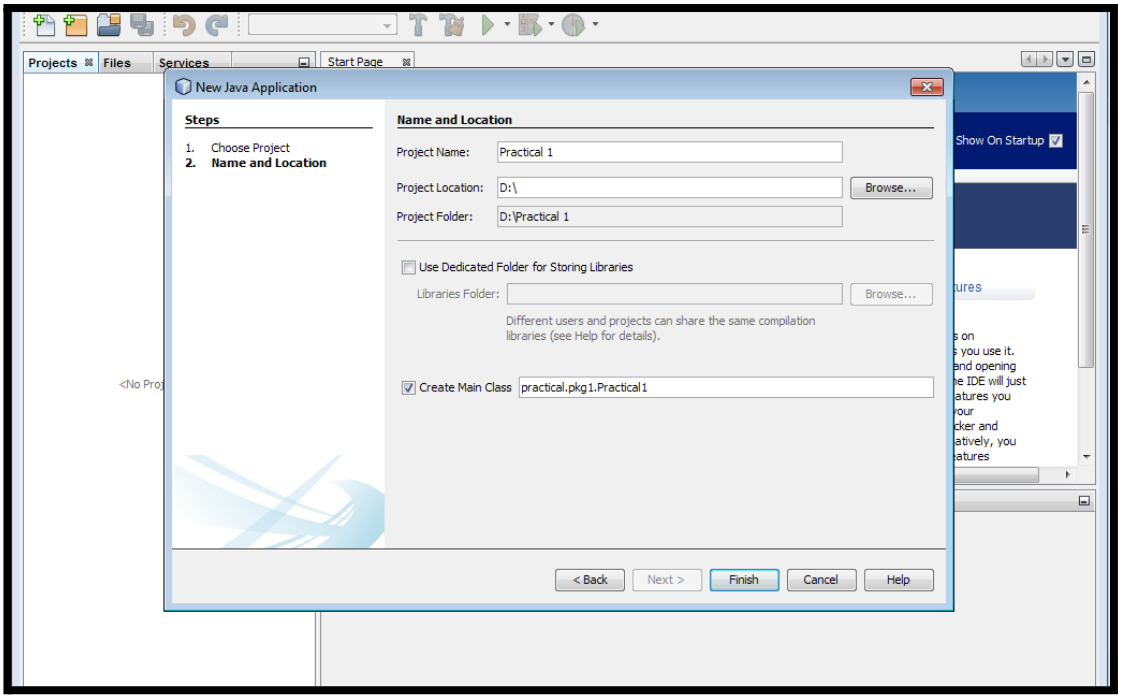
http://code.google.com/p/cloudsim/download

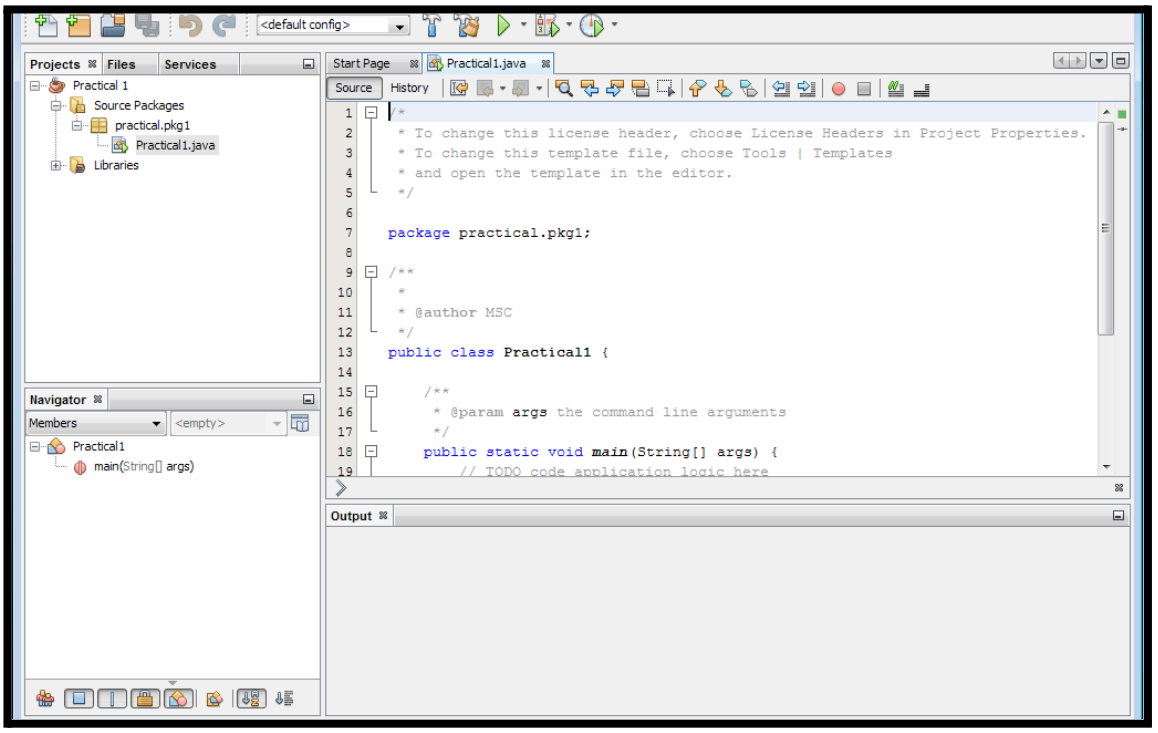


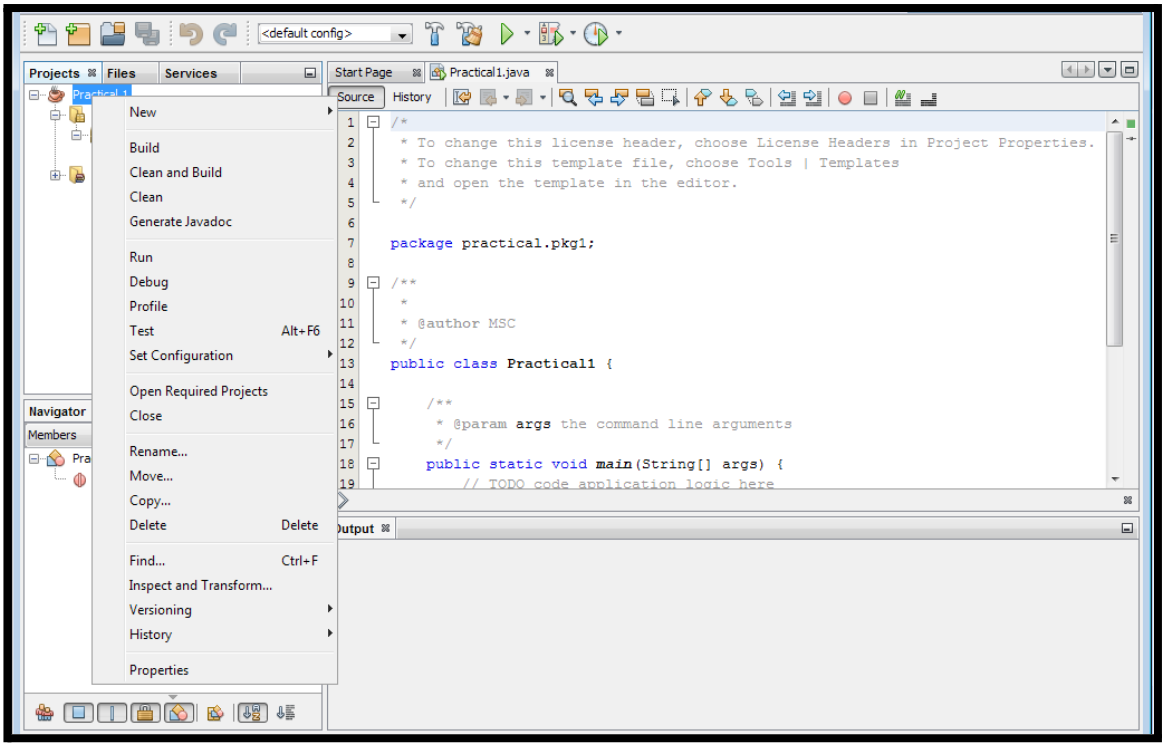
Select 1st option

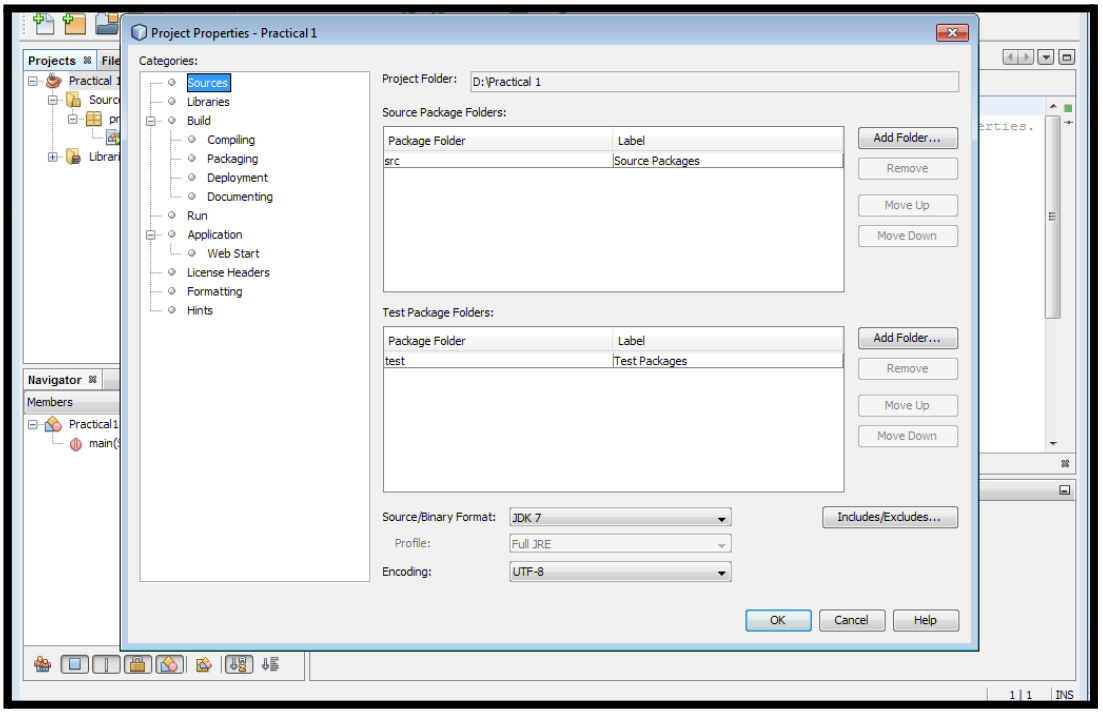


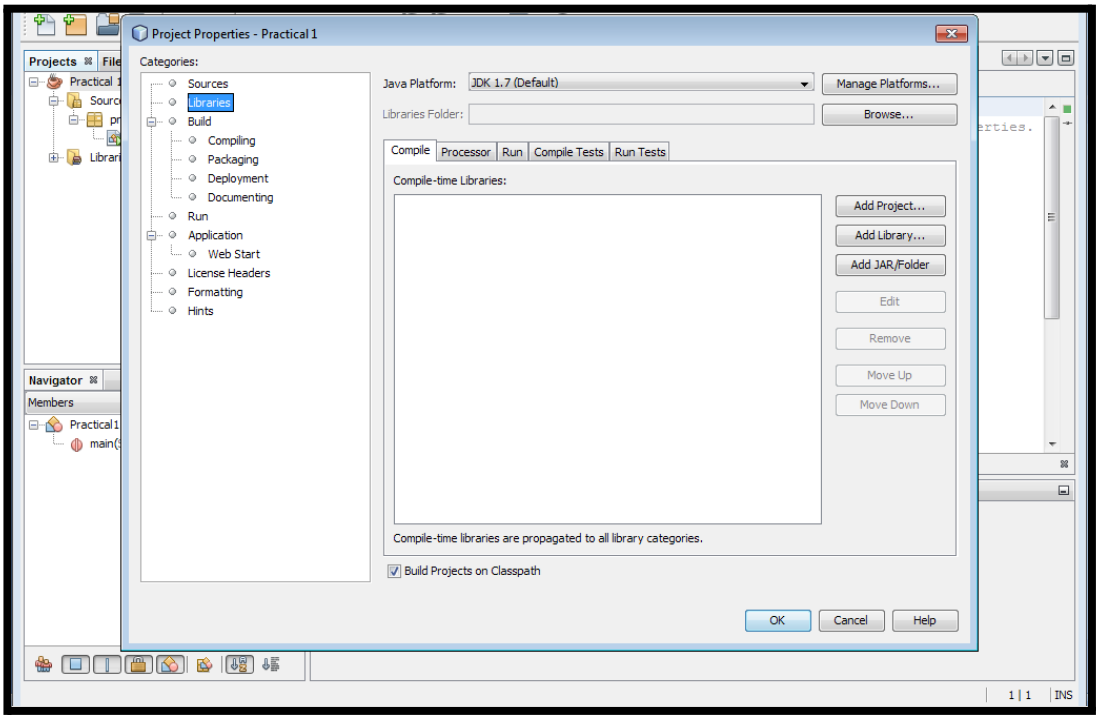

Projects
Files
Services
Start Page
New Project
Steps
1. Choose Project
2. ...
Choose Project
Filter:
Categories:
Java
JavaFX
Java Web
Java EE
HTML5
Java Card
Java ME
Maven
PHP
Groovy
C/C++
Projects:
Java Application
Java Class Library
Java Project with Existing Sources
Java Free-Form Project
Description:
Creates a new Java SE application in a standard IDE project. You can also generate a main class in the project. Standard projects use an IDE-generated Ant build script to build, run, and debug your project.
< Back
Next >
Finish
Cancel
Help

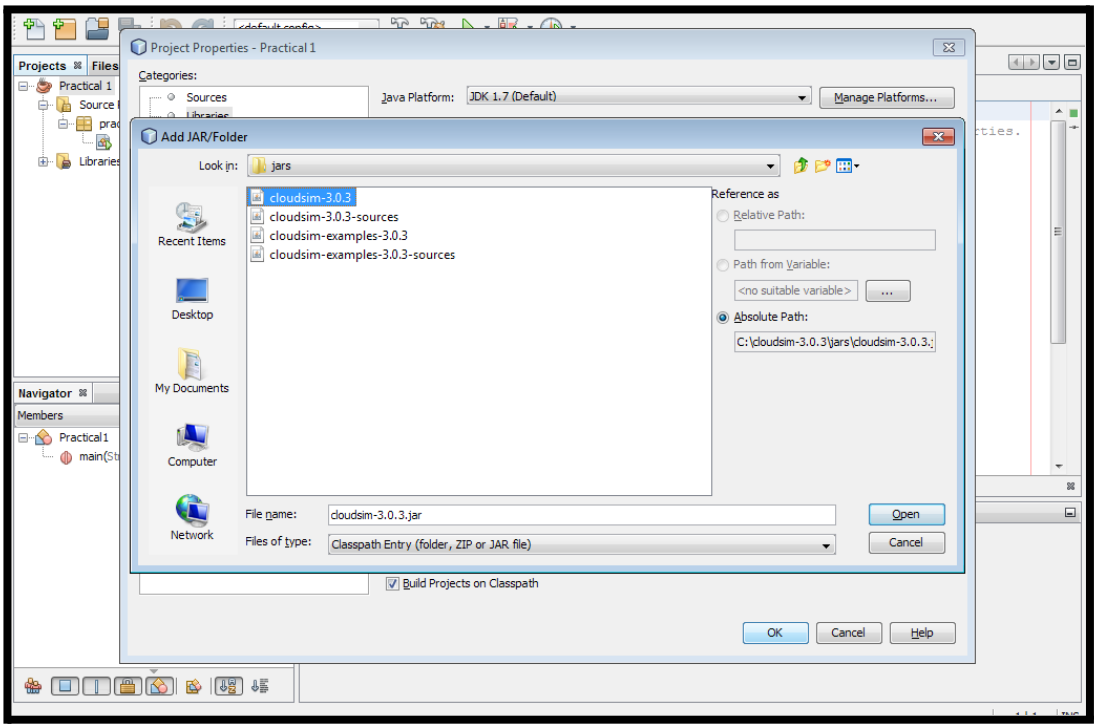

Projects
Files
Services
Start Page
New Java Application
Steps
1. Choose Project
2. Name and Location
Name and Location
Project Name: Practical 1
Project Location: D:\
Browse...
Project Folder: D:\Practical 1
Use Dedicated Folder for Storing Libraries
Libraries Folder:
Browse...
Different users and projects can share the same compilation libraries (see Help for details).
Create Main Class practical.pkg1.Practical1
< Back
Next >
Finish
Cancel
Help

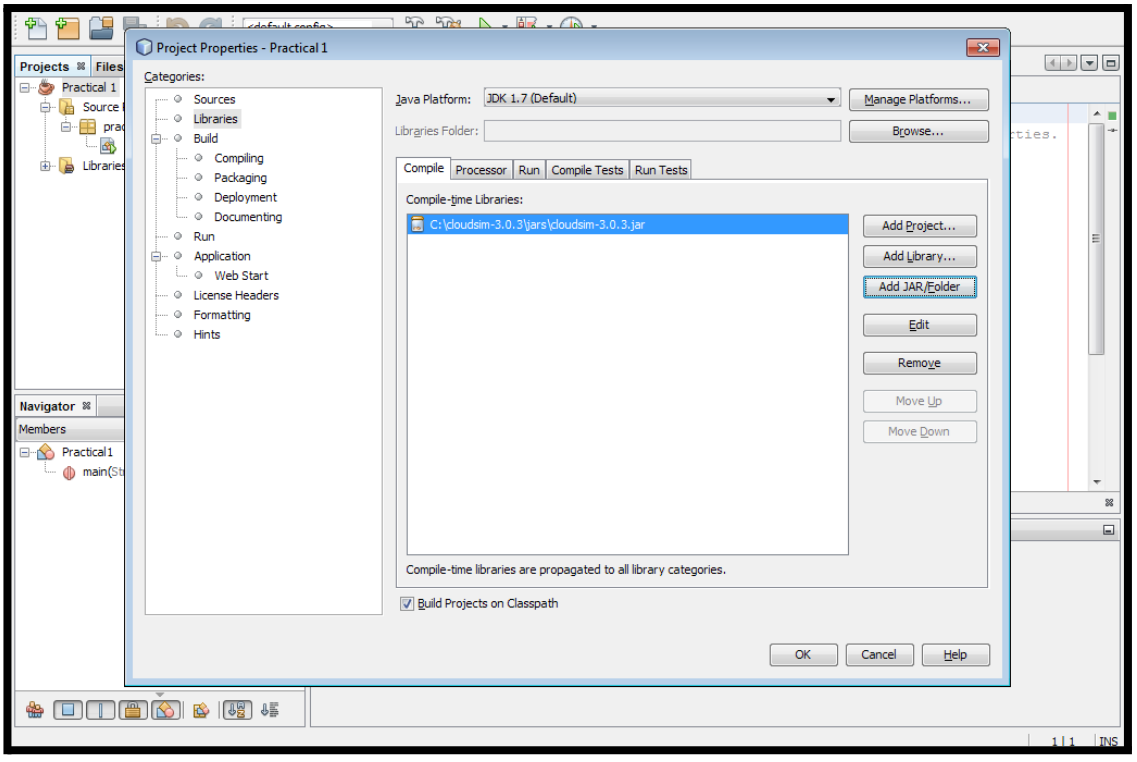

Projects
Files
Services
Practical 1
Source Packages
practical.pkg1
Practical1.java
Libraries
Navigator
Members
<empty>
Practical1
main(String[] args)
Start Page
Practical1.java
Source
History
/*
* To change this license header, choose License Headers in Project Properties.
* To change this template file, choose Tools | Templates
* and open the template in the editor.
*/
package practical.pkg1;
/**
*
* @author MSC
*/
public class Practical1 {
/**
* @param args the command line arguments
*/
public static void main(String[] args) {
// TODO code application logic here
Output

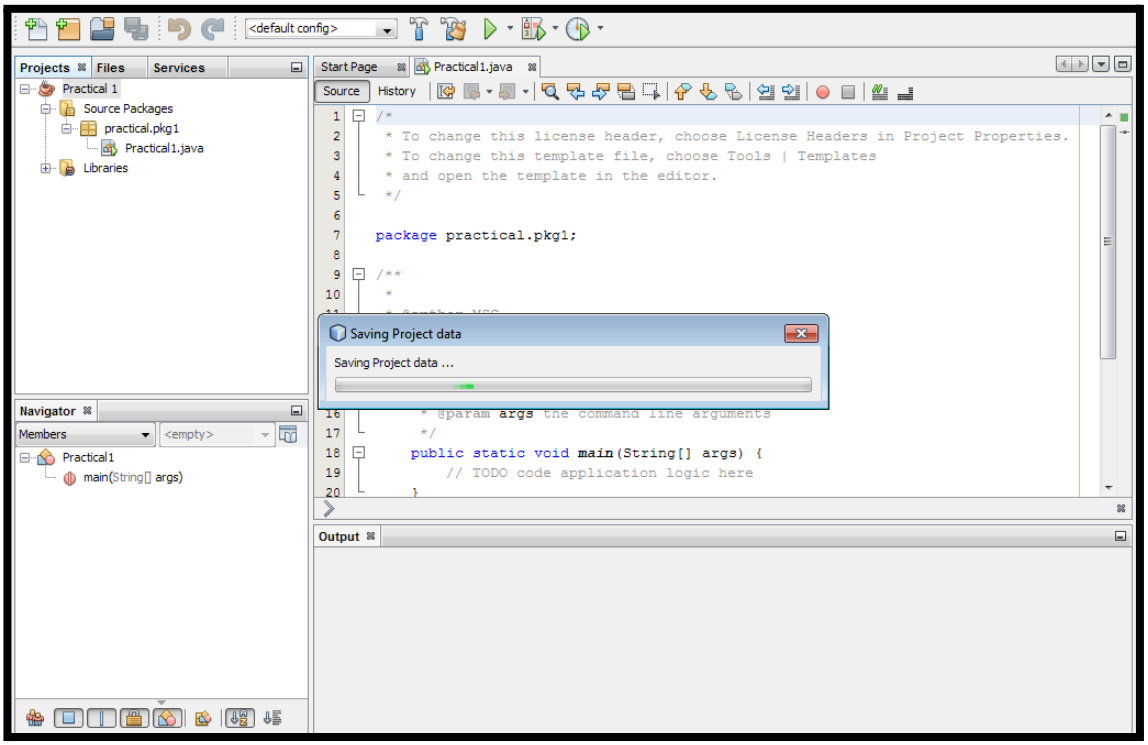

Projects
Files
Services
New
Build
Clean and Build
Clean
Generate Javadoc
Run
Debug
Profile
Test
Alt+F6
Set Configuration
Open Required Projects
Close
Rename...
Move...
Copy...
Delete
Delete
Find...
Ctrl+F
Inspect and Transform...
Versioning
History
Properties
Navigator
Members
Start Page
Practical1.java
Source
History
/*
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* @param args the command line arguments
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Project Properties - Practical 1
Categories:
Sources
Libraries
Build
Compiling
Packaging
Deployment
Documenting
Run
Application
Web Start
License Headers
Formatting
Hints
Project Folder:
D:\Practical 1
Source Package Folders:
Package Folder
Label
src
Source Packages
Add Folder...
Remove
Move Up
Move Down
Test Package Folders:
Package Folder
Label
test
Test Packages
Add Folder...
Remove
Move Up
Move Down
Source/Binary Format:
JDK 7
Profile:
Full JRE
Encoding:
UTF-8
Includes/Excludes...
OK
Cancel
Help


Project Properties - Practical 1
Categories:
Sources
Libraries
Build
Compiling
Packaging
Deployment
Documenting
Run
Application
Web Start
License Headers
Formatting
Hints
Java Platform:
JDK 1.7 (Default)
Manage Platforms...
Libraries Folder:
Browse...
Compile
Processor
Run
Compile Tests
Run Tests
Compile-time Libraries:
Add Project...
Add Library...
Add JAR/Folder
Edit
Remove
Move Up
Move Down
Compile-time libraries are propagated to all library categories.
Build Projects on Classpath
OK
Cancel
Help


Project Properties - Practical 1
Categories:
Sources
Java Platform:
JDK 1.7 (Default)
Manage Platforms...
Add JAR/Folder
Look in:
jars
cloudsim-3.0.3
cloudsim-3.0.3-sources
cloudsim-examples-3.0.3
cloudsim-examples-3.0.3-sources
Recent Items
Desktop
My Documents
Computer
Network
Reference as
Relative Path:
Path from Variable:
<no suitable variable>
Absolute Path:
C:\cloudsim-3.0.3\jars\cloudsim-3.0.3.
File name:
cloudsim-3.0.3.jar
Files of type:
Classpath Entry (folder, ZIP or JAR file)
Open
Cancel
Build Projects on Classpath
OK
Cancel
Help


Project Properties - Practical 1
Categories:
Sources
Libraries
Build
Compiling
Packaging
Deployment
Documenting
Run
Application
Web Start
License Headers
Formatting
Hints
Java Platform:
JDK 1.7 (Default)
Manage Platforms...
Libraries Folder:
Browse...
Compile
Processor
Run
Compile Tests
Run Tests
Compile-time Libraries:
C:\cloudsim-3.0.3\jars\cloudsim-3.0.3.jar
Add Project...
Add Library...
Add JAR/Folder
Edit
Remove
Move Up
Move Down
Compile-time libraries are propagated to all library categories.
Build Projects on Classpath
OK
Cancel
Help



Code:

```
package practical.pkg1;

import java.text.DecimalFormat;

import java.util.ArrayList;

import java.util.Calendar;

import java.util.LinkedList;

import java.util.List;

import org.cloudbus.cloudsim.Cloudlet;

import org.cloudbus.cloudsim.CloudletSchedulerTimeShared;

import org.cloudbus.cloudsim.Datacenter;

import org.cloudbus.cloudsim.DatacenterBroker;

import org.cloudbus.cloudsim.DatacenterCharacteristics;

import org.cloudbus.cloudsim.Host;

import org.cloudbus.cloudsim.Log;

import org.cloudbus.cloudsim.Pe;

import org.cloudbus.cloudsim.Storage;

import org.cloudbus.cloudsim.UtilizationModel;
```

```
import org.cloudbus.cloudsim.UtilizationModelFull;
import org.cloudbus.cloudsim.Vm;
import org.cloudbus.cloudsim.VmAllocationPolicySimple;
import org.cloudbus.cloudsim.VmSchedulerTimeShared;
import org.cloudbus.cloudsim.core.CloudSim;
import org.cloudbus.cloudsim.provisioners.BwProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.PeProvisionerSimple;
import org.cloudbus.cloudsim.provisioners.RamProvisionerSimple;

/**
 * A simple example showing how to create a datacenter with one host and run one
 * cloudlet on it.
 */
public class Practical1 {

	/** The cloudlet list. */
	private static List<Cloudlet> cloudletList;

	/** The vmlist. */
	private static List<Vm> vmlist;

	/**
	 * Creates main() to run this example.
	 *
	 * @param args the args
	 */
	@SuppressWarnings("unused")
	public static void main(String[] args) {
		Log.printLine("Starting CloudSimExample1...");
		try {
			// First step: Initialize the CloudSim package. It should be called
			// before creating any entities.
```

```
                int num_user = 1; // number of cloud users
                Calendar calendar = Calendar.getInstance();
                boolean trace_flag = false; // mean trace events
                // Initialize the CloudSim library
                CloudSim.init(num_user, calendar, trace_flag);
                // Second step: Create Datacenters
                // Datacenters are the resource providers in CloudSim. We need at
                // list one of them to run a CloudSim simulation
                Datacenter datacenter0 = createDatacenter("Datacenter_0");
                // Third step: Create Broker
                DatacenterBroker broker = createBroker();
                int brokerId = broker.getId();
                // Fourth step: Create one virtual machine
                vmlist = new ArrayList<Vm>();
                // VM description
                int vmid = 0;
                int mips = 1000;
                long size = 10000; // image size (MB)
                int ram = 512; // vm memory (MB)
                long bw = 1000;
                int pesNumber = 1; // number of cpus
                String vmm = "Xen"; // VMM name
                // create VM
                Vm vm = new Vm(vmid, brokerId, mips, pesNumber, ram, bw, size, vmm, new
CloudletSchedulerTimeShared());
                // add the VM to the vmList
                vmlist.add(vm);
                // submit vm list to the broker
                broker.submitVmList(vmlist);
                // Fifth step: Create one Cloudlet
                cloudletList = new ArrayList<Cloudlet>();
                // Cloudlet properties
```

```
                int id = 0;
                long length = 400000;
                long fileSize = 300;
                long outputSize = 300;
                UtilizationModel utilizationModel = new UtilizationModelFull();
                Cloudlet cloudlet = new Cloudlet(id, length, pesNumber, fileSize, outputSize, utilizationModel,
utilizationModel, utilizationModel);
                cloudlet.setUserId(brokerId);
                cloudlet.setVmId(vmid);
                // add the cloudlet to the list
                cloudletList.add(cloudlet);
                // submit cloudlet list to the broker
                broker.submitCloudletList(cloudletList);
                // Sixth step: Starts the simulation
                CloudSim.startSimulation();
                CloudSim.stopSimulation();
                //Final step: Print results when simulation is over
                List<Cloudlet> newList = broker.getCloudletReceivedList();
                printCloudletList(newList);

                Log.printLine("CloudSimExample1 finished!");
            } catch (Exception e) {
                e.printStackTrace();
                Log.printLine("Unwanted errors happen");
            }
        }
        /**
         * Creates the datacenter.
         *
         * @param name the name
         *
         * @return the datacenter
```

```
 */
private static Datacenter createDatacenter(String name) {

        // Here are the steps needed to create a PowerDatacenter:
        // 1. We need to create a list to store
        // our machine
        List<Host> hostList = new ArrayList<Host>();

        // 2. A Machine contains one or more PEs or CPUs/Cores.
        // In this example, it will have only one core.
        List<Pe> peList = new ArrayList<Pe>();
        int mips = 1000;
        // 3. Create PEs and add these into a list.
        peList.add(new Pe(0, new PeProvisionerSimple(mips))); // need to store Pe id and MIPS Rating
        // 4. Create Host with its id and list of PEs and add them to the list
        // of machines
        int hostId = 0;
        int ram = 2048; // host memory (MB)
        long storage = 1000000; // host storage
        int bw = 10000;
        hostList.add(
                new Host(
                        hostId,
                        new RamProvisionerSimple(ram),
                        new BwProvisionerSimple(bw),
                        storage,
                        peList,
                        new VmSchedulerTimeShared(peList)
                )
        ); // This is our machine
        // 5. Create a DatacenterCharacteristics object that stores the
        // properties of a data center: architecture, OS, list of
```

```
                // Machines, allocation policy: time- or space-shared, time zone
                // and its price (G$/Pe time unit).
                String arch = "x86"; // system architecture
                String os = "Linux"; // operating system
                String vmm = "Xen";
                double time_zone = 10.0; // time zone this resource located
                double cost = 3.0; // the cost of using processing in this resource
                double costPerMem = 0.05; // the cost of using memory in this resource
                double costPerStorage = 0.001; // the cost of using storage in this
                                                                        // resource
                double costPerBw = 0.0; // the cost of using bw in this resource
                LinkedList<Storage> storageList = new LinkedList<Storage>(); // we are not adding SAN
                                                                        // devices by now

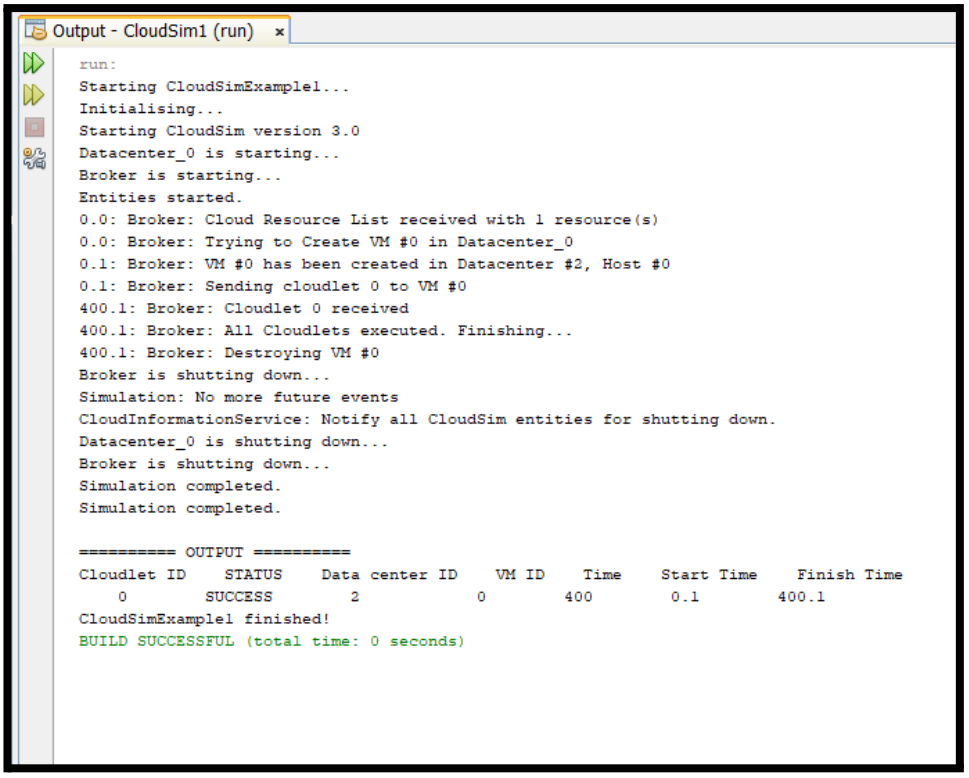

                DatacenterCharacteristics characteristics = new DatacenterCharacteristics(
                                arch, os, vmm, hostList, time_zone, cost, costPerMem,
                                costPerStorage, costPerBw);
                // 6. Finally, we need to create a PowerDatacenter object.
                Datacenter datacenter = null;
                try {
                        datacenter = new Datacenter(name, characteristics, new
VmAllocationPolicySimple(hostList), storageList, 0);
                } catch (Exception e) {
                        e.printStackTrace();
                }
                return datacenter;}
        // We strongly encourage users to develop their own broker policies, to
        // submit vms and cloudlets according
        // to the specific rules of the simulated scenario
        /**
         * Creates the broker.
         *
         * @return the datacenter broker
```

```
 */
private static DatacenterBroker createBroker() {
	DatacenterBroker broker = null;
	try {
		broker = new DatacenterBroker("Broker");
	} catch (Exception e) {
		e.printStackTrace();
		return null;
	}
	return broker;
}
// Prints the Cloudlet objects.
//@param list list of Cloudlets

private static void printCloudletList(List<Cloudlet> list) {
	int size = list.size();
	Cloudlet cloudlet;
	String indent = "    ";
	Log.printLine();
	Log.printLine("========== OUTPUT ==========");
	Log.printLine("Cloudlet ID" + indent + "STATUS" + indent
			+ "Data center ID" + indent + "VM ID" + indent + "Time" + indent
			+ "Start Time" + indent + "Finish Time");
	DecimalFormat dft = new DecimalFormat("###.##");
	for (int i = 0; i < size; i++) {
		cloudlet = list.get(i);
		Log.print(indent + cloudlet.getCloudletId() + indent + indent);
		if (cloudlet.getCloudletStatus() == Cloudlet.SUCCESS) {
			Log.print("SUCCESS");
			Log.printLine(indent + indent + cloudlet.getResourceId()
					+ indent + indent + indent + cloudlet.getVmId()
					+ indent + indent
```

```
                                        + dft.format(cloudlet.getActualCPUTime()) + indent
                                        + indent + dft.format(cloudlet.getExecStartTime())
                                        + indent + indent
                                        + dft.format(cloudlet.getFinishTime()));
                }}}}
```

Output:



Conclusion: Successfully executed and checked the performance of algorithm using CloudSim.

Practical No: 2

Aim : Install a Cloud Analyst and Integrate with Eclipse/Netbeans. Monitor the performance of Existing Algorithms.

Theory:

CloudAnalyst

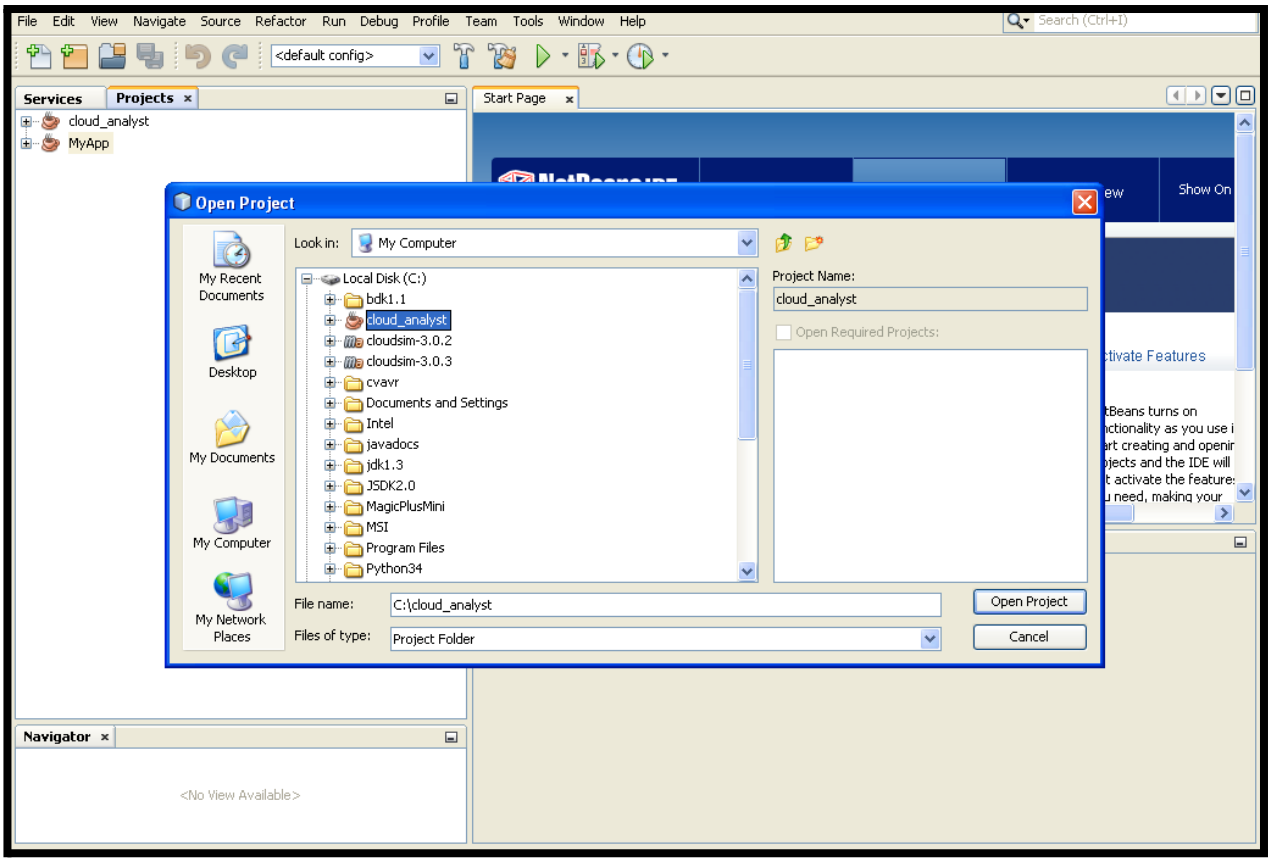
Cloud Analyst is a tool developed at the University of Melbourne whose goal is to support evaluation of social networks tools according to geographic distribution of users and data centres. In this tool, communities of users and data centres supporting the social networks are characterised and, based on their location; parameters such as user experience while using the social network application and load on the data centre are obtained/logged

Installation is just download cloud_analyst from

https://sourceforge.net/projects/cloudanalystnetbeans/

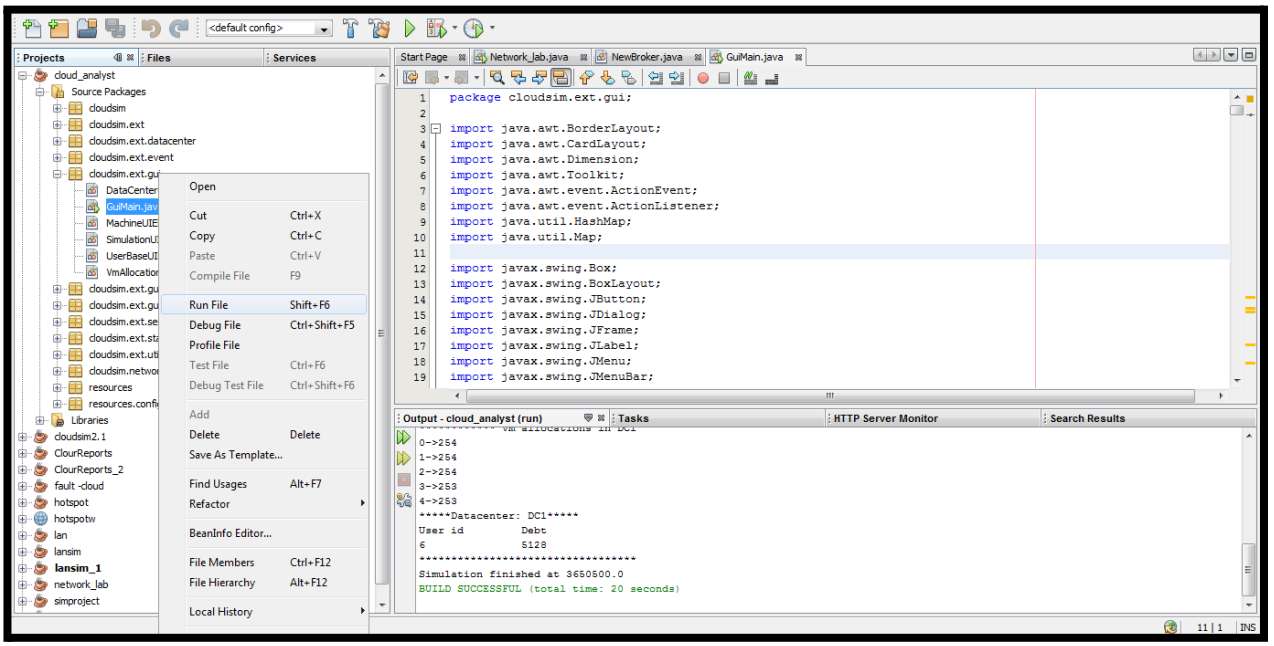
and copy in drive .....then unzip it to use

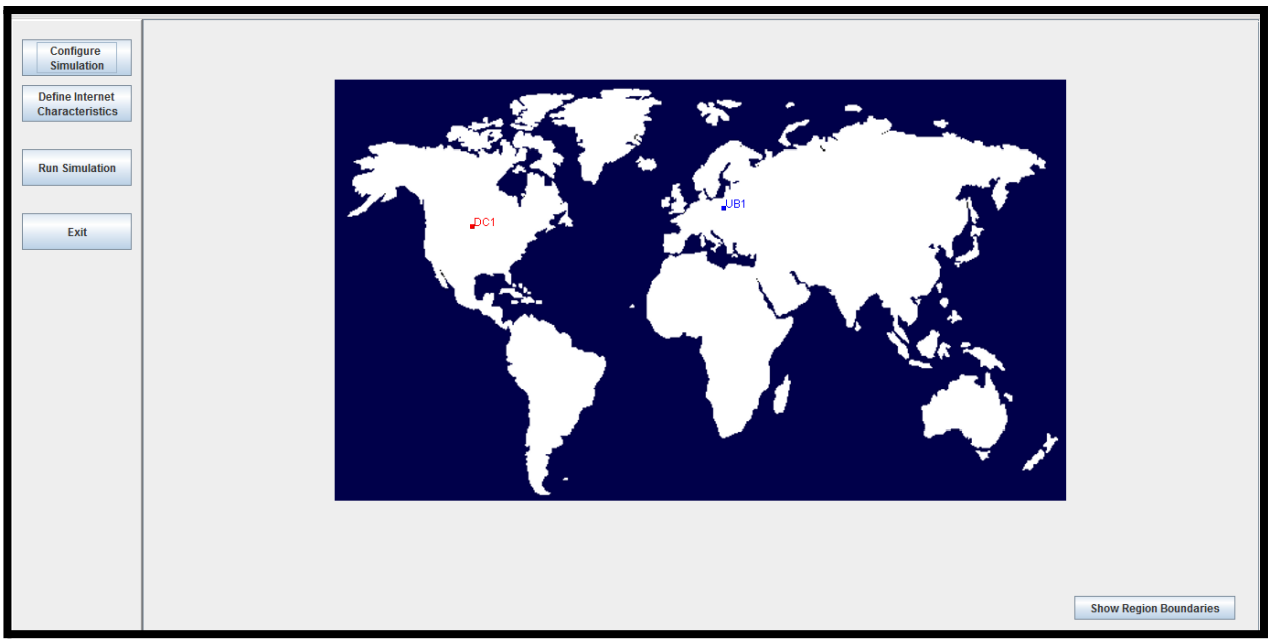
After installation.....In netbeans7.4.....To run it simply go to file->open project --> simply browse the unziped folder



Expand source package folder inside which open cloudsim.ext.gui

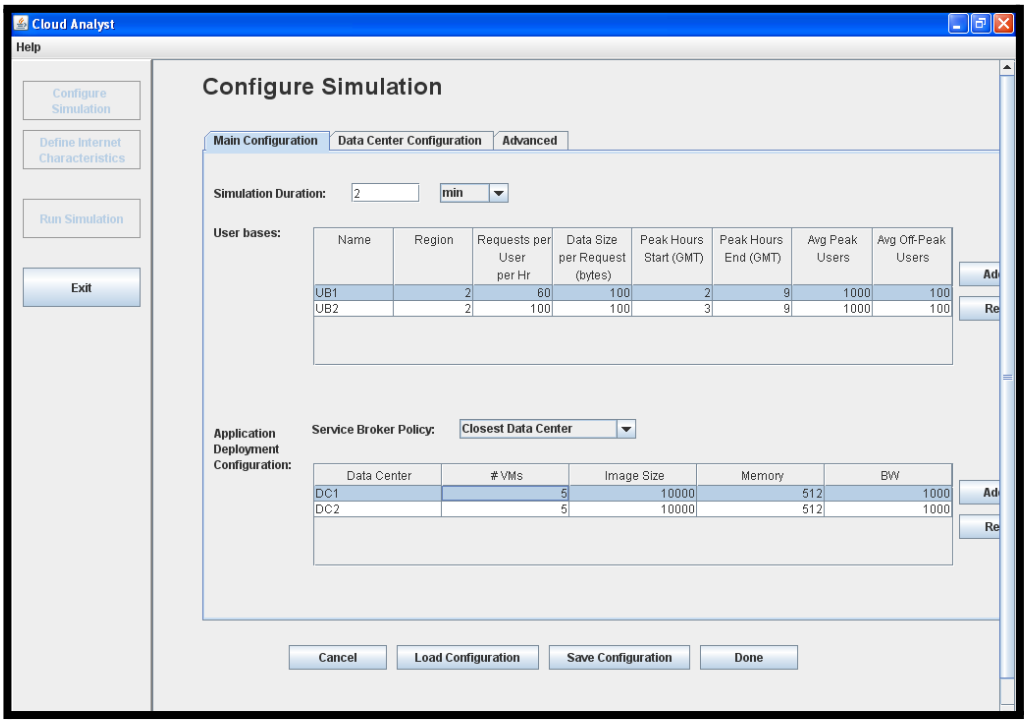
right click on the GuiMain.java and click run



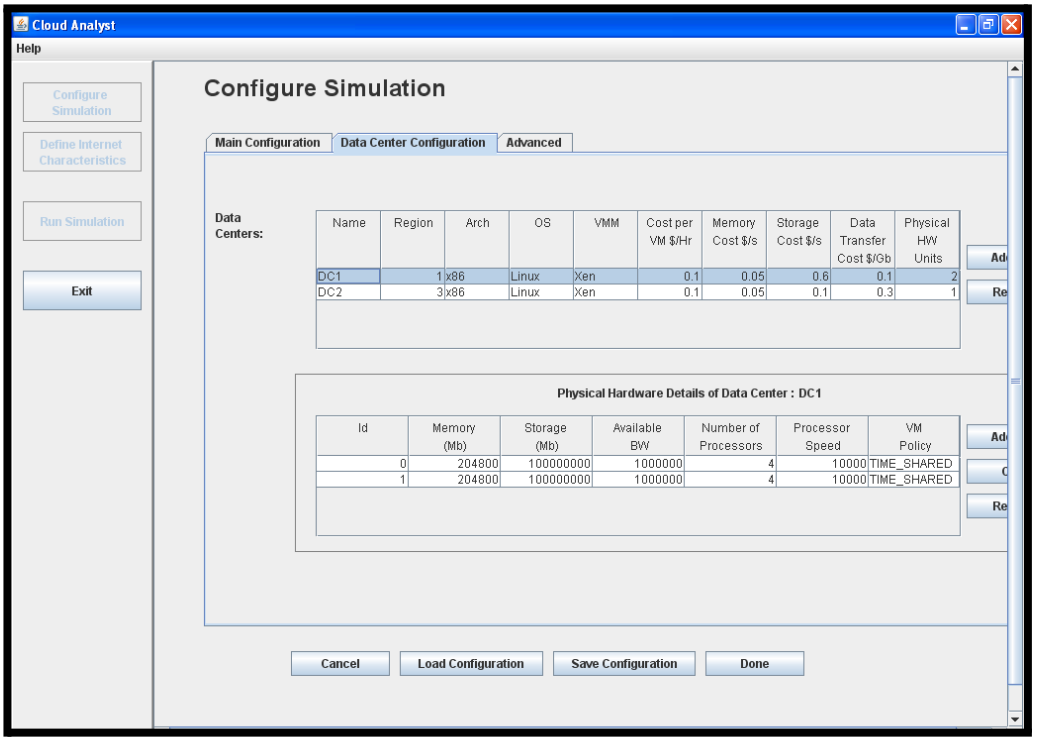


Now we are ready for checking algorithm performance

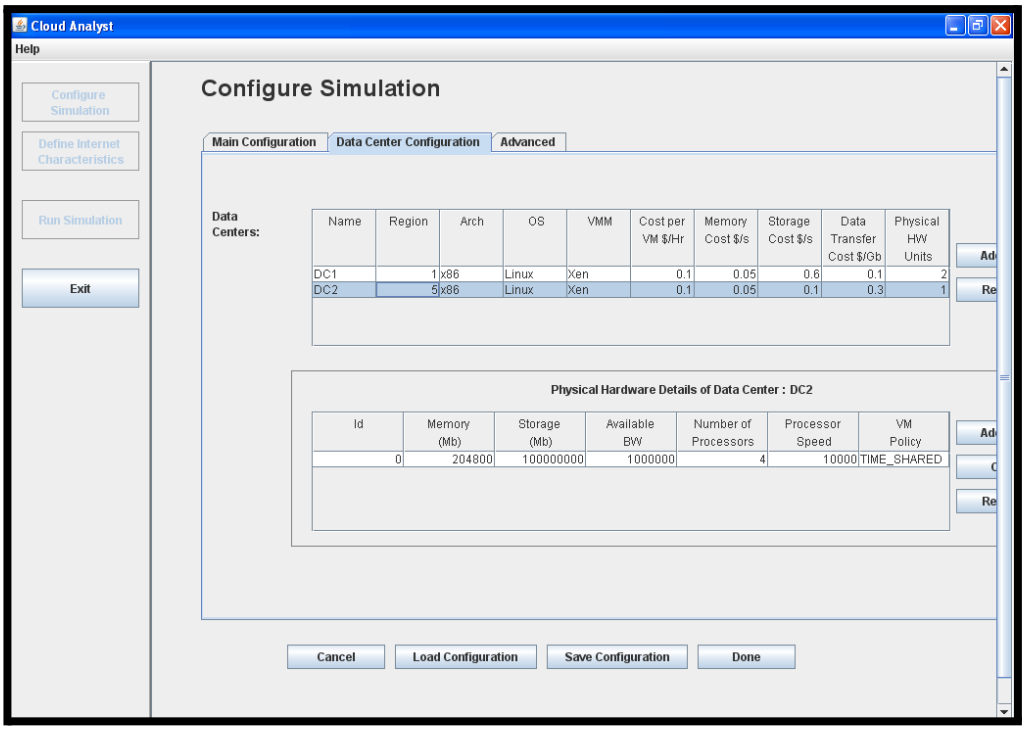
In Configure simulation window -> Main configuration tab set the values to



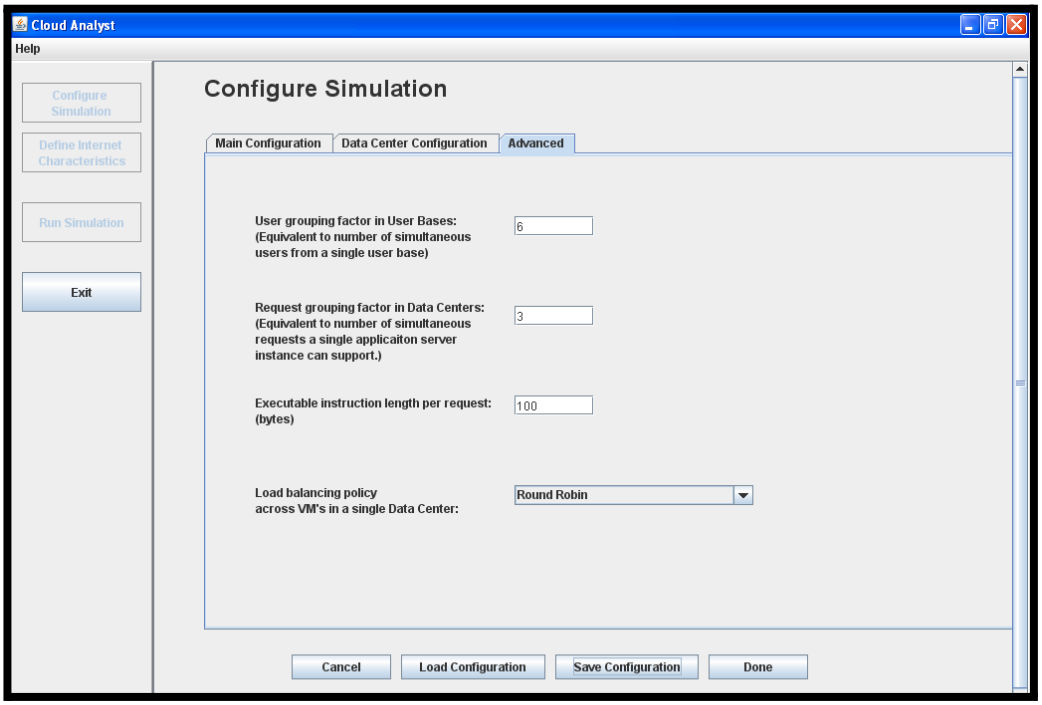
In data center configuration tab

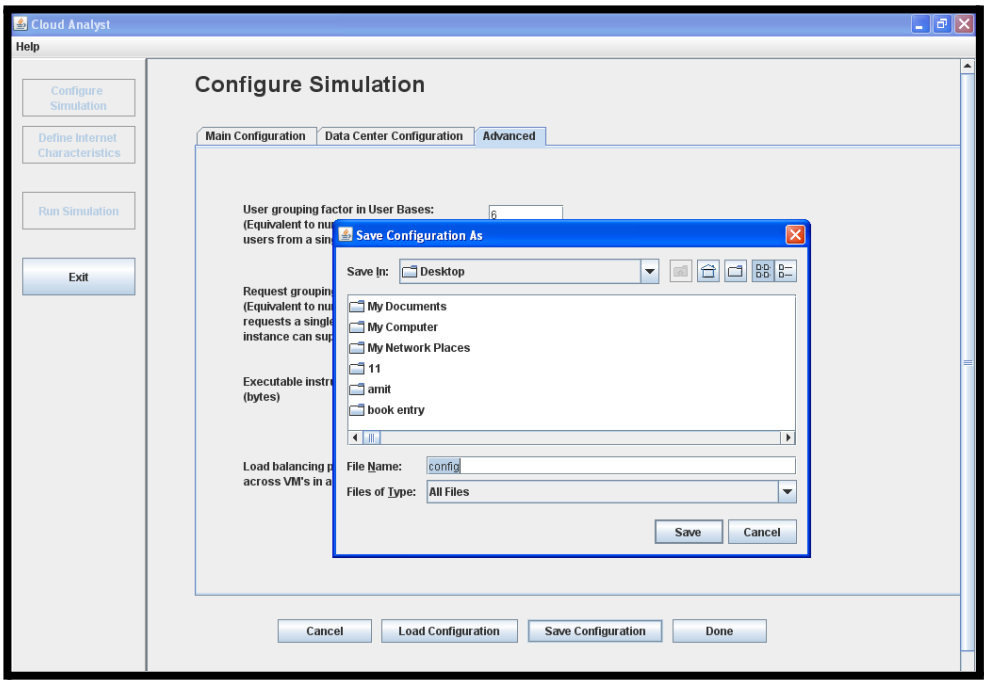


And

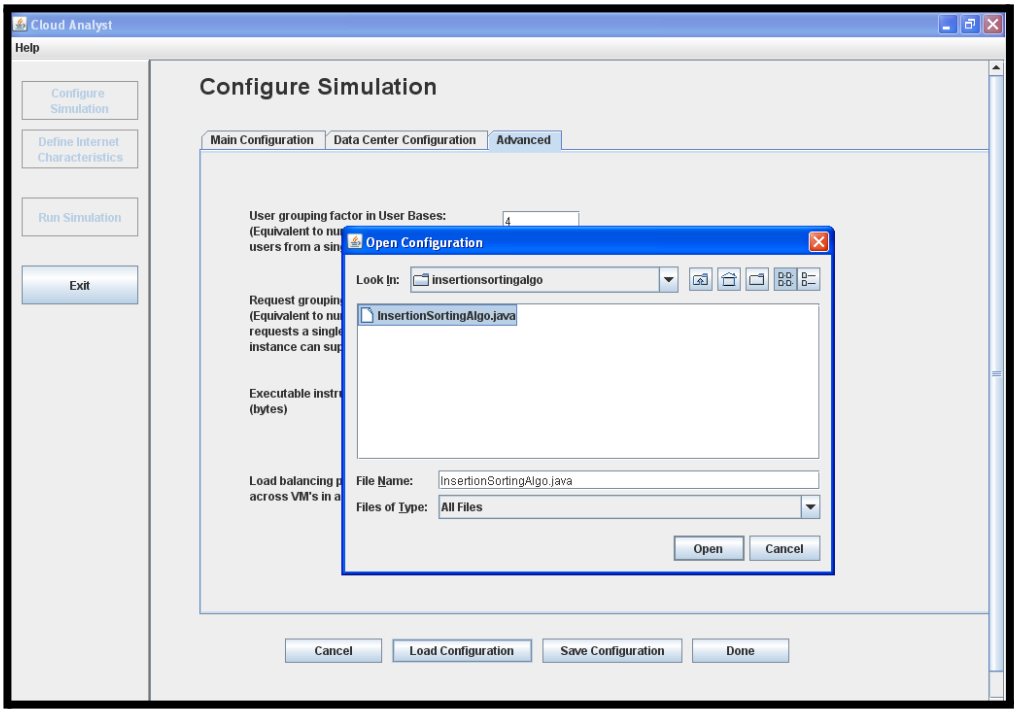


In advanced tab

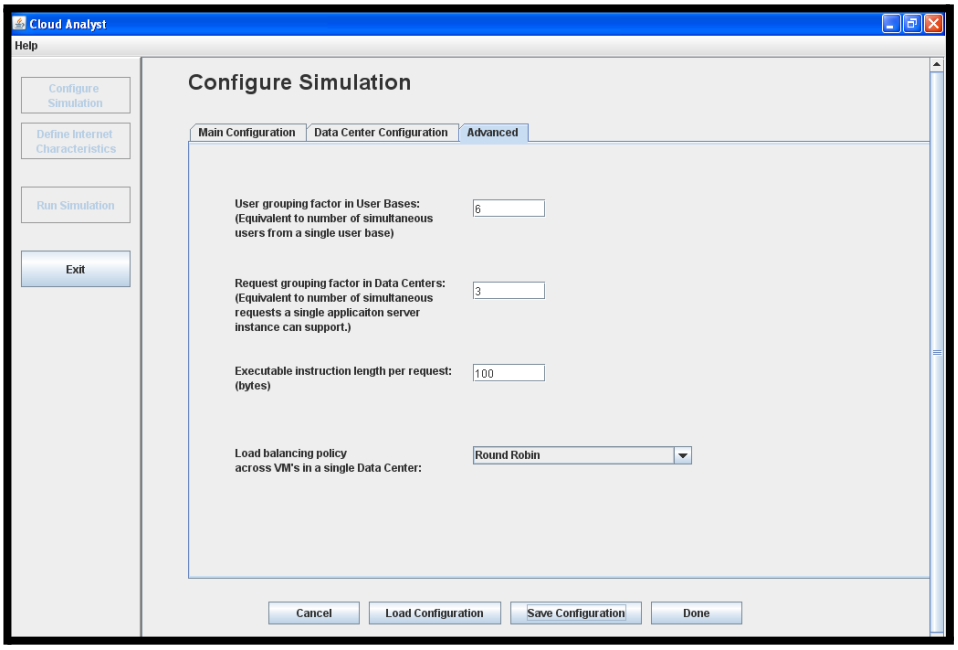




then click on load configuration button and select algorithm as

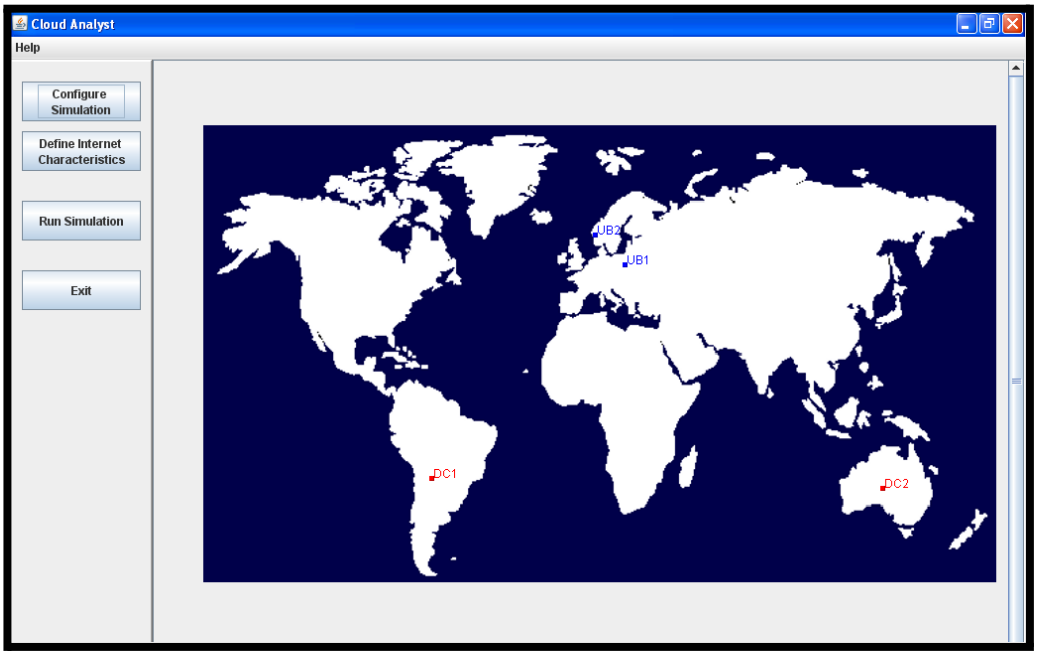


then save configuration as

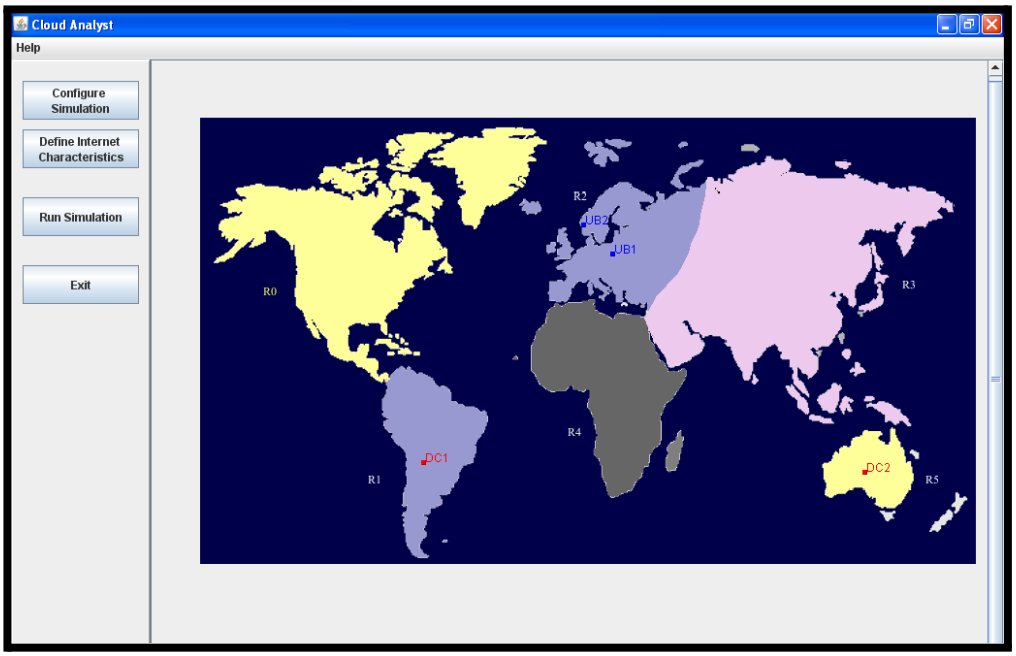


.sim file will saved to given location

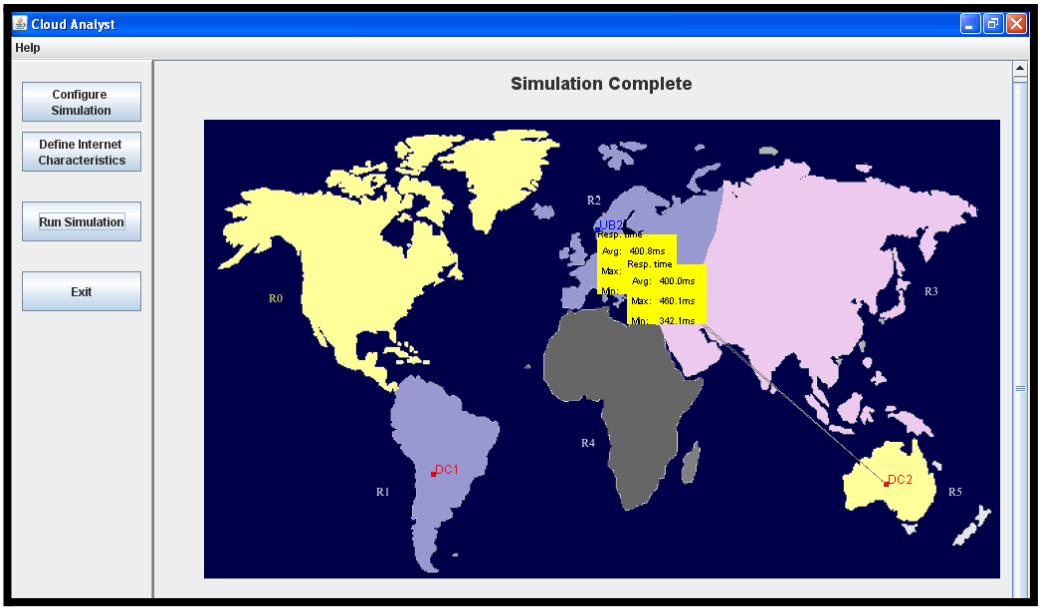
then click on done

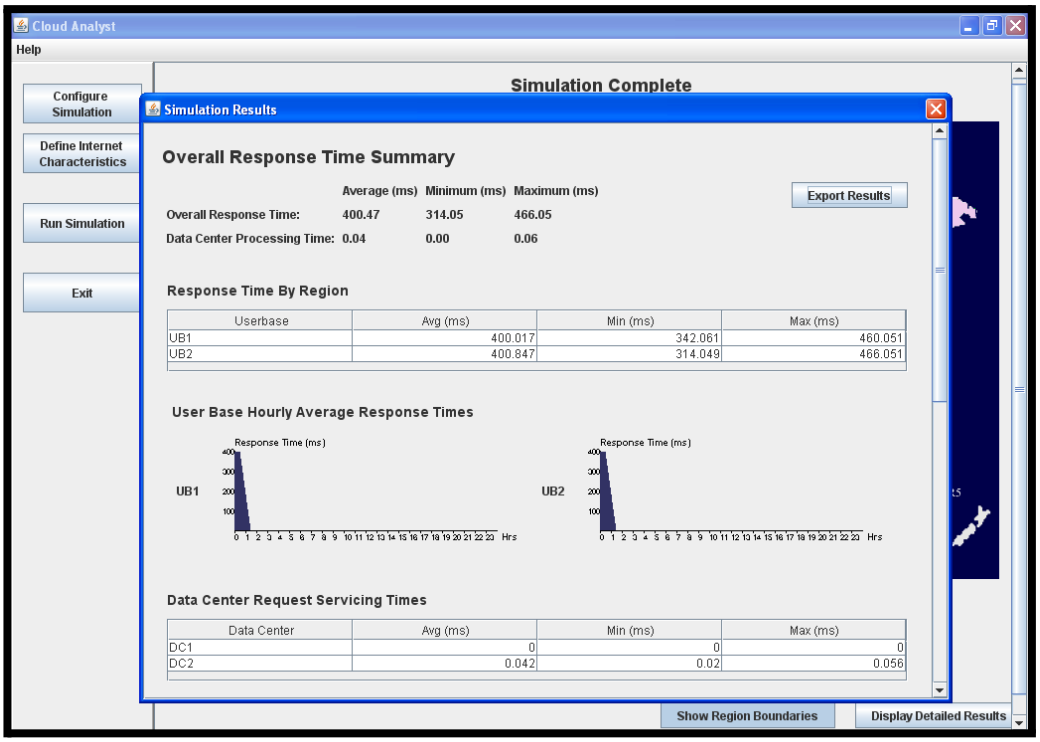


click on show region boundaries



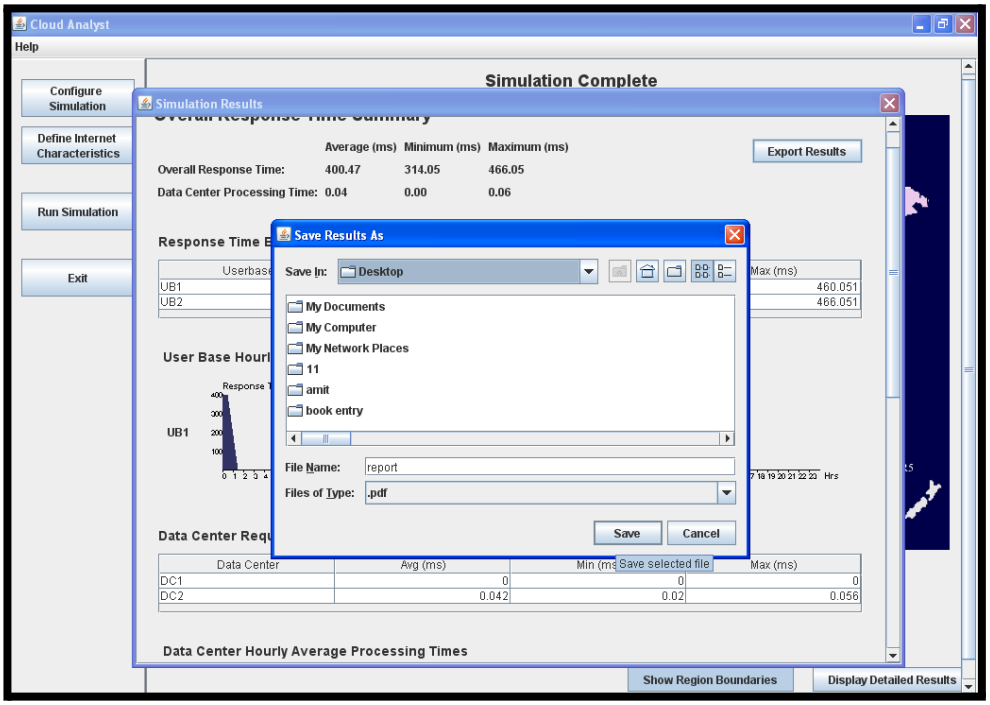
then click run simulation





then click on export result

pdf will be saved to given location



Output:

simulation time =120000.0ms

Starting Simulation...

Initialising...

Creating new broker DC1-Broker

Creating new broker DC2-Broker

0.0 Creating new user base UB1

0.0 Creating new user base UB2

Starting GridSim version 4.2

Entities started.

Starting user base 9 UB1

Starting broker 8 name=DC2-Broker

Starting user base 11 UB2

Starting broker 6 name=DC1-Broker

Starting internet 13

5.0: DC1-Broker: Cloud Resource List received with 2 resource(s)

5.0: DC2-Broker: Cloud Resource List received with 2 resource(s)

5.0: DC2-Broker: Trying to Create VM #0

5.0: DC2-Broker: Trying to Create VM #1

5.0: DC2-Broker: Trying to Create VM #2

5.0: DC2-Broker: Trying to Create VM #3

5.0: DC2-Broker: Trying to Create VM #4

5.0: DC1-Broker: Trying to Create VM #0

5.0: DC1-Broker: Trying to Create VM #1

5.0: DC1-Broker: Trying to Create VM #2

5.0: DC1-Broker: Trying to Create VM #3

5.0: DC1-Broker: Trying to Create VM #4

Gathering simulation data.

DC1-Broker finalizing, submitted cloudlets=0 processing cloudlets=0 ,allRequestsProcessed=0

Got response for 900047 but it seems to be completed.

DC2-Broker finalizing, submitted cloudlets=226 processing cloudlets=0 ,allRequestsProcessed=670

UB1 finalizing. Messages sent:52, Received:52

```
UB1 requests sent=302 , received=302

UB2 finalizing. Messages sent:63, Received:63

UB2 requests sent=368 , received=368

Simulation completed.

********** Vm allocations in DC1

0->2

1->1

2->1

3->1

4->1

********** Vm allocations in DC2

0->92

1->92

2->92

3->91

4->91

*****Datacenter: DC1*****

User id                 Debt

6                       30128

**********************************

*****Datacenter: DC2*****

User id                 Debt

8                       5128

**********************************

Simulation finished at 145540.0

BUILD SUCCESSFUL (total time: 37 minutes 1 second)
```

Conclusion: Successfully executed and checked the performance of algorithm using CloudSim

Practical No : 3

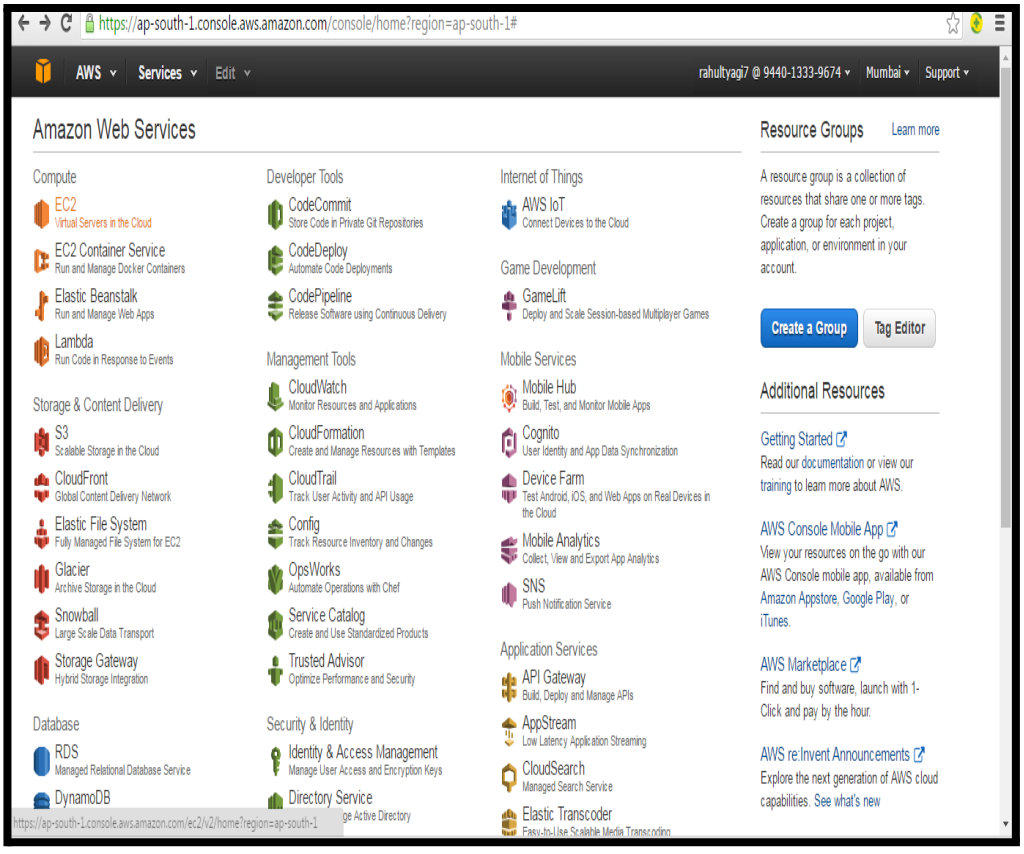
Aim : Build an application on private cloud.

Note :

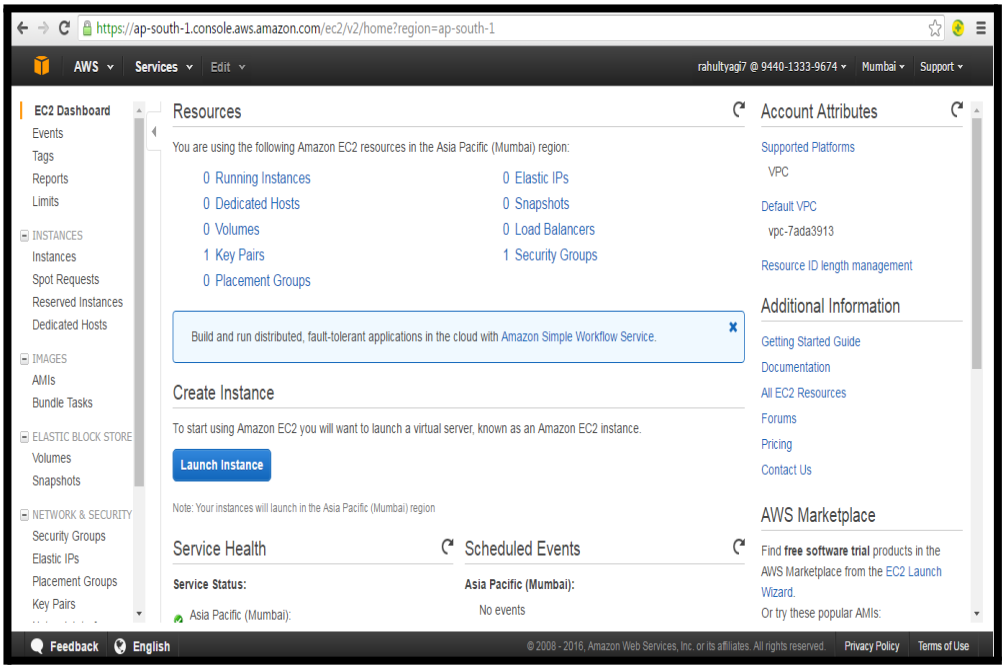
1. Internet Connectivity Required
2. Preferably use Mozilla Firefox as browser
3. Install putty and puttygen and then start with practical
4. AWS Demo Account is required

Login to the aws manager account at link https://aws.amazon.com/

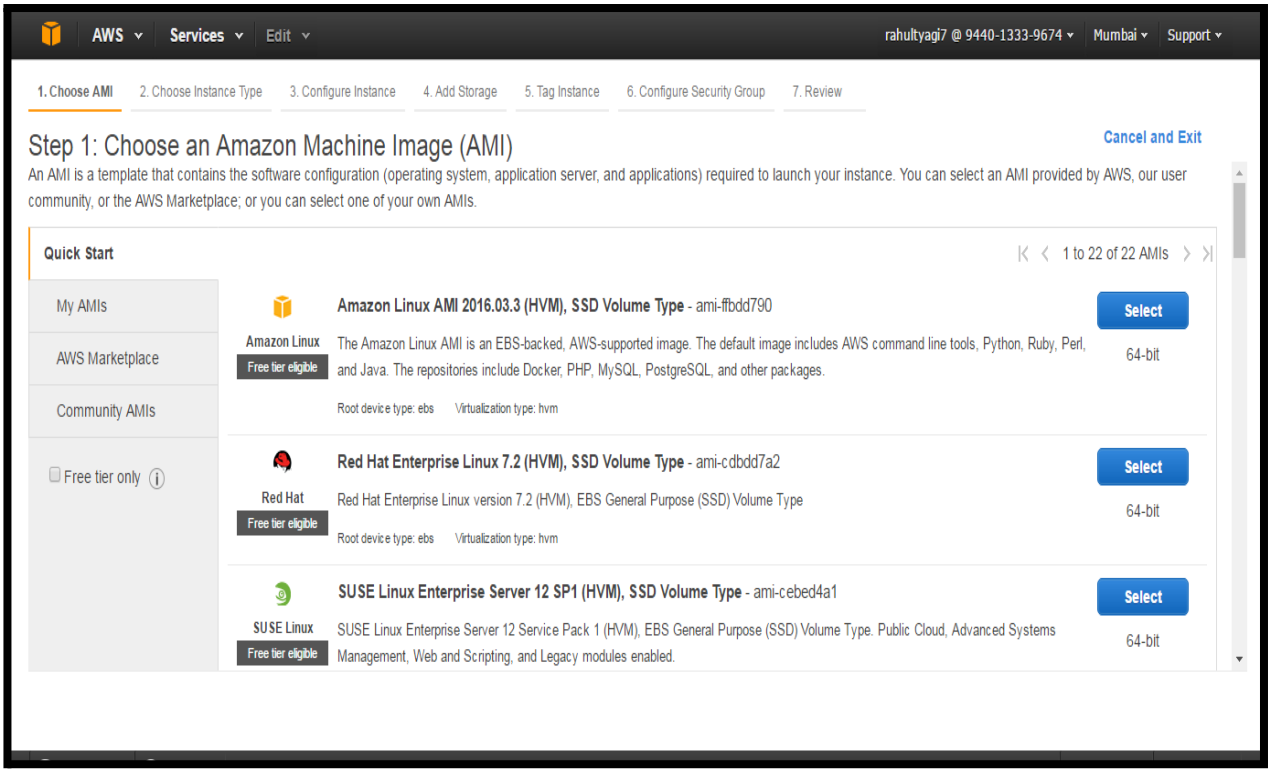
Select EC2



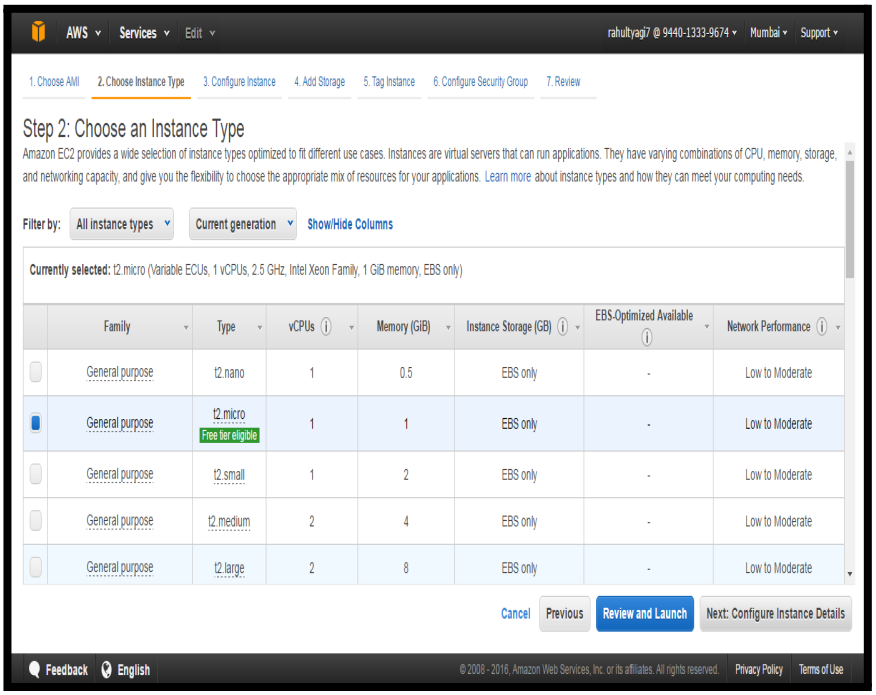
Click on EC2 and open it ,then select Launch INSTANCE



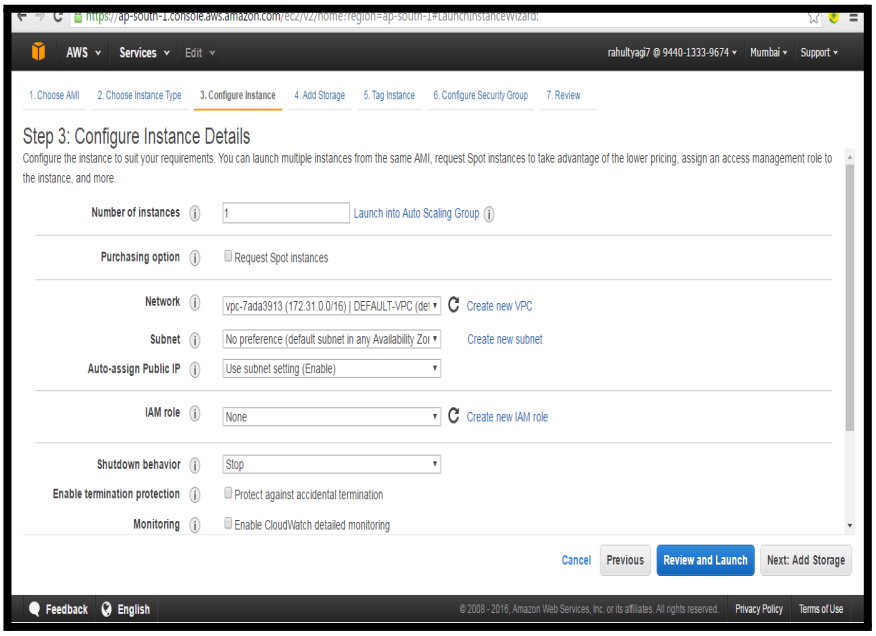
Select Linux AMI



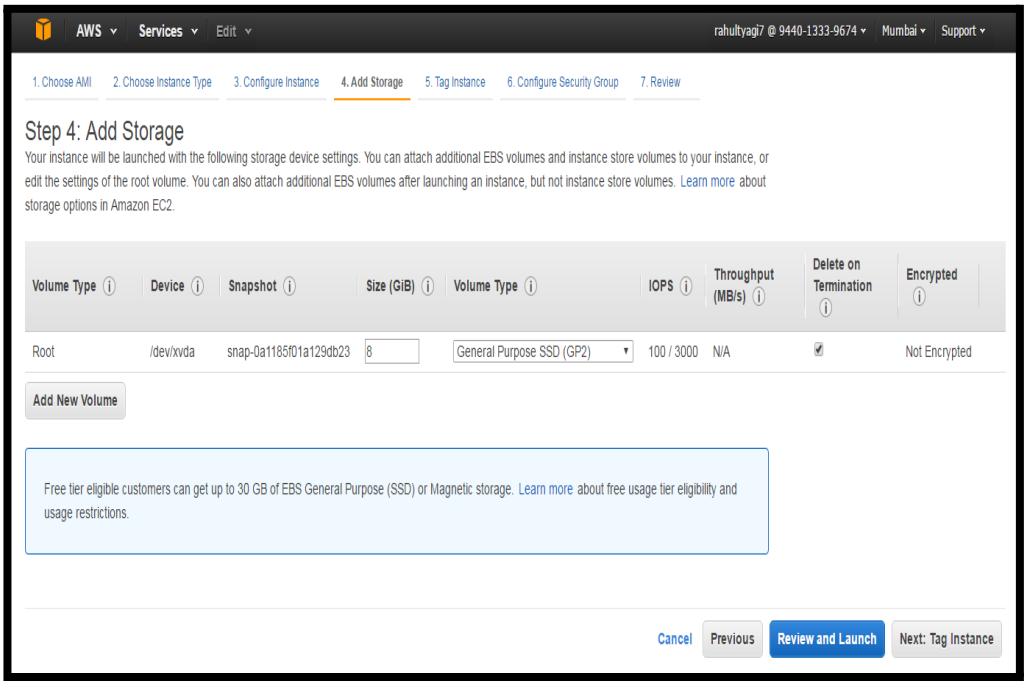
Chose the instance type



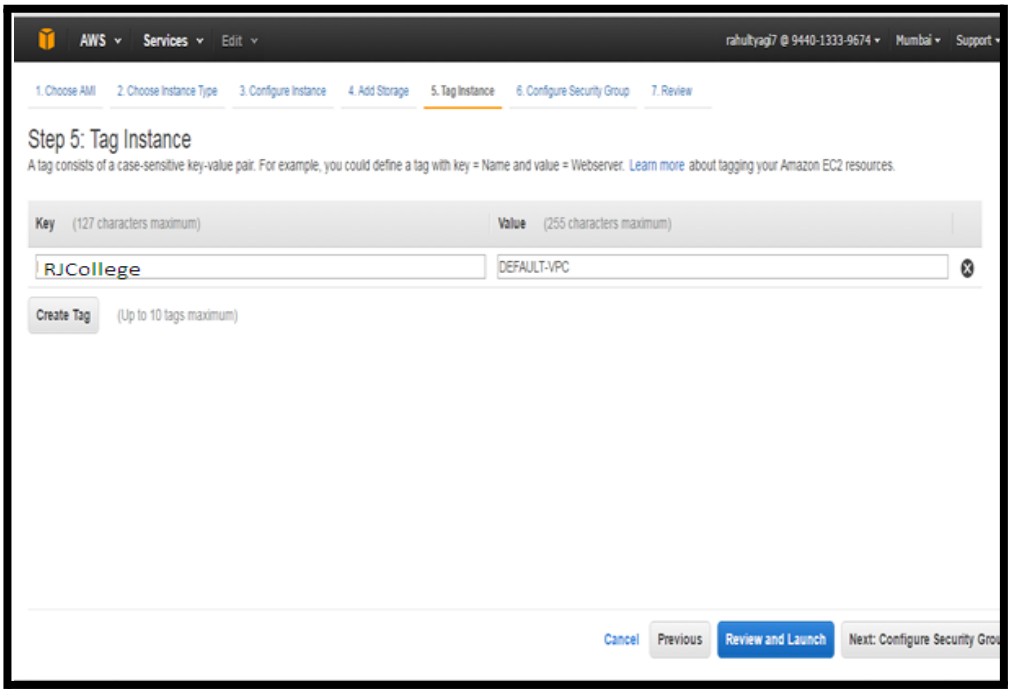
Configure the Instance



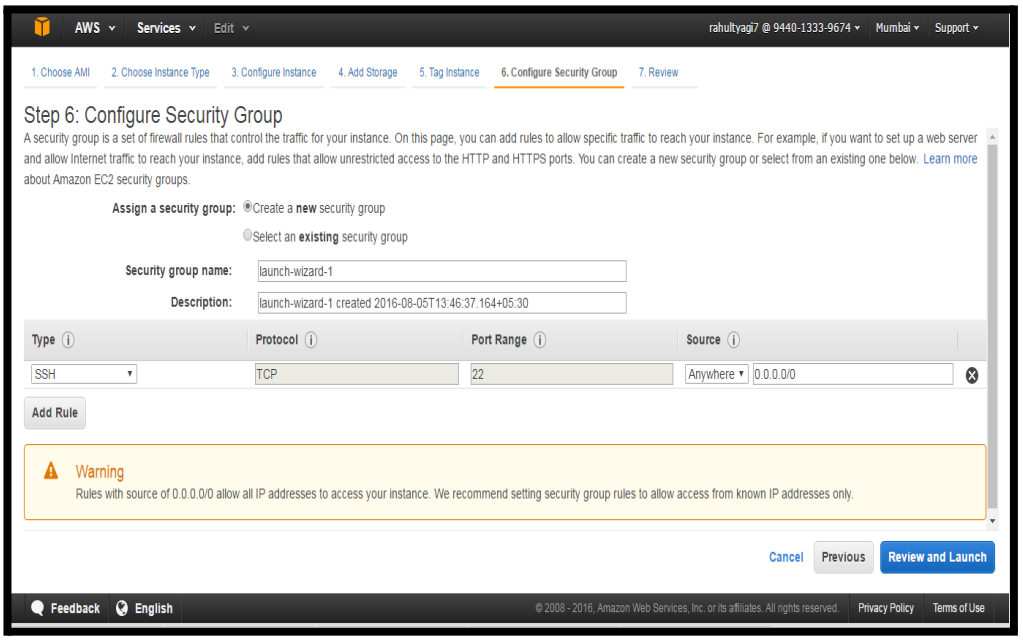
Add the Storage type .according to the Requirement



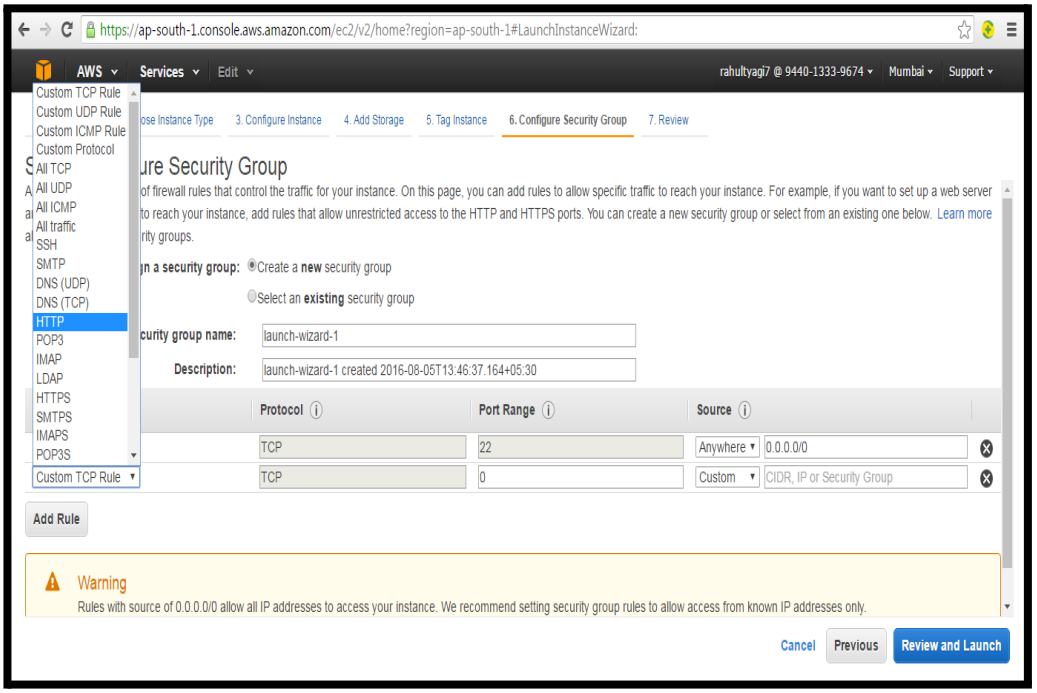
Tag Instance .key and the value



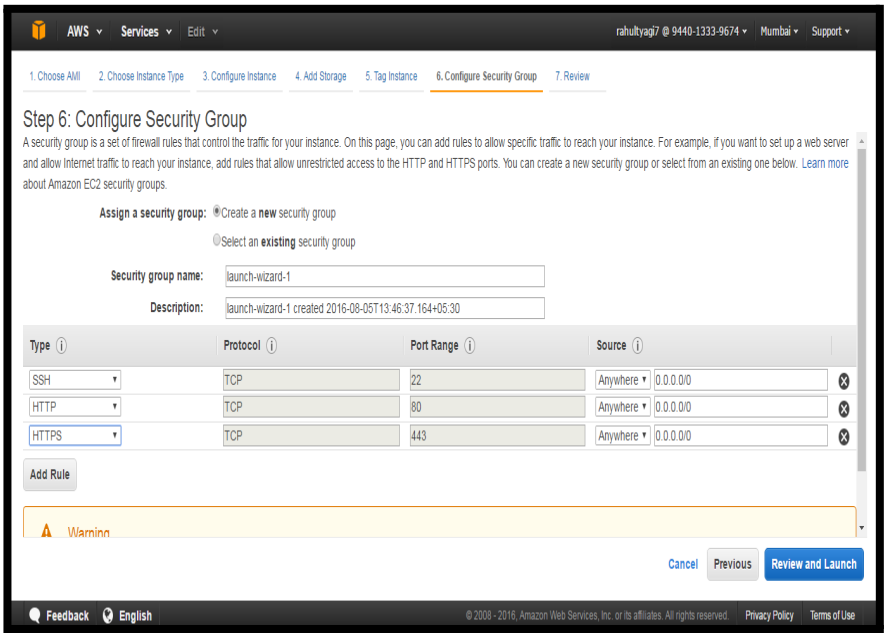
Configure the security group by adding some rule..



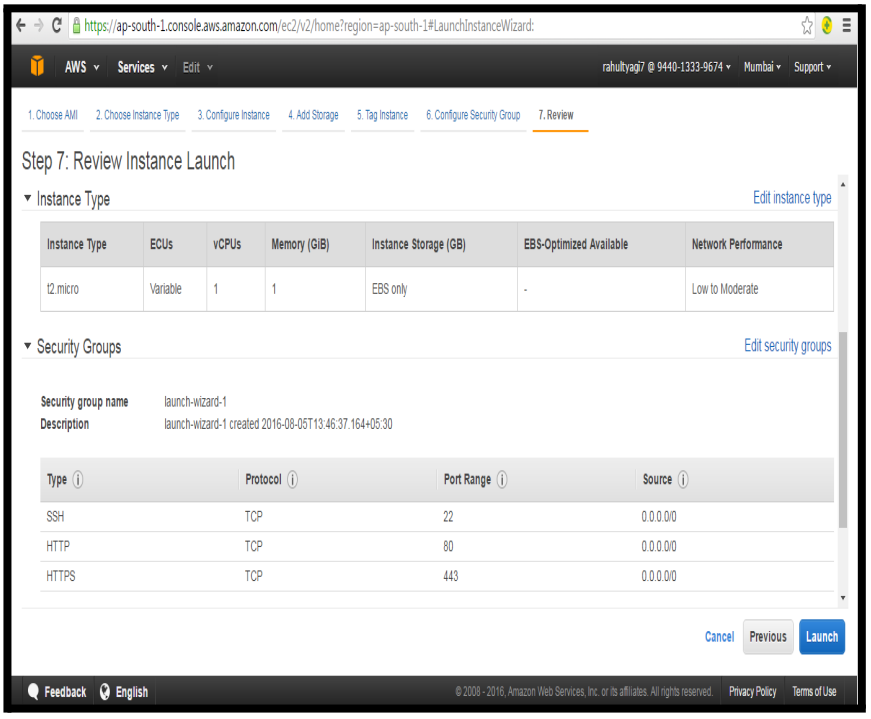
Add the first rule Http.



Add the second rule Https

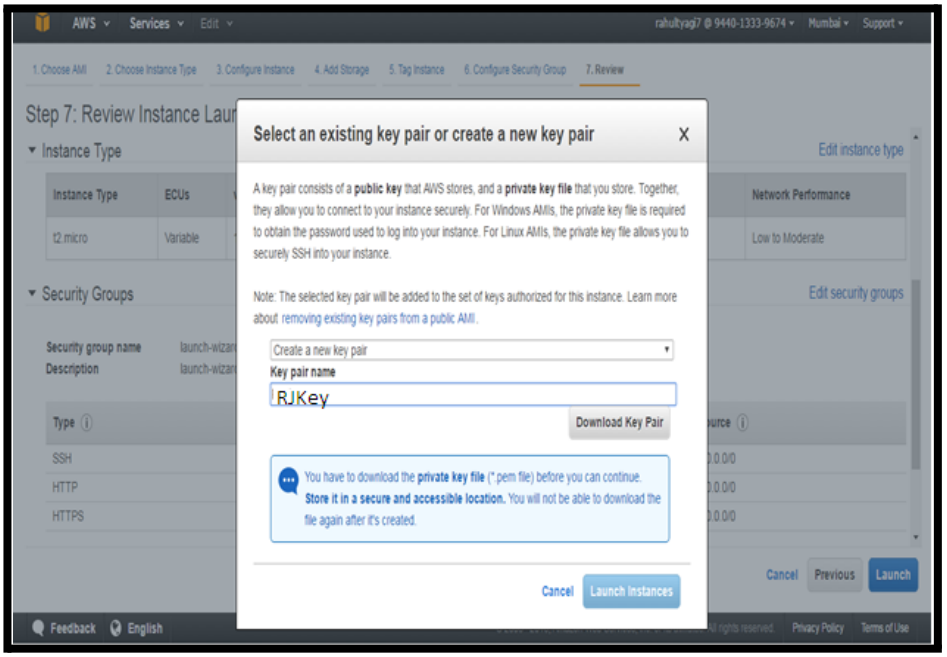


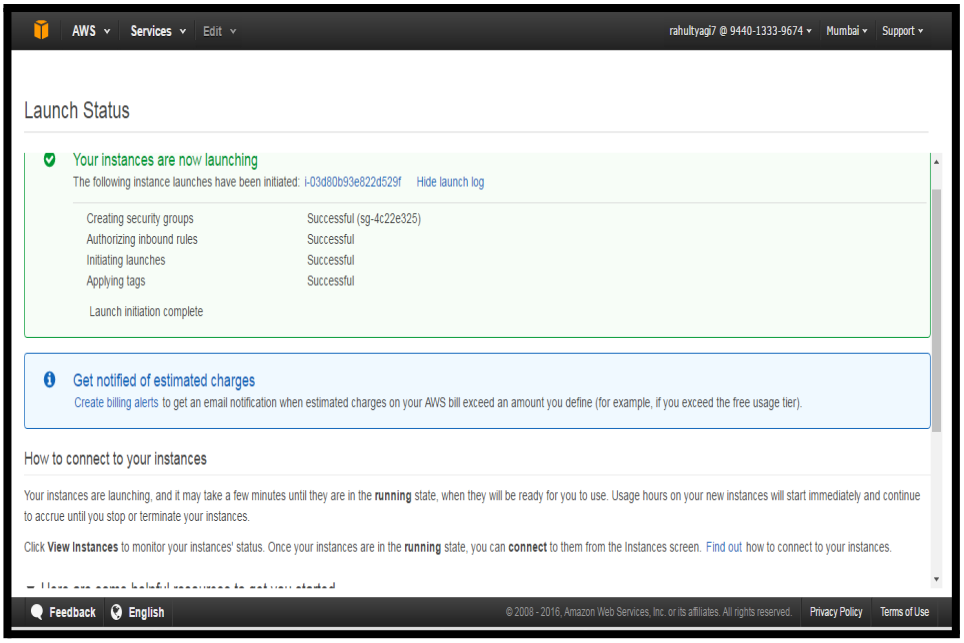
Review the Instance before launch



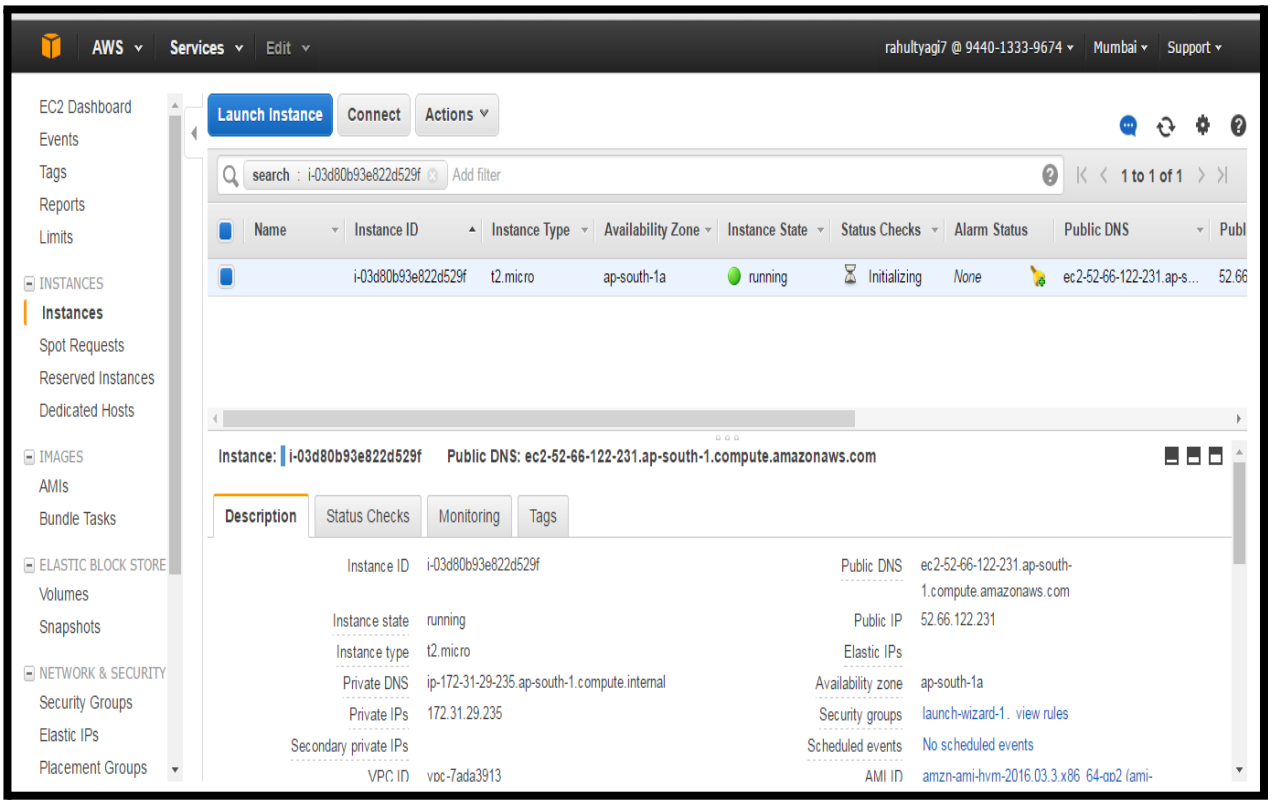
Once by Launching instance.thedailoug box is pop up.by the massage select key creating option.

And Download it.





Once you launch the instance it will take some minutes to Launch it…the Public DNS address will be launch

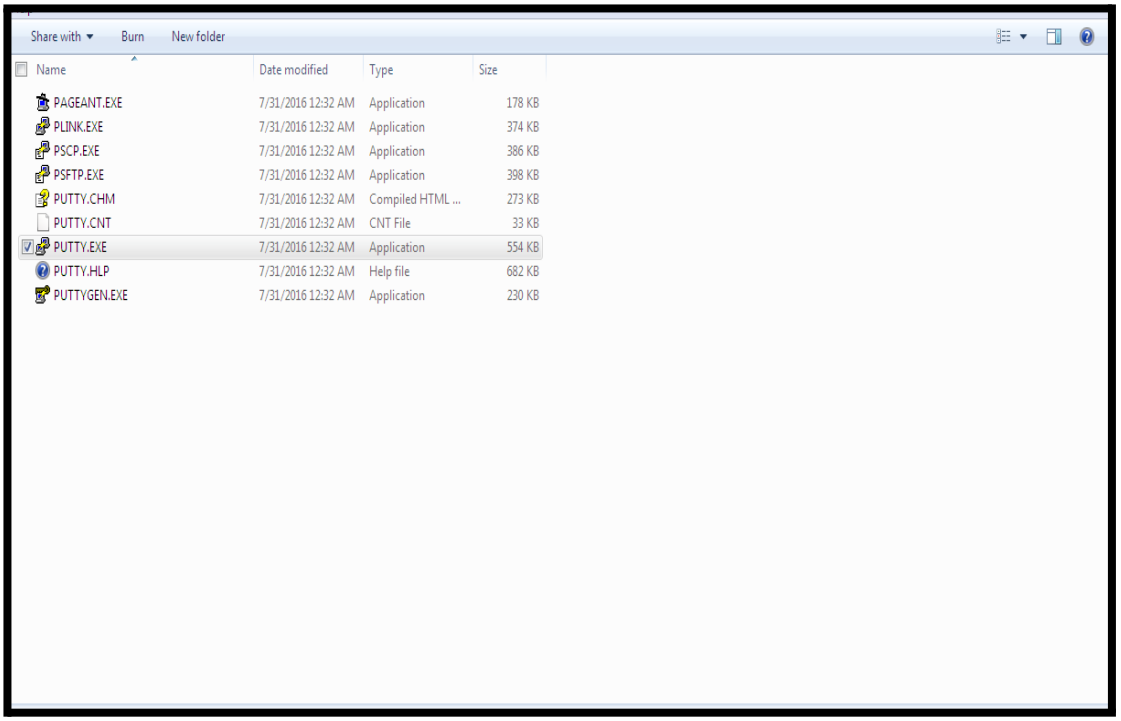


Here we have completed with instance creation.....

Now we have to connect this created instance with LAMP Server.....

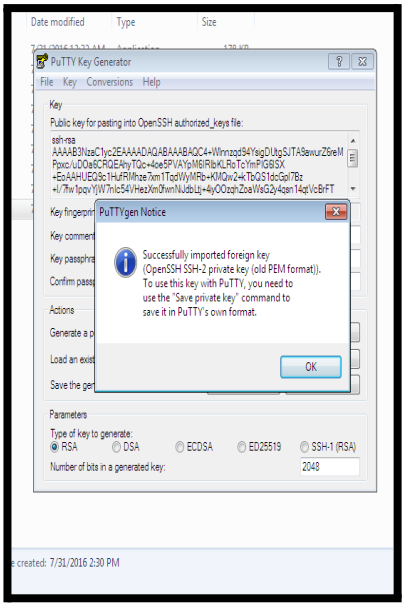
Download the puttygen

and putty for the key configuration..

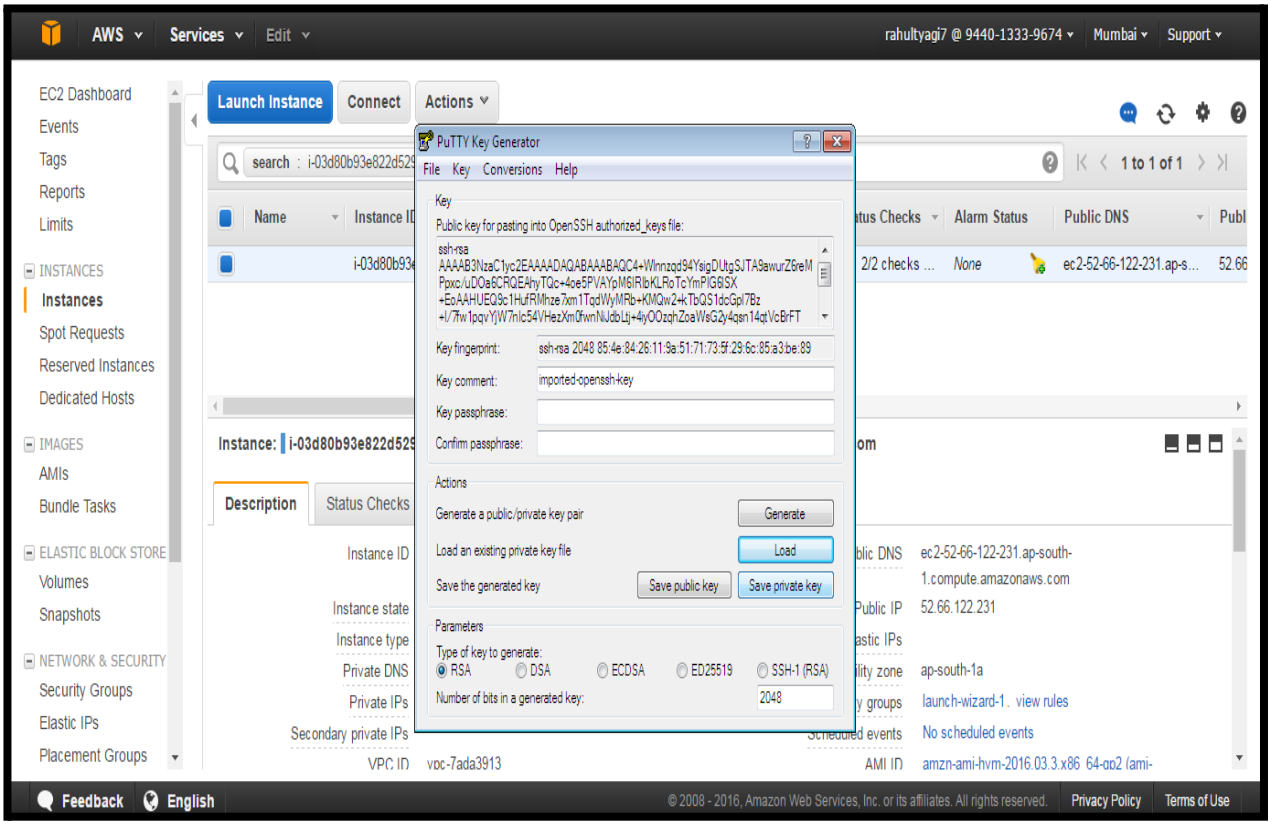


Open the puttygen-> load-><go to address of public key> -> open the public key you had save..

After that save the Private keywhich is use for verification for secure connection.



Save the private key.



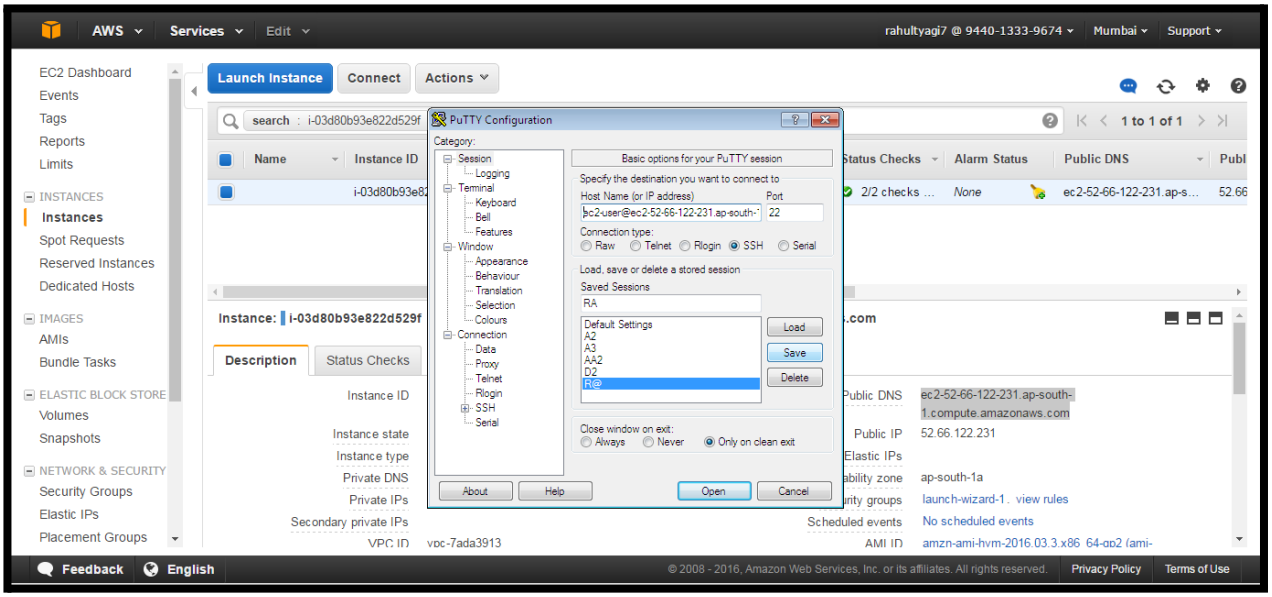
Now open the putty for the key configuration.

In Host Name(Or IP address) write

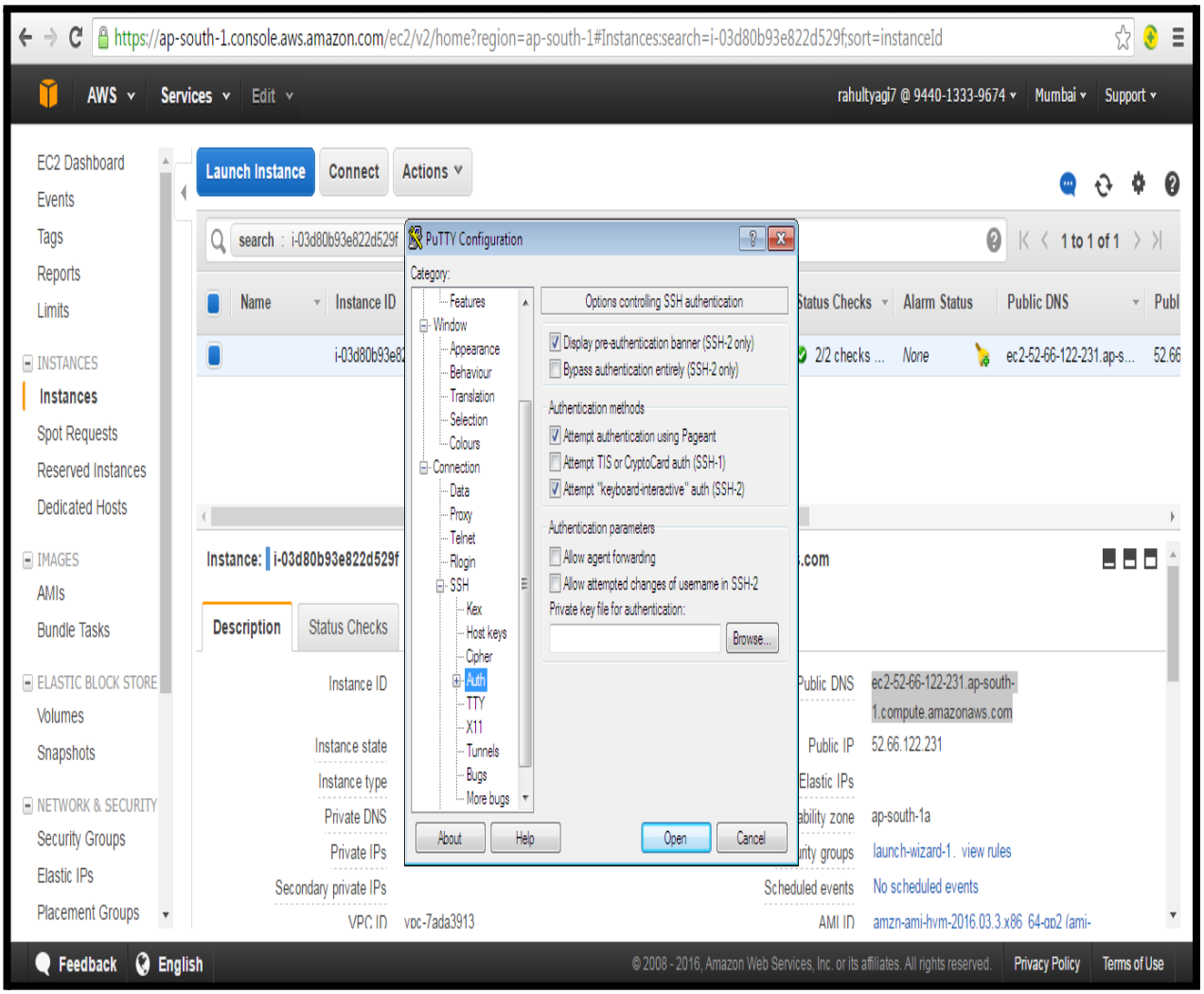
Ec2-user@public DNS.

In saved Session option provide the name and save it.

Now at left side select SSH ->Auth ->Browse(Right side)->select private key from where had it saved.....

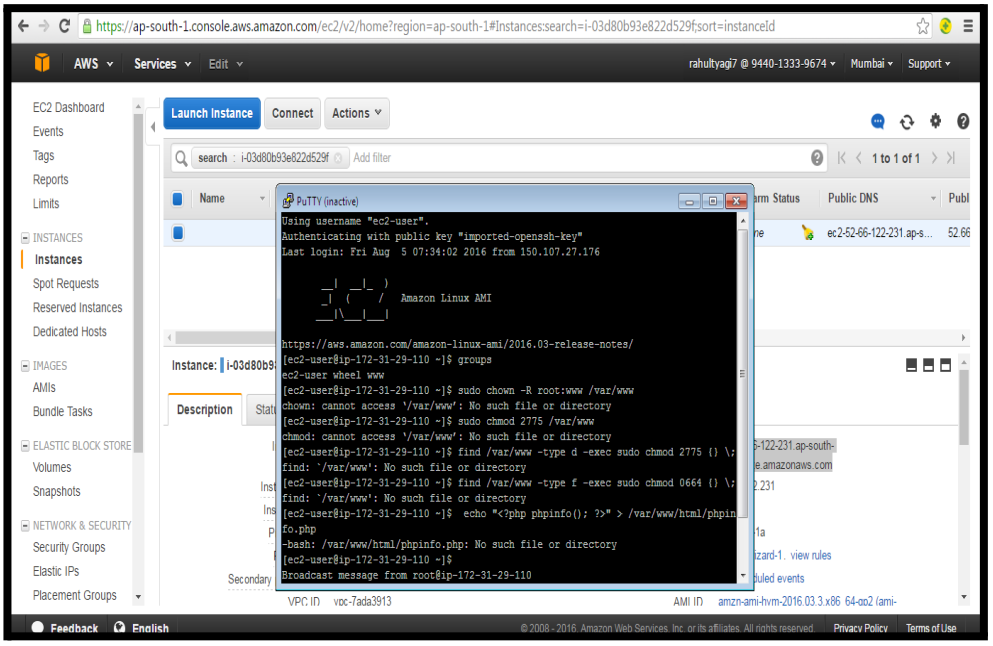


Now select the private key you have save.

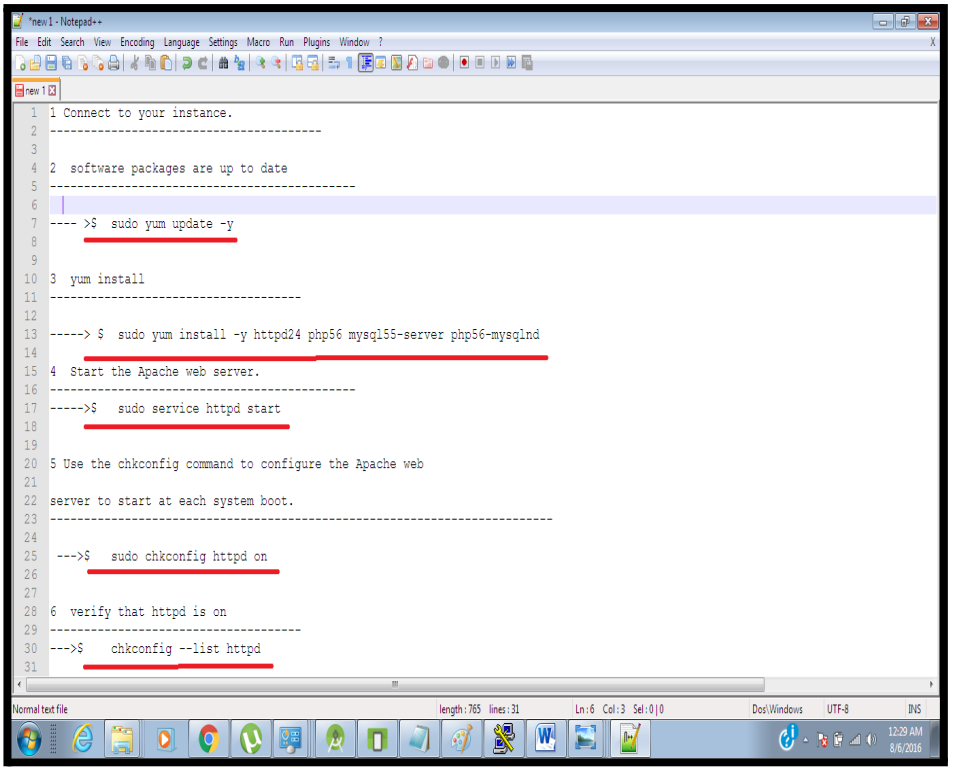


By clicking on the open, the Ec2 Amazon Linux will be launch...

For the configuration use the Command provided in putty.....



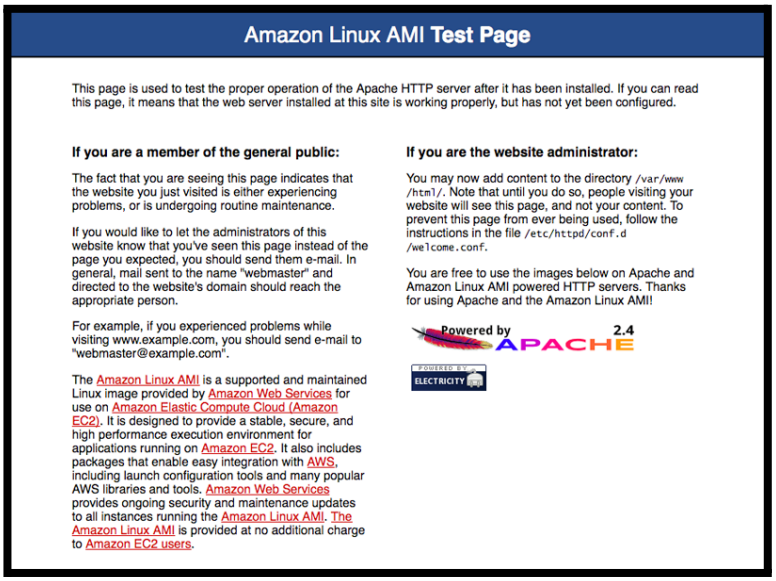
Command for launching Lamp server.



Now minimize putty.....

Copy Public DNS in new tab of browser.....

If the following page appears means LAMP server is on.....



Amazon Linux AMI Test Page

This page is used to test the proper operation of the Apache HTTP server after it has been installed. If you can read this page, it means that the web server installed at this site is working properly, but has not yet been configured.

If you are a member of the general public:

The fact that you are seeing this page indicates that the website you just visited is either experiencing problems, or is undergoing routine maintenance.

If you would like to let the administrators of this website know that you've seen this page instead of the page you expected, you should send them e-mail. In general, mail sent to the name "webmaster" and directed to the website's domain should reach the appropriate person.

For example, if you experienced problems while visiting www.example.com, you should send e-mail to "webmaster@example.com".

The Amazon Linux AMI is a supported and maintained Linux image provided by Amazon Web Services for use on Amazon Elastic Compute Cloud (Amazon EC2). It is designed to provide a stable, secure, and high performance execution environment for applications running on Amazon EC2. It also includes packages that enable easy integration with AWS, including launch configuration tools and many popular AWS libraries and tools. Amazon Web Services provides ongoing security and maintenance updates to all instances running the Amazon Linux AMI. The Amazon Linux AMI is provided at no additional charge to Amazon EC2 users.

If you are the website administrator:

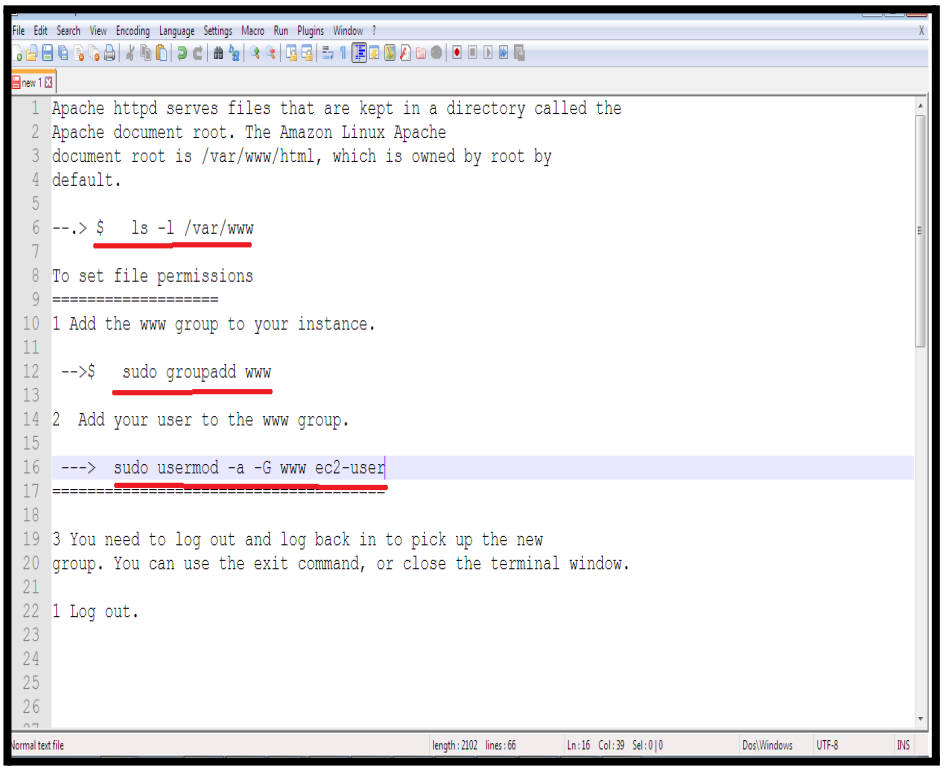
You may now add content to the directory `/var/www/html/`. Note that until you do so, people visiting your website will see this page, and not your content. To prevent this page from ever being used, follow the instructions in the file `/etc/httpd/conf.d/welcome.conf`.

You are free to use the images below on Apache and Amazon Linux AMI powered HTTP servers. Thanks for using Apache and the Amazon Linux AMI!

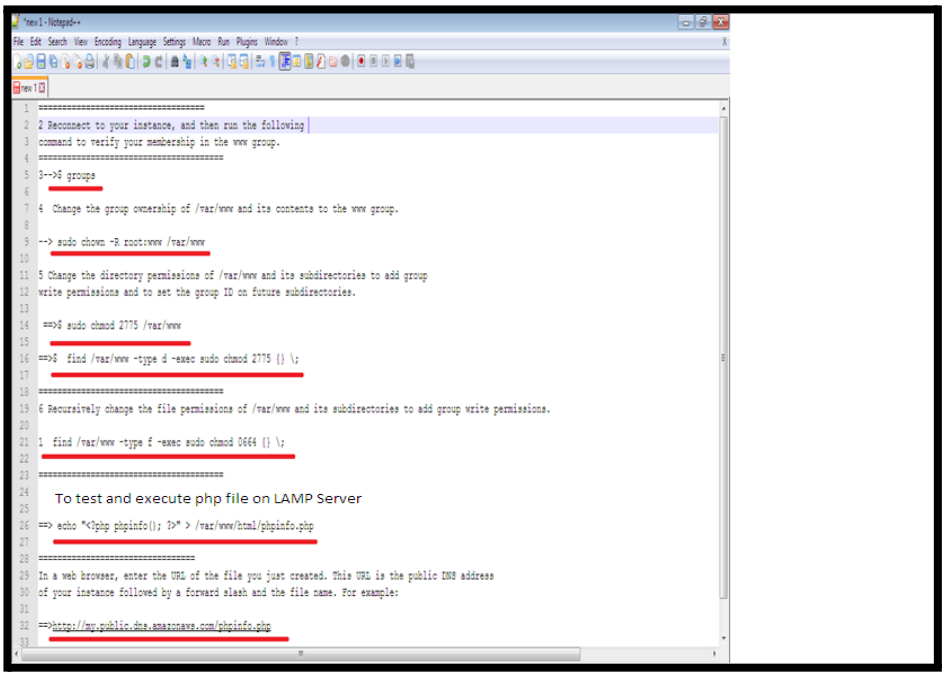
Powered by APACHE 2.4

POWERED BY ELECTRICITY

Minimize it and go to putty to Configure Apache server for PHP

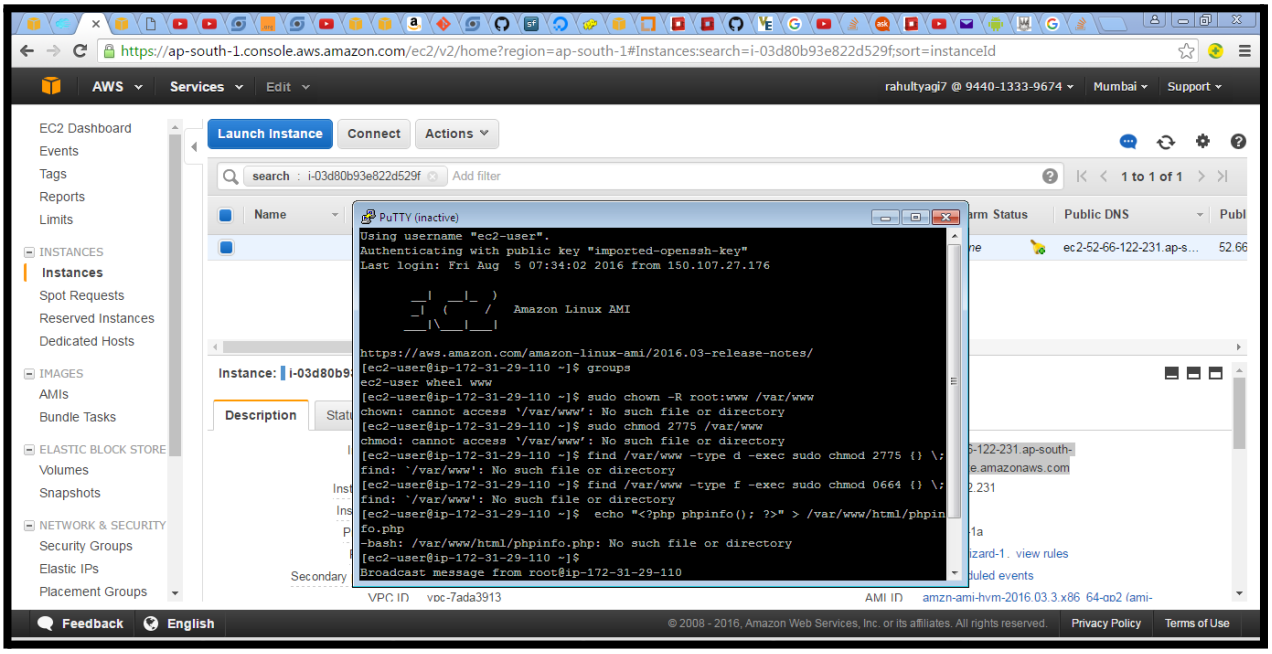


```
1  Apache httpd serves files that are kept in a directory called the
2  Apache document root. The Amazon Linux Apache
3  document root is /var/www/html, which is owned by root by
4  default.
5
6  --.> $    ls -l /var/www
7
8  To set file permissions
9  ===================
10 1 Add the www group to your instance.
11
12  -->$    sudo groupadd www
13
14 2  Add your user to the www group.
15
16  --->  sudo usermod -a -G www ec2-user
17 ======================================
18
19 3 You need to log out and log back in to pick up the new
20 group. You can use the exit command, or close the terminal window.
21
22 1 Log out.
23
24
25
26
```



Now test php file.....

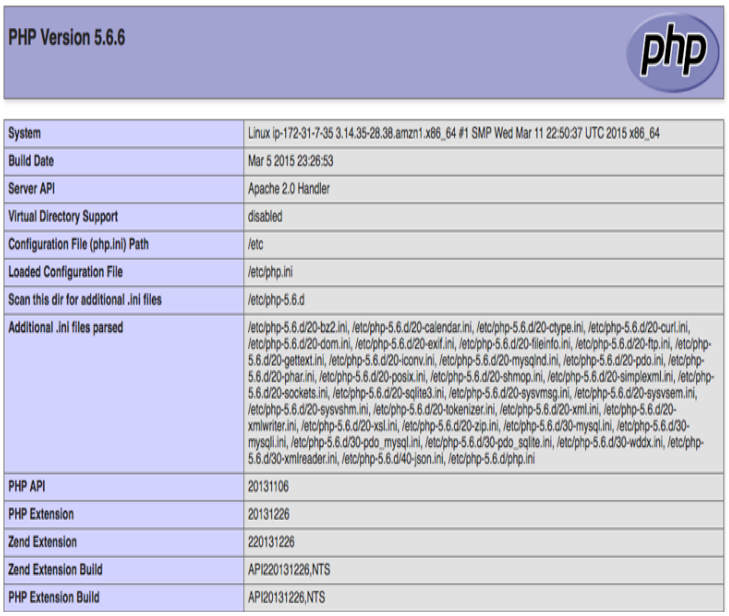
$ echo "<?phpphpinfo(); ?>" > /var/www/html/phpinfo.php



Now

In a web browser, enter the URL of the file you just created. This URL is the public DNS address of your instance followed by a forward slash and the file name. For example:

http://my.public.dns.amazonaws.com/phpinfo.php



PHP Version 5.6.6

System	Linux ip-172-31-7-35 3.14.35-28.38.amzn1.x86_64 #1 SMP Wed Mar 11 22:50:37 UTC 2015 x86_64
Build Date	Mar 5 2015 23:26:53
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc
Loaded Configuration File	/etc/php.ini
Scan this dir for additional .ini files	/etc/php-5.6.d
Additional .ini files parsed	/etc/php-5.6.d/20-bz2.ini, /etc/php-5.6.d/20-calendar.ini, /etc/php-5.6.d/20-ctype.ini, /etc/php-5.6.d/20-curl.ini, /etc/php-5.6.d/20-dom.ini, /etc/php-5.6.d/20-exif.ini, /etc/php-5.6.d/20-fileinfo.ini, /etc/php-5.6.d/20-ftp.ini, /etc/php-5.6.d/20-gettext.ini, /etc/php-5.6.d/20-iconv.ini, /etc/php-5.6.d/20-mysqlnd.ini, /etc/php-5.6.d/20-pdo.ini, /etc/php-5.6.d/20-phar.ini, /etc/php-5.6.d/20-posix.ini, /etc/php-5.6.d/20-shmop.ini, /etc/php-5.6.d/20-simplexml.ini, /etc/php-5.6.d/20-sockets.ini, /etc/php-5.6.d/20-sqlite3.ini, /etc/php-5.6.d/20-sysvmsg.ini, /etc/php-5.6.d/20-sysvsem.ini, /etc/php-5.6.d/20-sysvshm.ini, /etc/php-5.6.d/20-tokenizer.ini, /etc/php-5.6.d/20-xml.ini, /etc/php-5.6.d/20-xmlwriter.ini, /etc/php-5.6.d/20-xsl.ini, /etc/php-5.6.d/20-zip.ini, /etc/php-5.6.d/30-mysql.ini, /etc/php-5.6.d/30-mysqli.ini, /etc/php-5.6.d/30-pdo_mysql.ini, /etc/php-5.6.d/30-pdo_sqlite.ini, /etc/php-5.6.d/30-wddx.ini, /etc/php-5.6.d/30-xmlreader.ini, /etc/php-5.6.d/40-json.ini, /etc/php-5.6.d/php.ini
PHP API	20131106
PHP Extension	20131226
Zend Extension	220131226
Zend Extension Build	API220131226,NTS
PHP Extension Build	API20131226,NTS

Now executephp file as....

echo "<?php echo 'Hello from Zeeshan ' ; ?>" > /var/www/html/RJ.php



Hello from Zeeshan

Conclusion: Successfully built an application on private cloud.

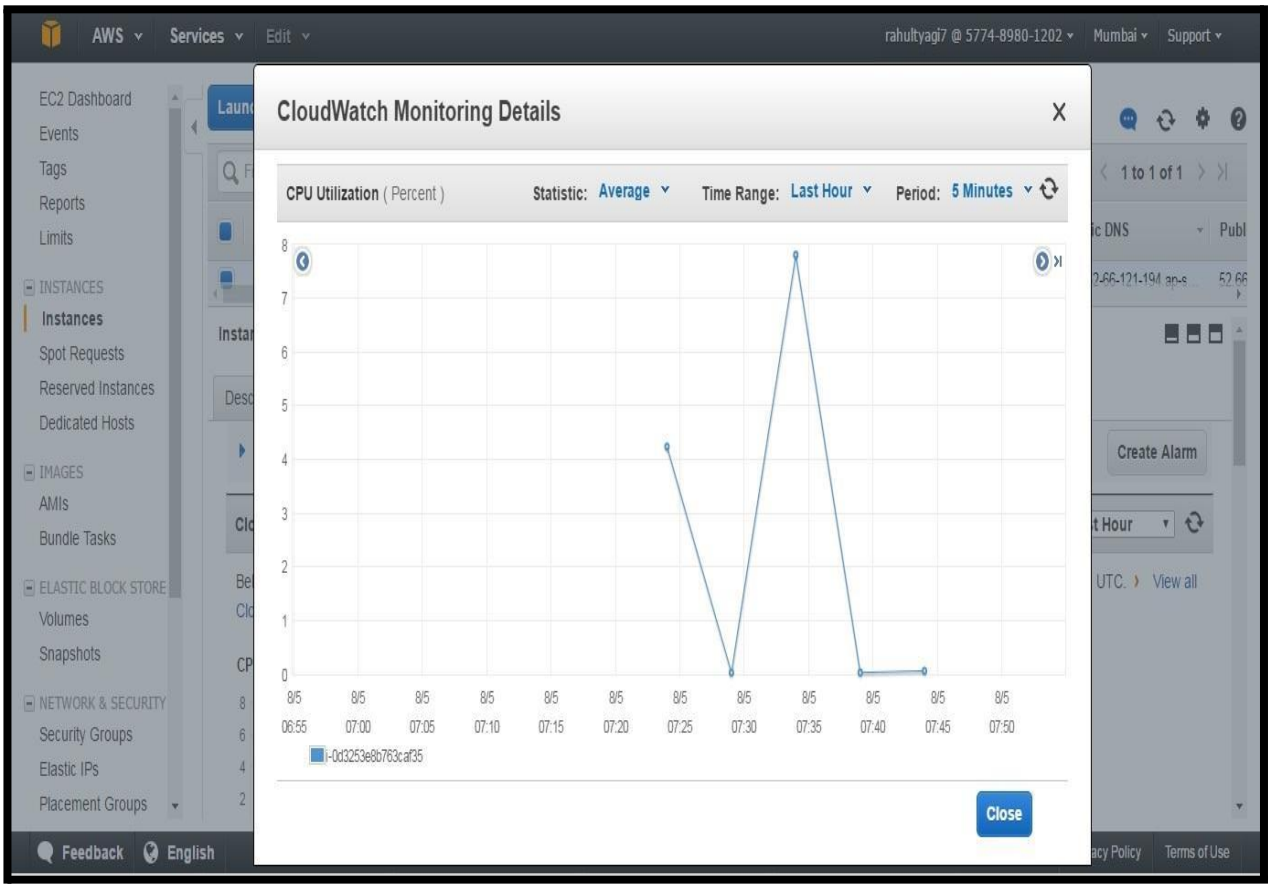
Practical No : 4

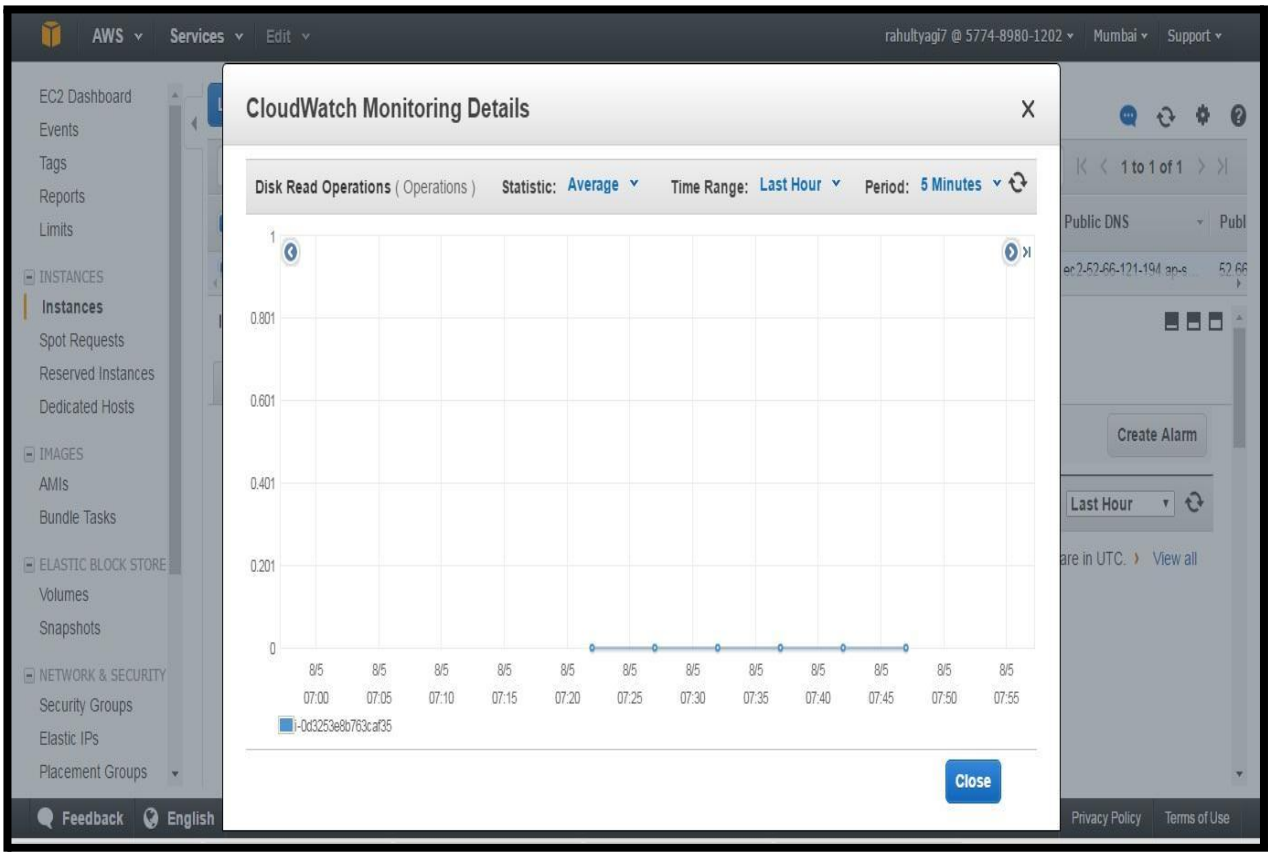
Aim : Demonstrate any Cloud Monitoring tool.

Theory:

Amazon CloudWatch is a monitoring and observability service built for DevOps engineers, developers, site reliability engineers (SREs), IT managers, and product owners. CloudWatch provides you with data and actionable insights to monitor your applications, respond to system-wide performance changes, and optimize resource utilization. CloudWatch collects monitoring and operational data in the form of logs, metrics, and events. You get a unified view of operational health and gain complete visibility of your AWS resources, applications, and services running on AWS and on-premises. You can use CloudWatch to detect anomalous behavior in your environments, set alarms, visualize logs and metrics side by side, take automated actions, troubleshoot issues, and discover insights to keep your applications running smoothly.

Now Monitor the graph of CPU..




AWS
Services
Edit
rahultyagi7 @ 5774-8980-1202
Mumbai
Support
EC2 Dashboard
Events
Tags
Reports
Limits
INSTANCES
Instances
Spot Requests
Reserved Instances
Dedicated Hosts
IMAGES
AMIs
Bundle Tasks
ELASTIC BLOCK STORE
Volumes
Snapshots
NETWORK & SECURITY
Security Groups
Elastic IPs
Placement Groups
Feedback
English
CloudWatch Monitoring Details
X
Disk Read Operations (Operations)
Statistic: Average
Time Range: Last Hour
Period: 5 Minutes
1
0.801
0.601
0.401
0.201
0
8/5 07:00
8/5 07:05
8/5 07:10
8/5 07:15
8/5 07:20
8/5 07:25
8/5 07:30
8/5 07:35
8/5 07:40
8/5 07:45
8/5 07:50
8/5 07:55
i-0d3253e8b763caf35
Close
1 to 1 of 1
Public DNS
Create Alarm
Last Hour
are in UTC.
View all
Privacy Policy
Terms of Use

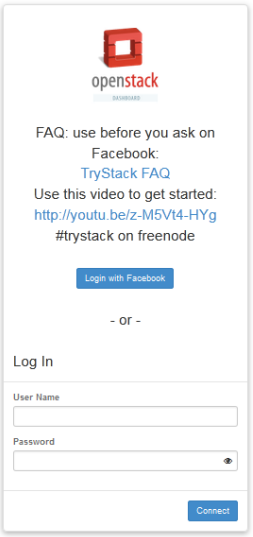
Practical No: 5

Aim : Evaluate a Private IAAS Cloud using TryStack

Note :

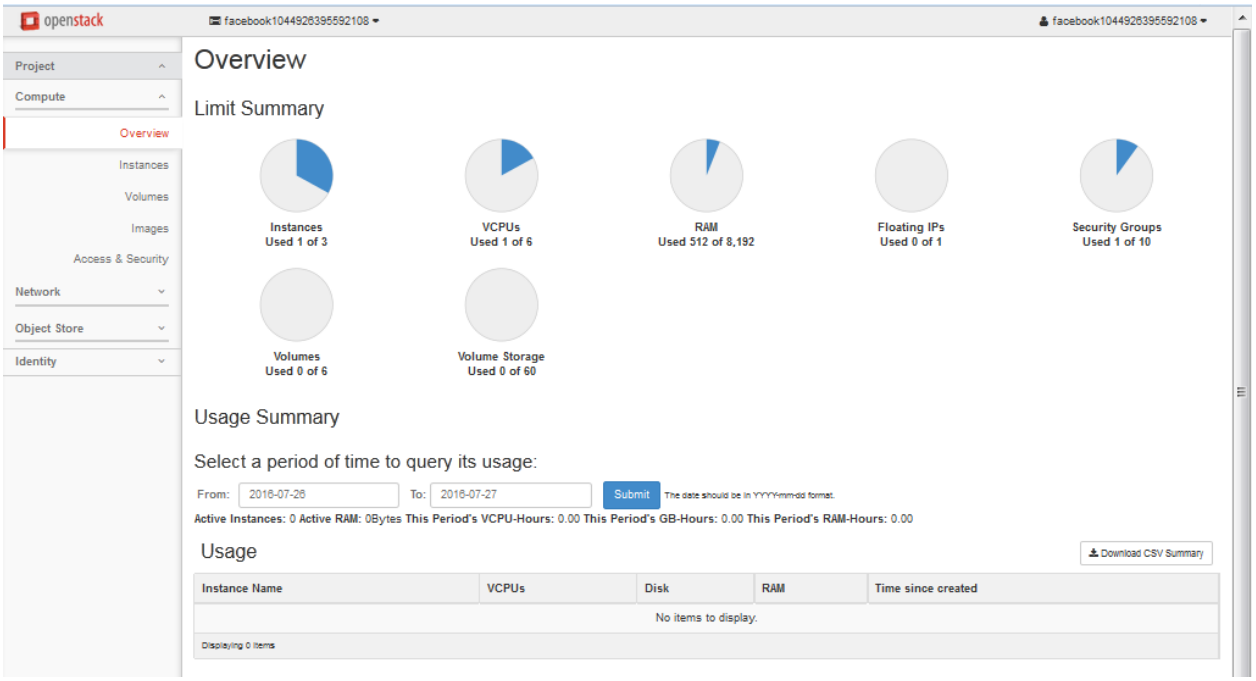
1. Install cygwin and then start with practical.

Go to link
https://x86.trystack.org/dashboard/auth/login/?next=/dashboard/

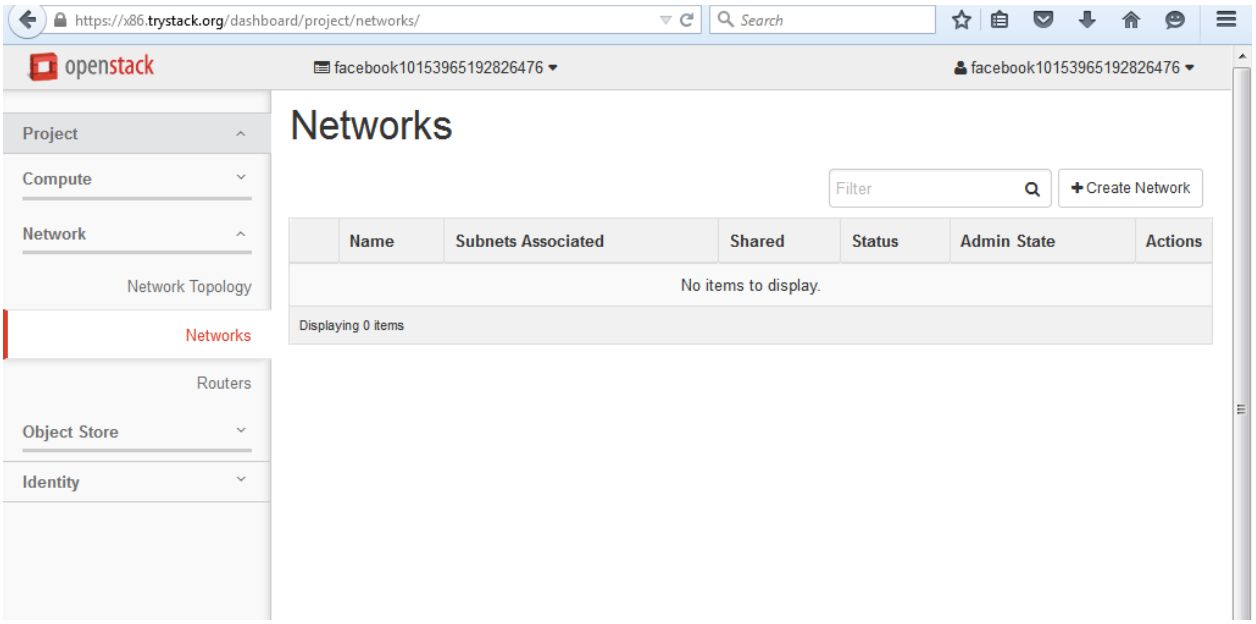


Login with facebook account....

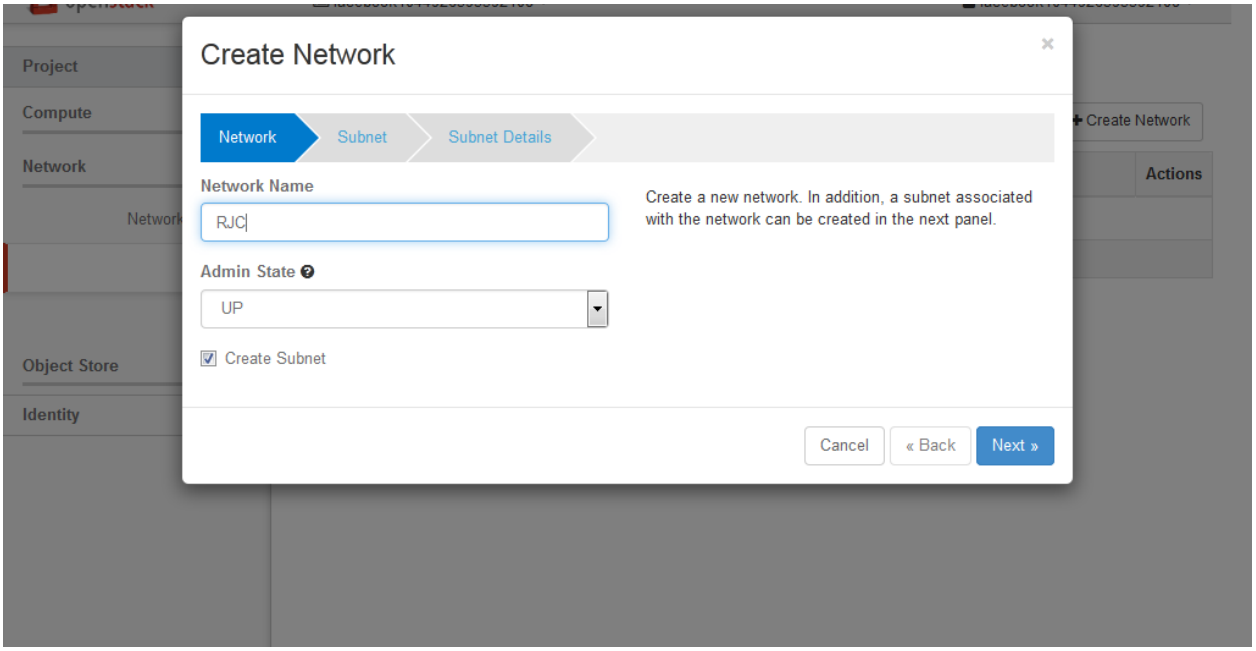
After login.....



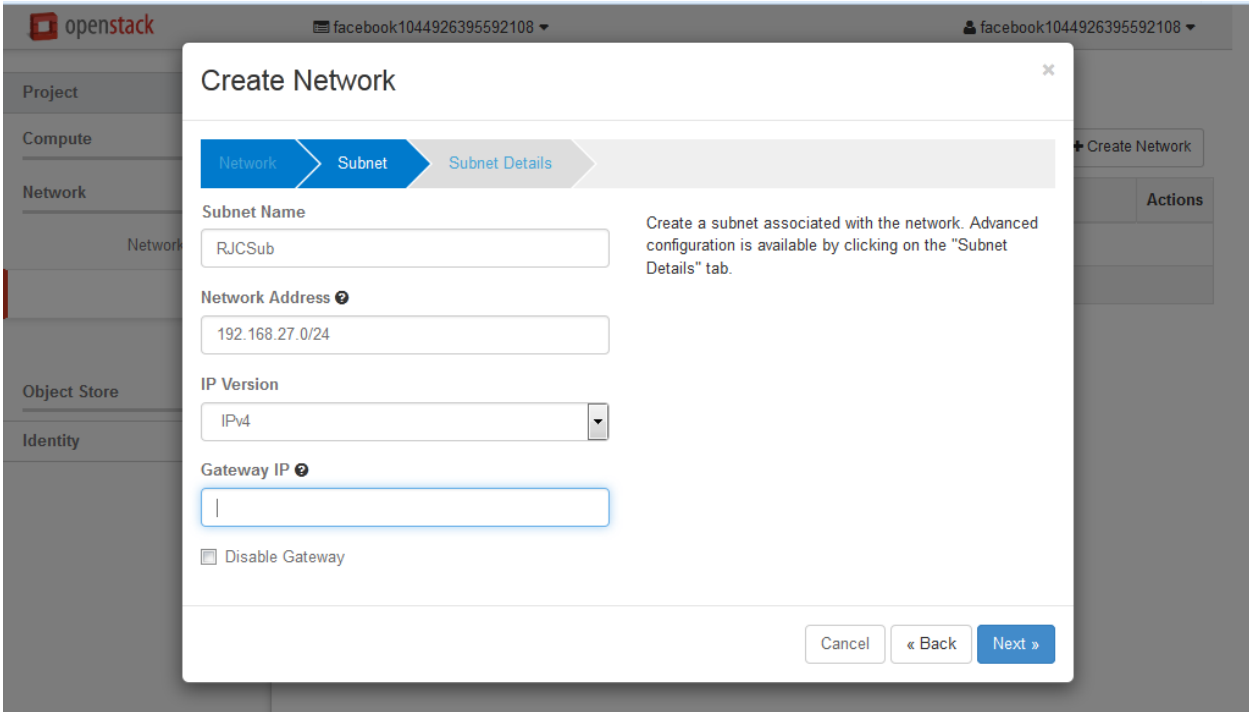
Go to Network(left side) ->



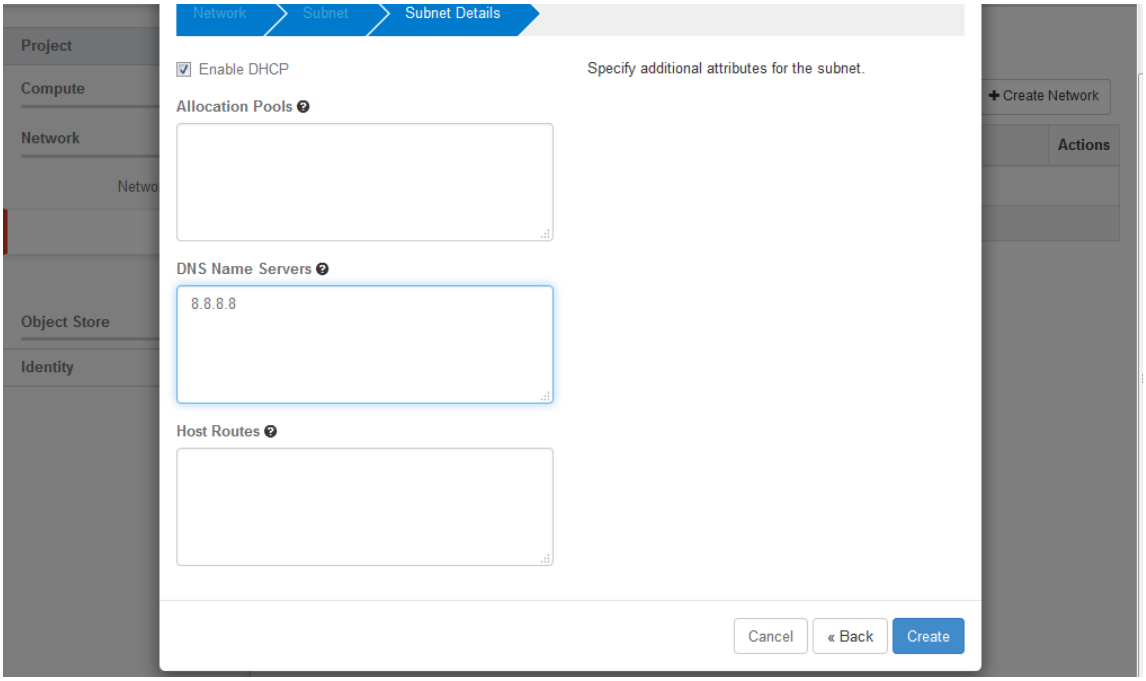
Click on Create Network(Right side tab)



At Subnet tab

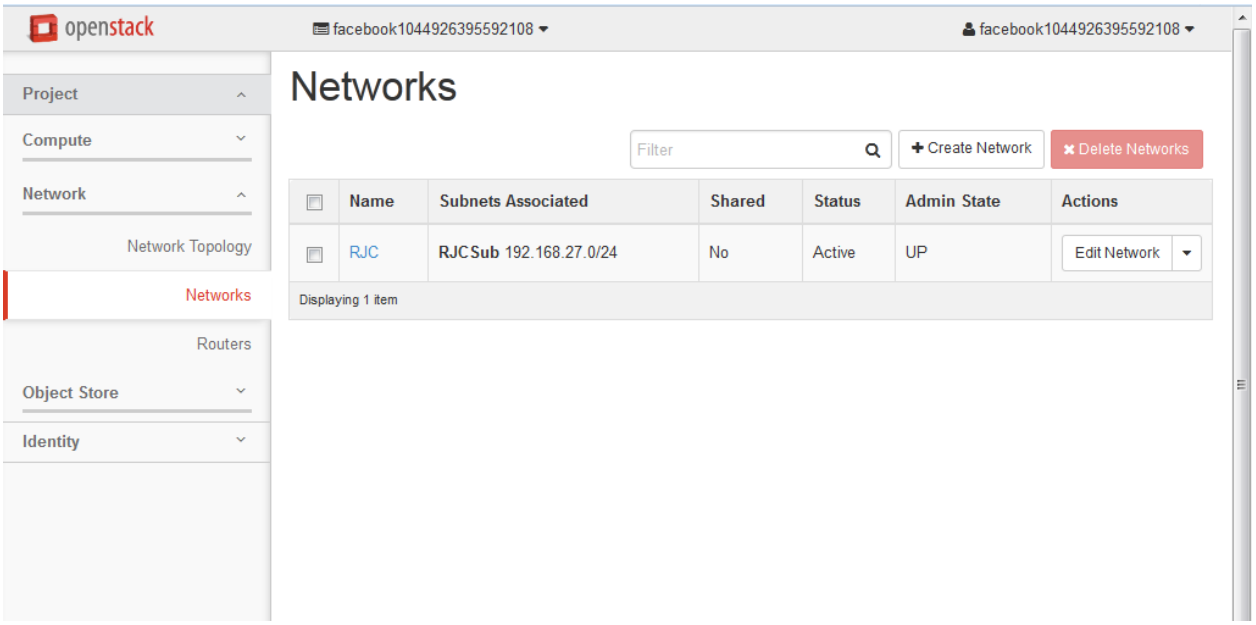


At subnet Details

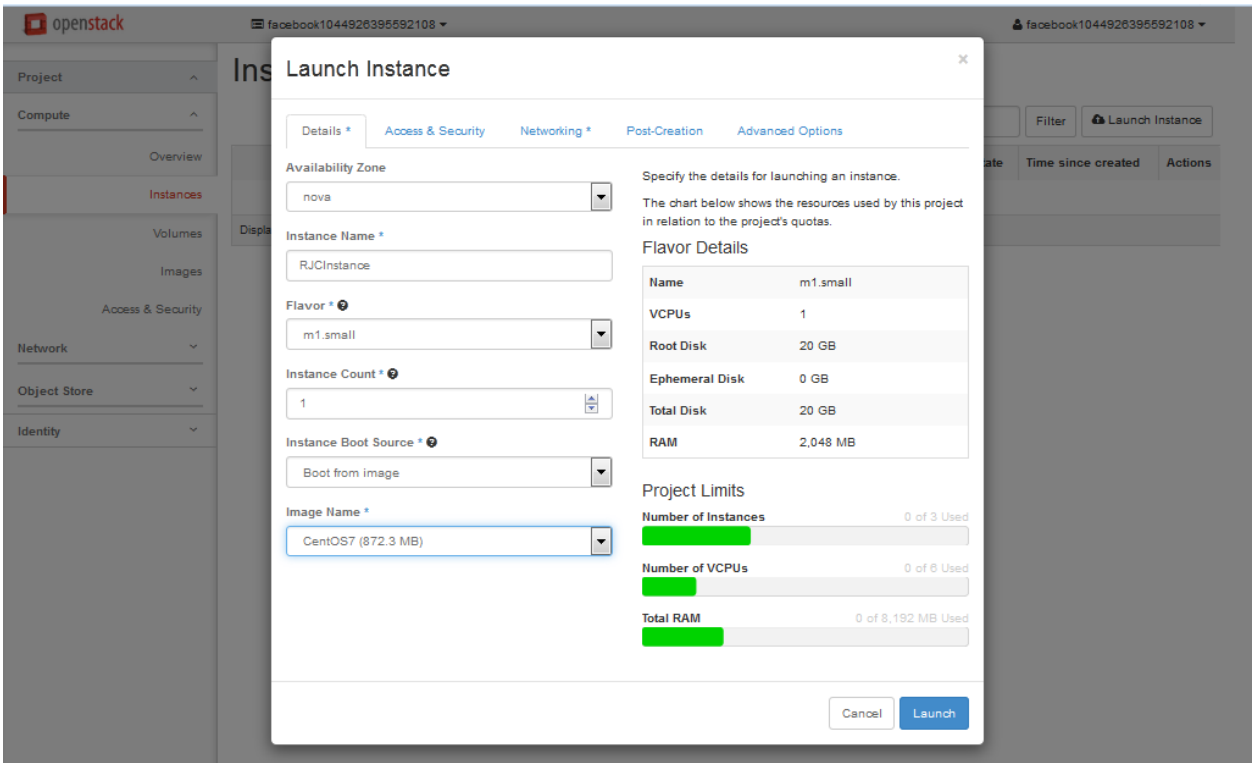


8.8.8.8 google's DNS.....Click on create

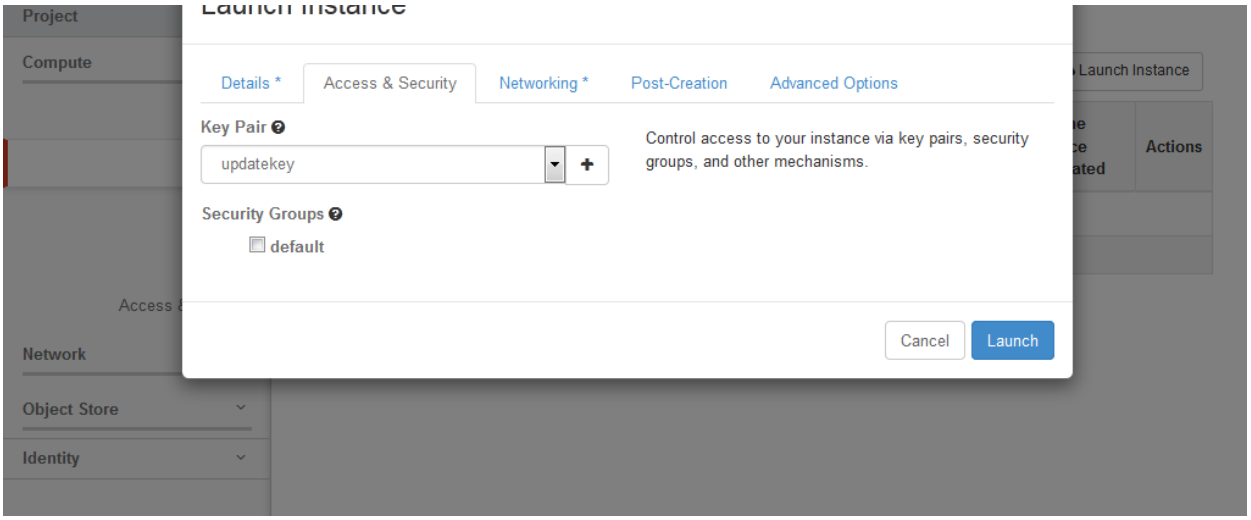
So we have created network



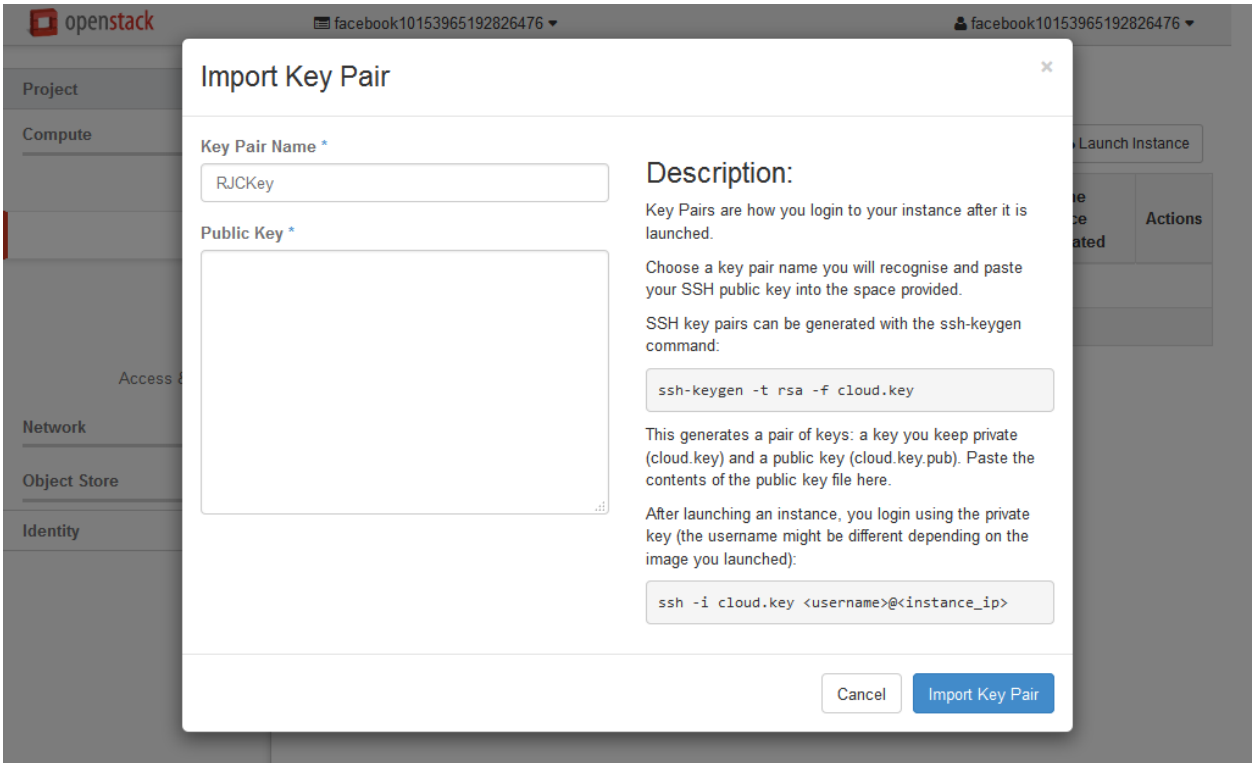
Go to compute(left side) -> Instance-> Launch Instance(Right Side)



Go to Access & security



Click on “ + “ Sign



Install SSH Package of cygwin terminal for creating SSH key

Execute given command in cygwin

Two keys will be generated (1 Public & 1 Private).....import the public key

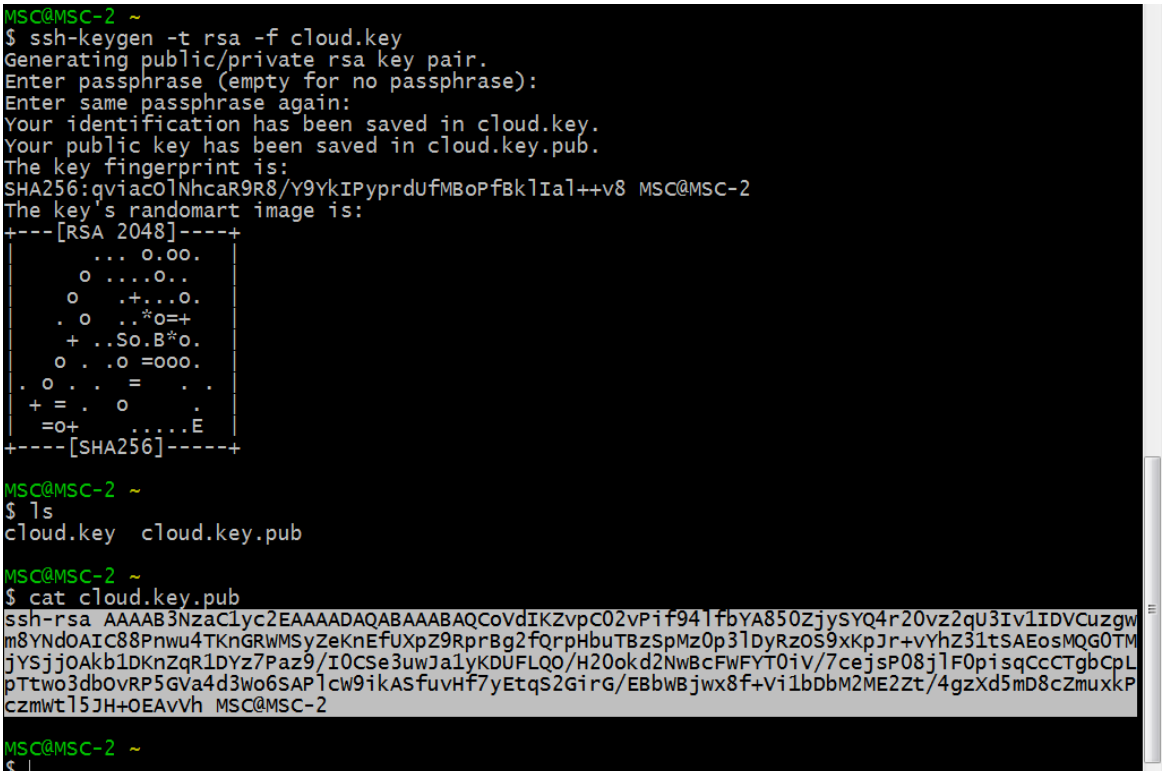
```
MSC@MSC-2 ~
$ ssh-keygen -t rsa -f cloud.key
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in cloud.key.
Your public key has been saved in cloud.key.pub.
The key fingerprint is:
SHA256:qviacOlNhcaR9R8/Y9YkIPyprdUfMBoPfBklIal++v8 MSC@MSC-2
The key's randomart image is:
+---[RSA 2048]----+
|      ... o.oo.  |
|     o ....o..   |
|    o   .+...o.  |
|   . o  ..*o=+   |
|    + ..So.B*o.  |
|   o . .o =ooo.  |
|. o . .  =   . . |
| + = .  o     .  |
|  =o+    .....E  |
+----[SHA256]-----+

MSC@MSC-2 ~
$ ls
cloud.key  cloud.key.pub

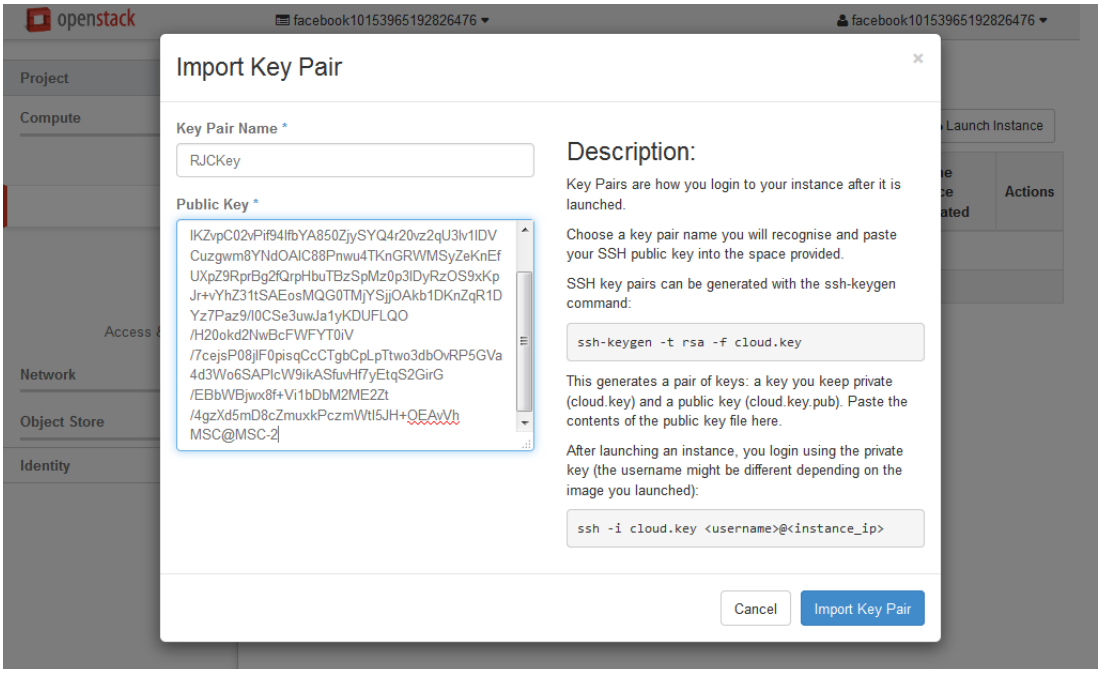
MSC@MSC-2 ~
$ cat cloud.key.pub
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABAQCoVdIKZvpC02vPif94lfbYA850ZjySYQ4r20vz2qU3Iv1IDVCuzgw
m8YNdOAIC88Pnwu4TKnGRWMSyZeKnEfUXpZ9RprBg2fQrpHbuTBzSpMzOp3lDyRzOS9xKpJr+vYhZ31tSAEosMQGOTM
jYSjjOAkb1DKnZqR1DYz7Paz9/IOCSe3uwJa1yKDUFLQO/H2Ookd2NwBcFWFYTOiV/7cejsP08jlFOpisqCcCTgbCpL
pTtwo3dbOvRP5GVa4d3Wo6SAPlcW9ikASfuvHf7yEtqS2GirG/EBbWBjwx8f+Vi1bDbM2ME2Zt/4gzXd5mD8cZmuxkP
czmWtl5JH+OEAvVh MSC@MSC-2

MSC@MSC-2 ~
$
```

Select public key from terminal as shown below

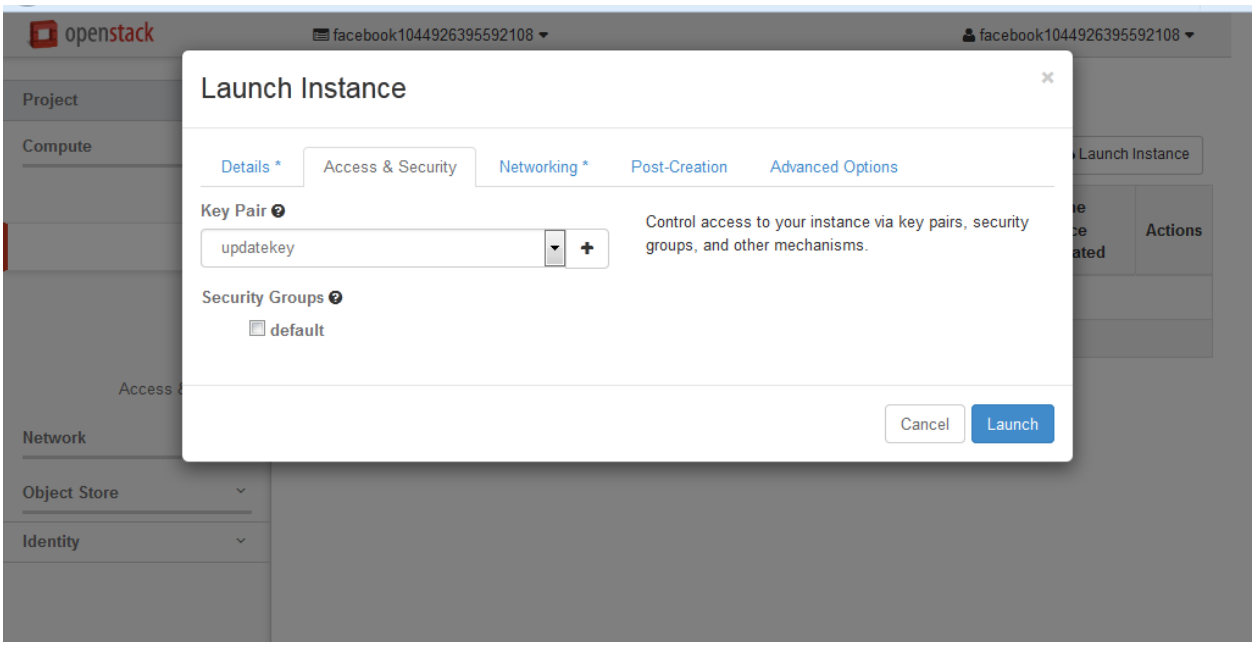


Now copy it as shown below

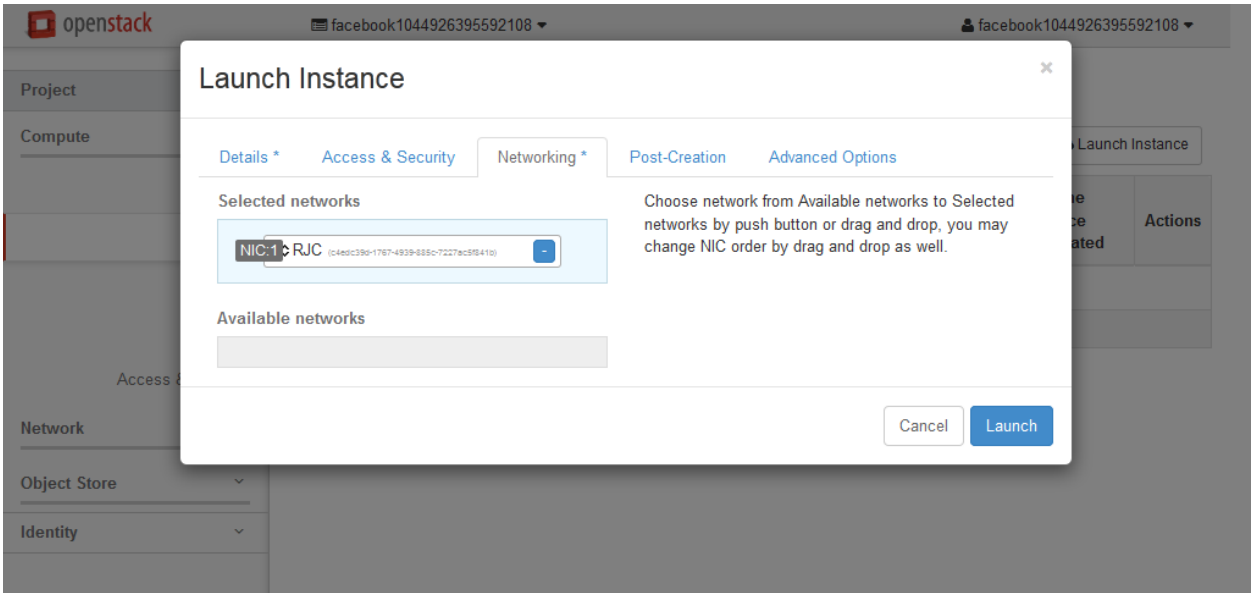


Click on Import key pair.....

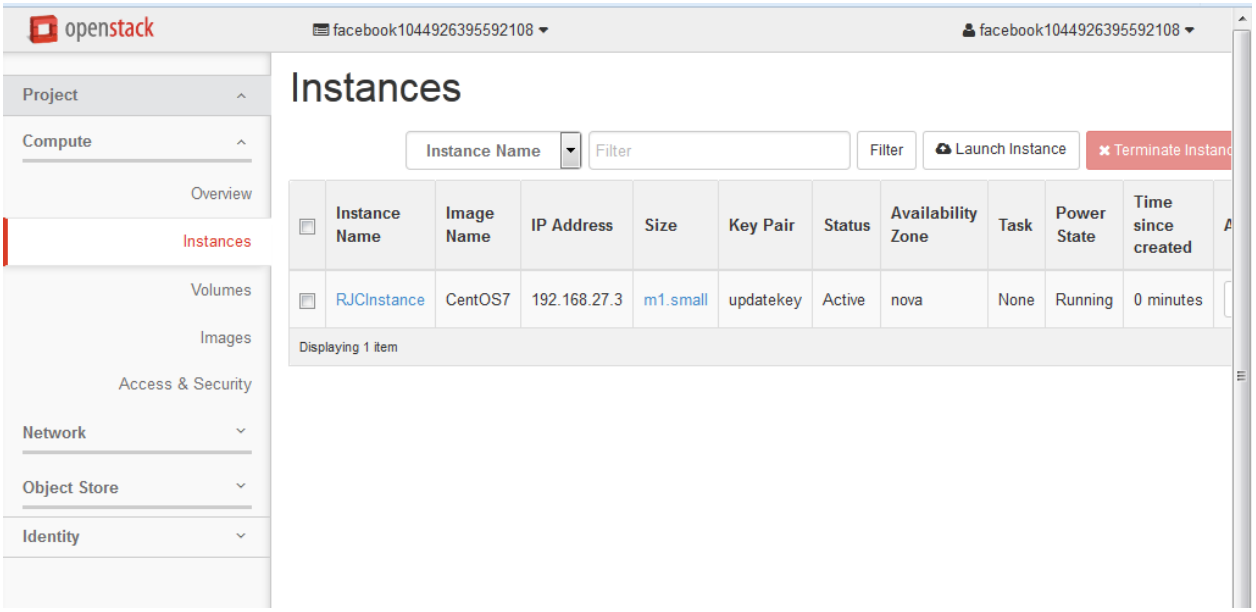
Select name of the key from dropdown.....



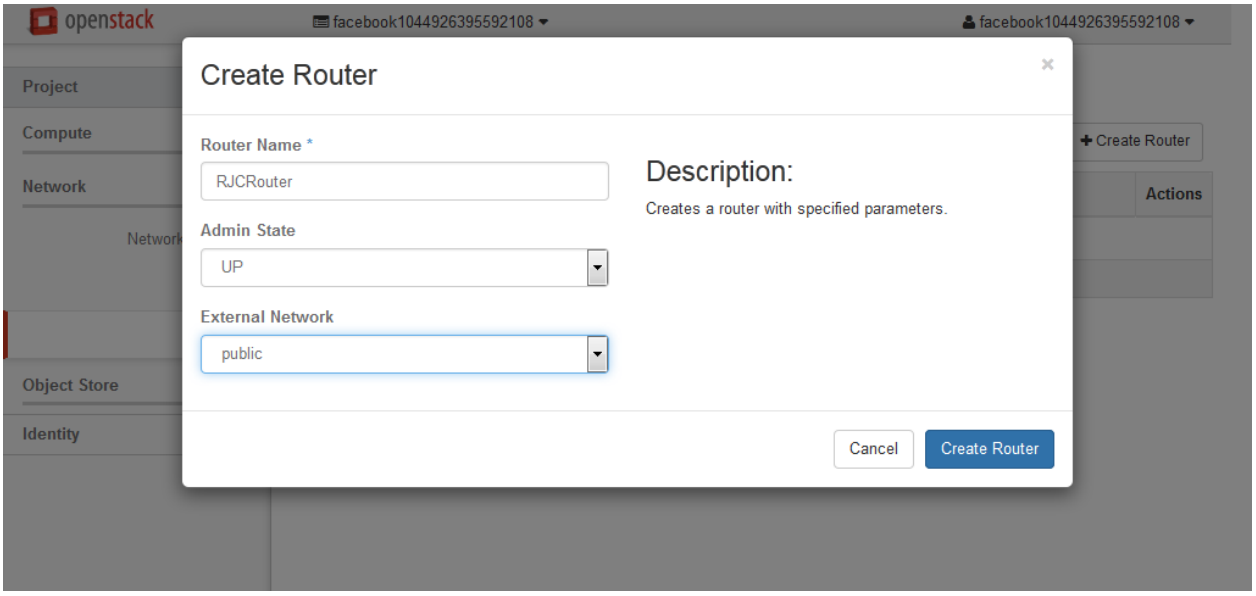
At networking tab



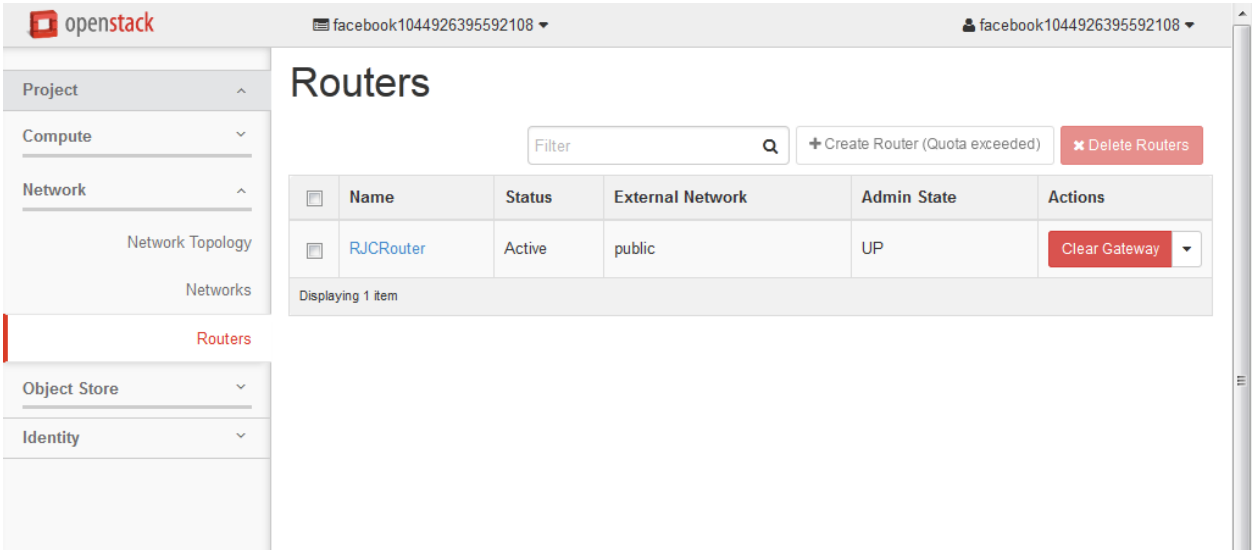
Click on Launch



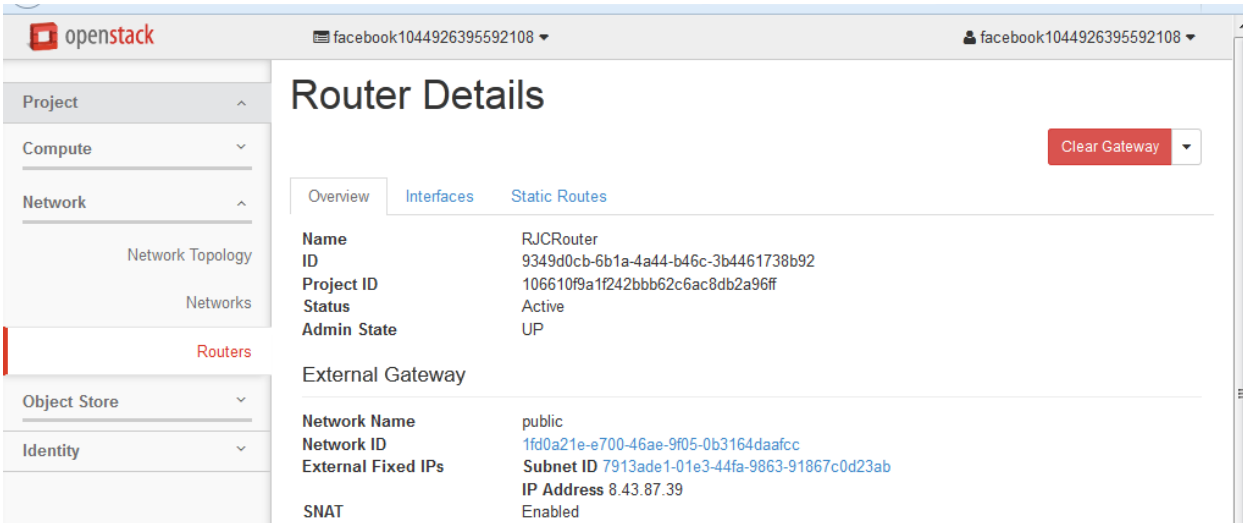
Go to Network -> Routers -> Create Routers



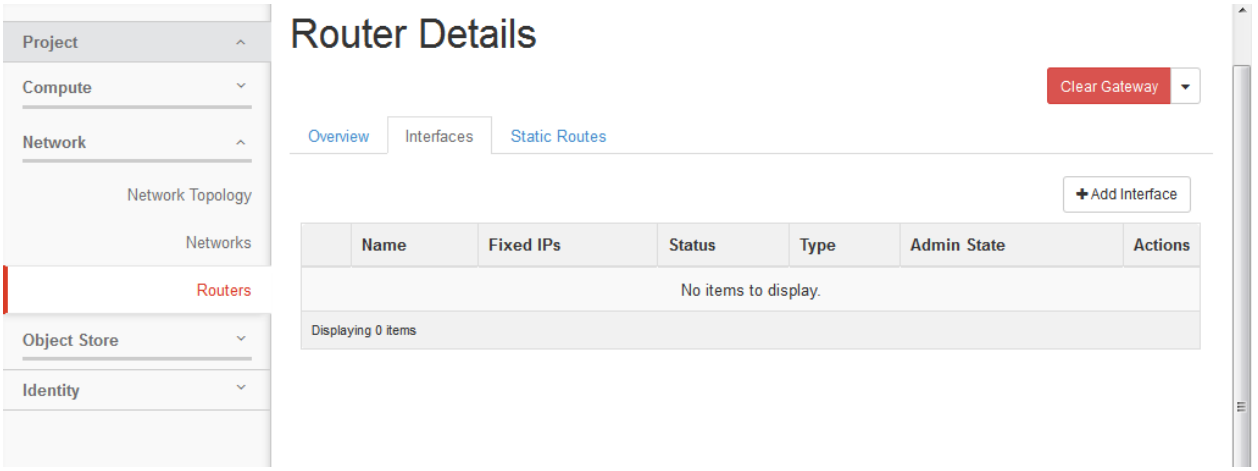
Click on Create Router



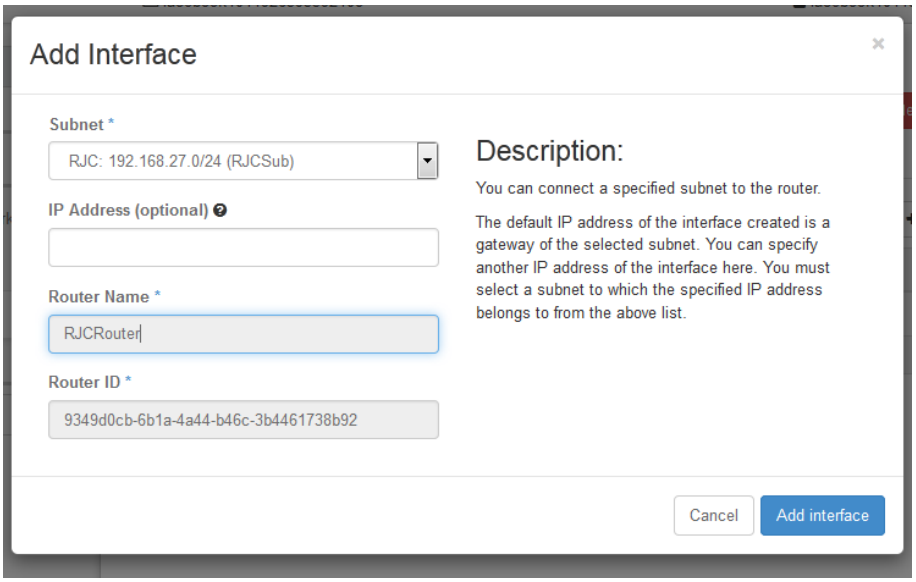
Click on Name of Router (RJCRouter)



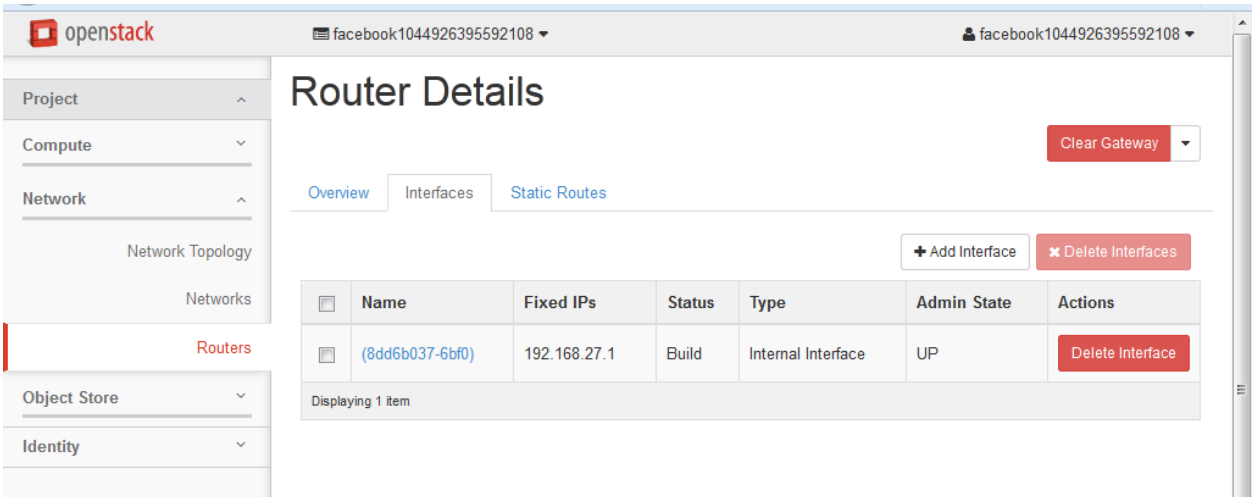
Go to Interfaces->



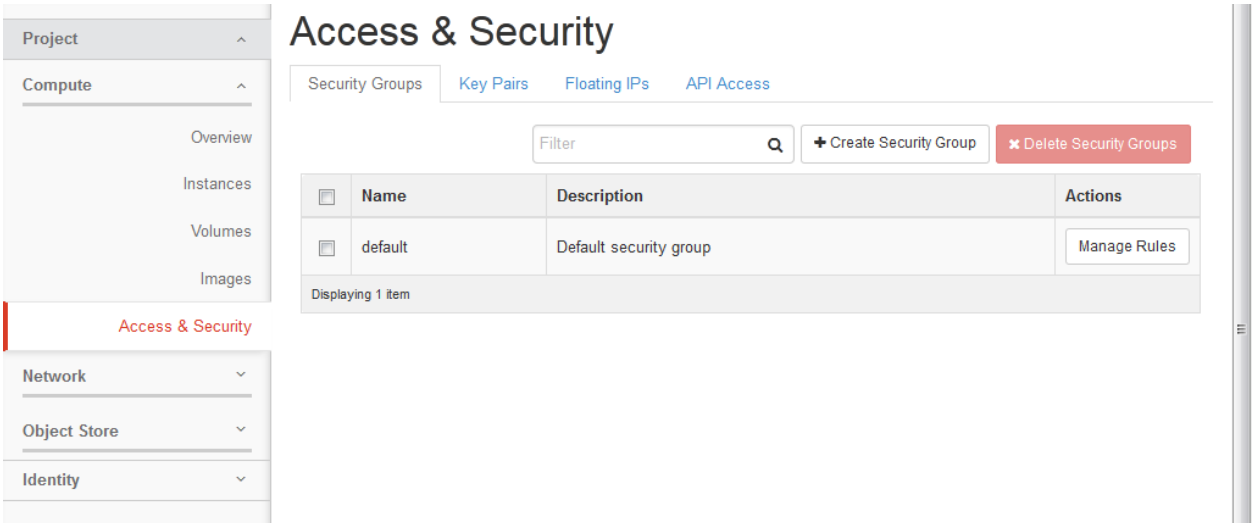
Add Interface



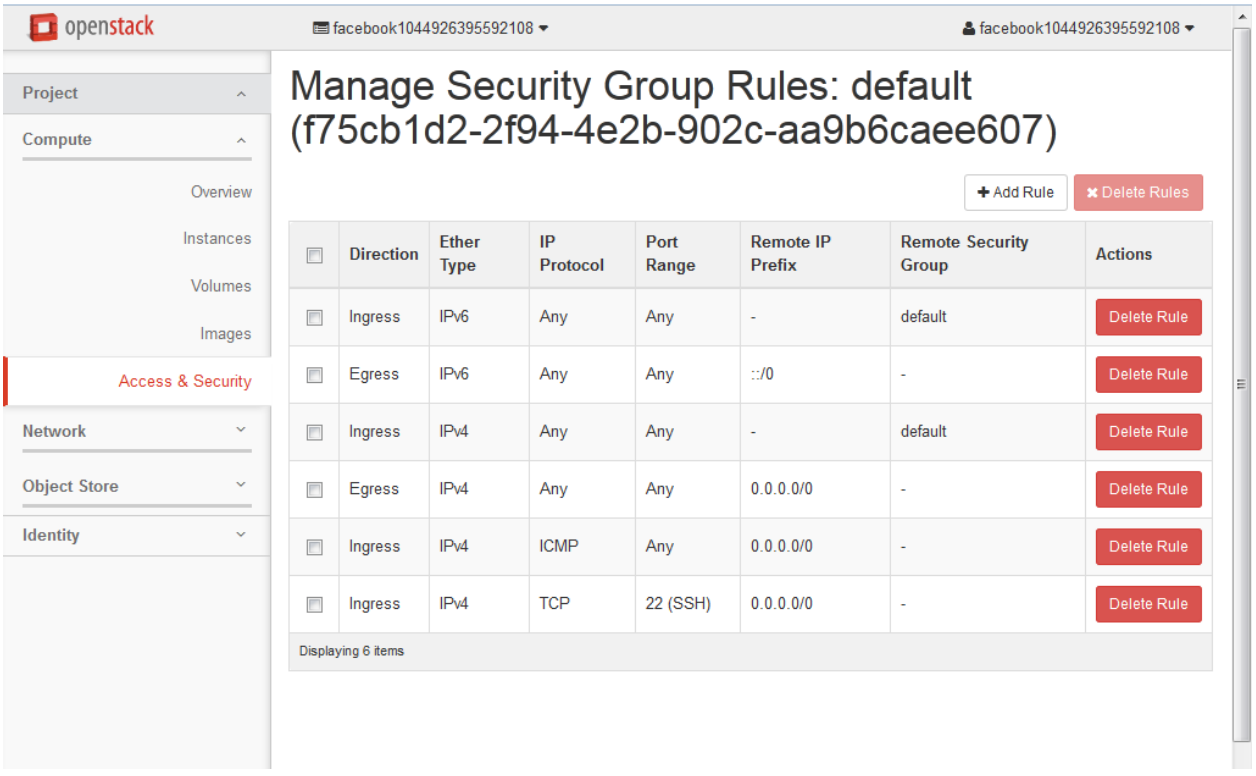
Click on Add Interface



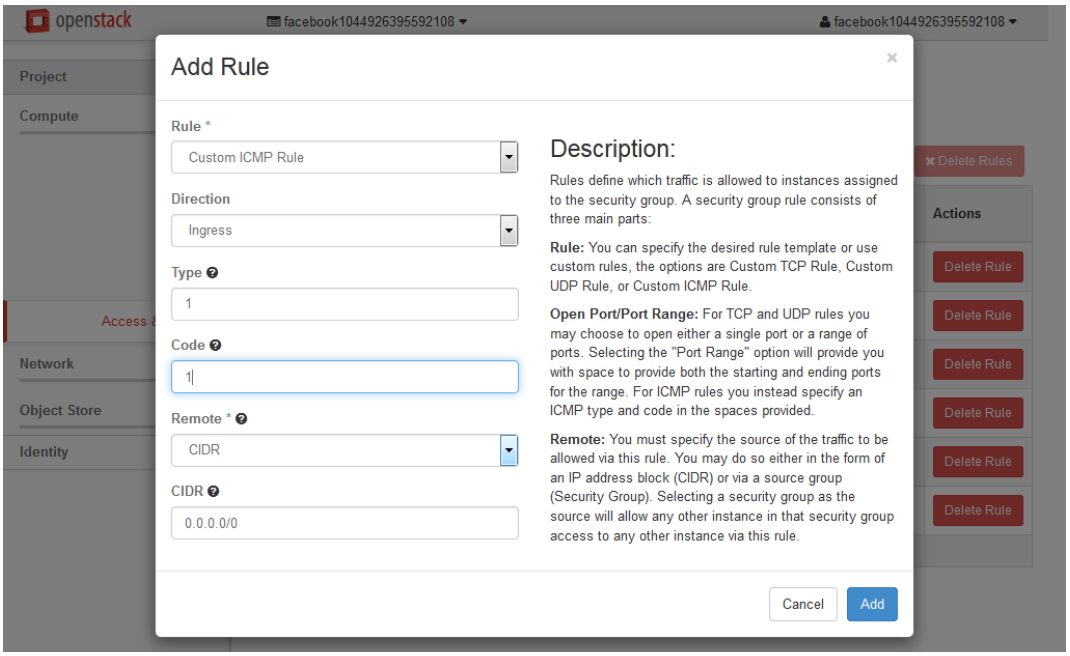
Go to Compute -> Acess & Security



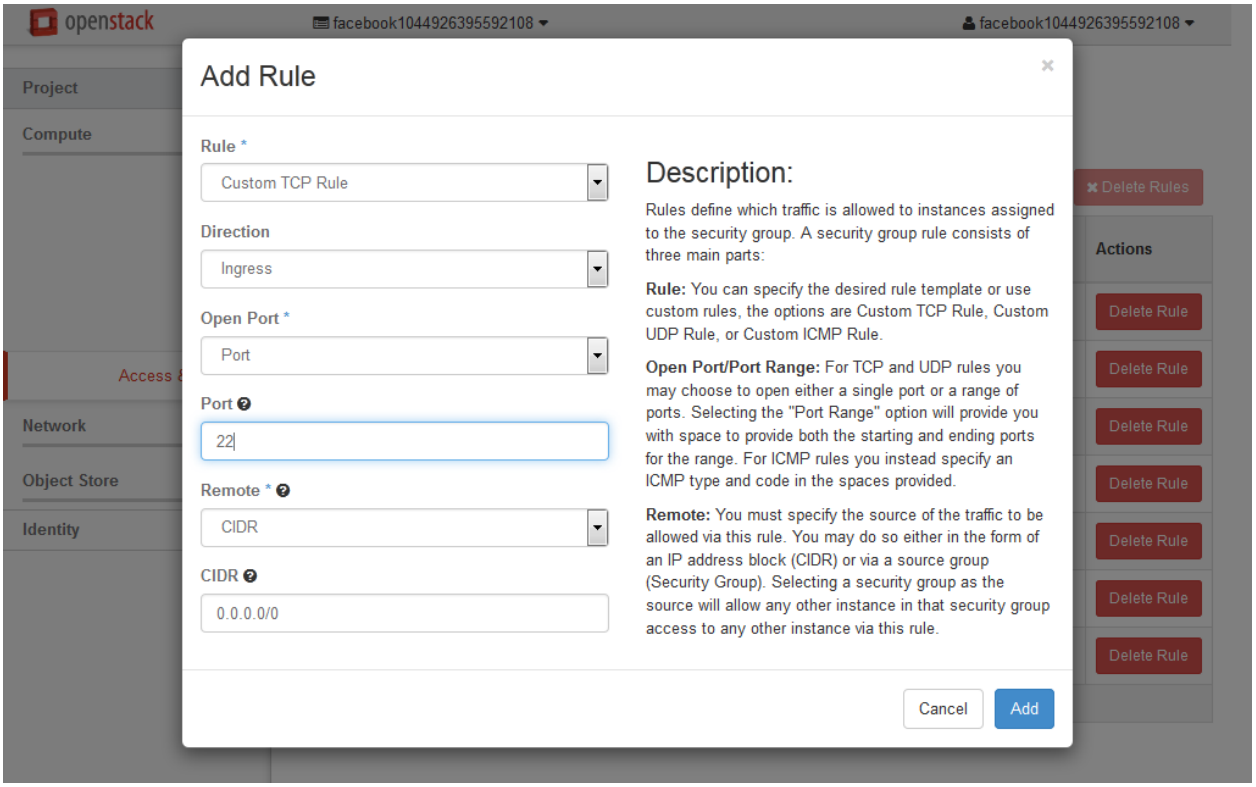
Click on Manage Rules



Click on AddRule & select following options

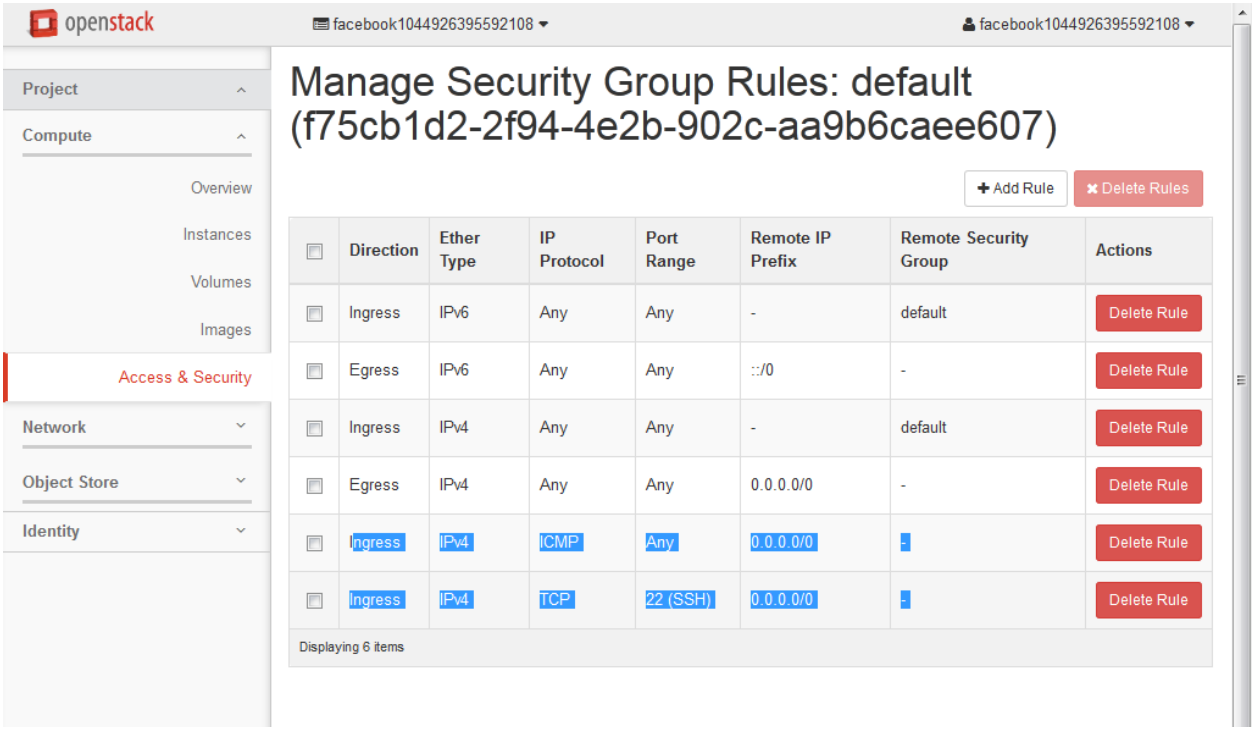


Again click on AddRule & now insert following options

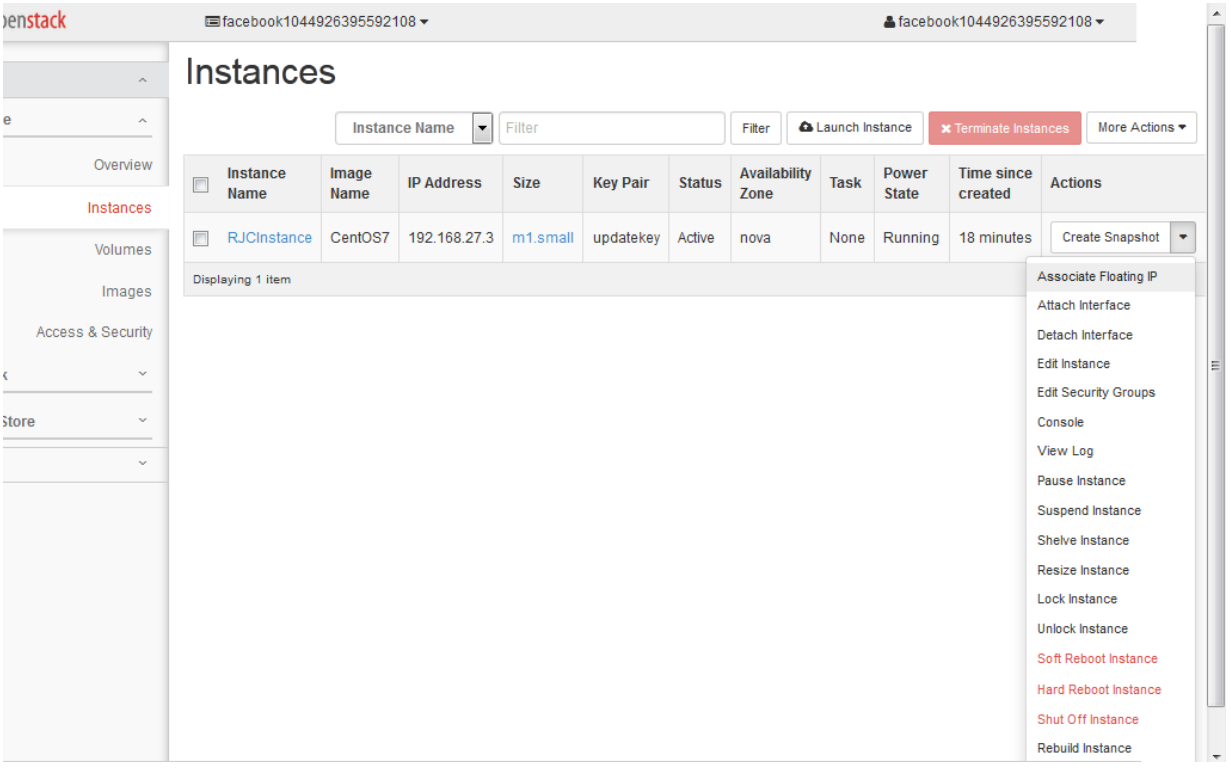


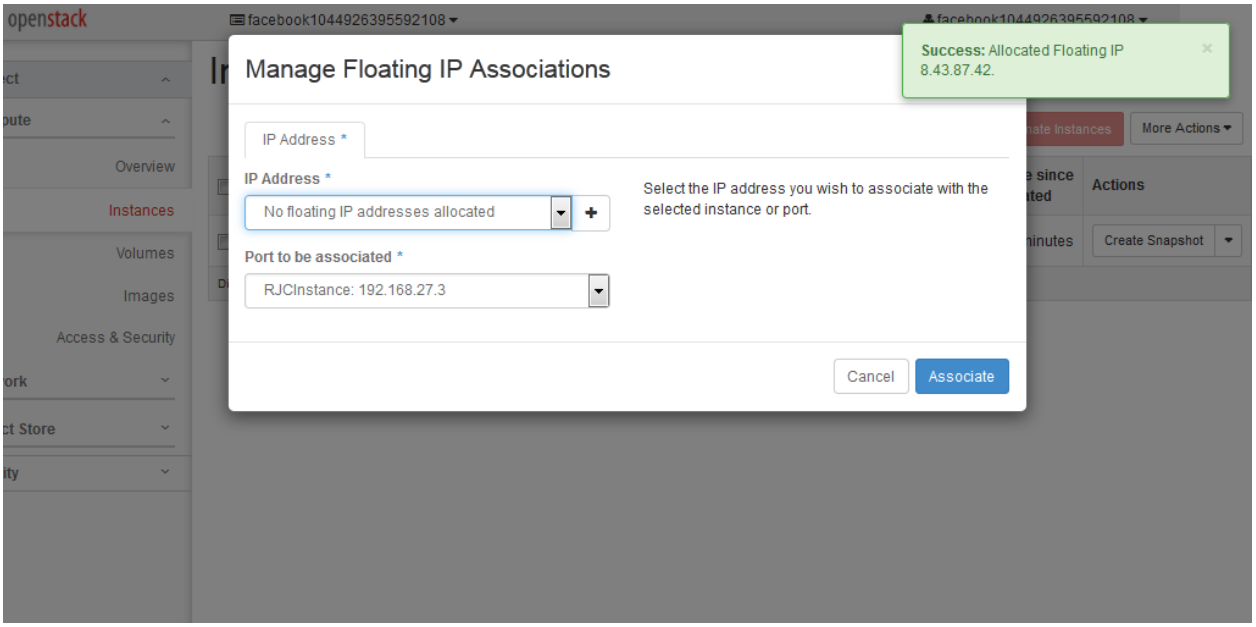
\

After adding 2 rules

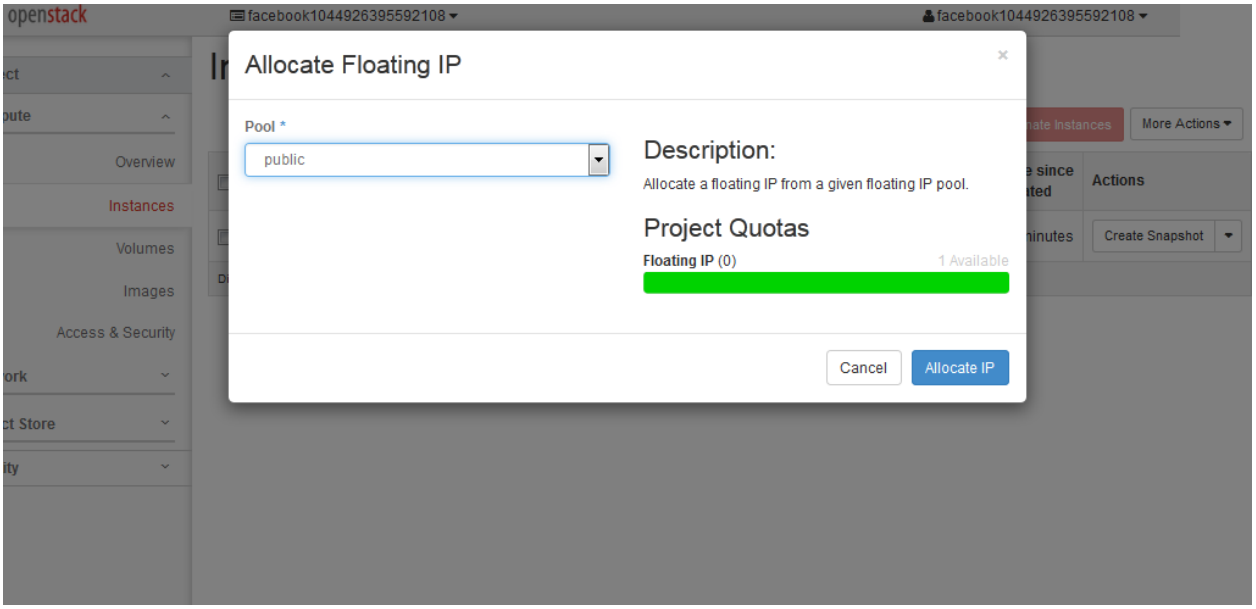


Go to Instance .....drop down create snapshot & select Associate floating IP

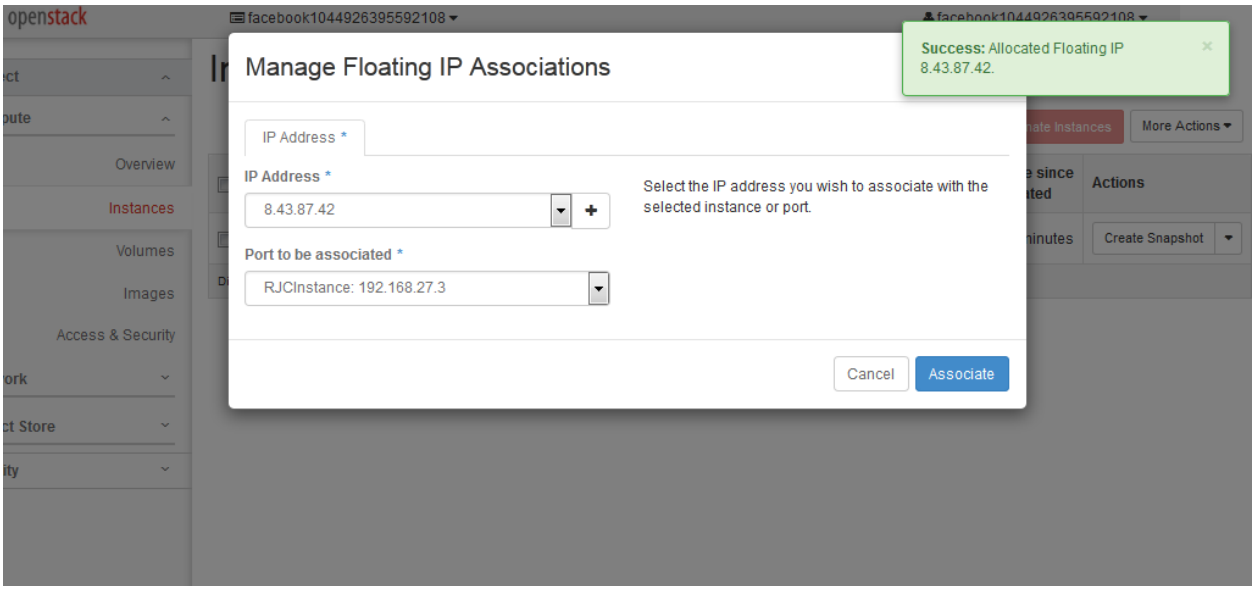




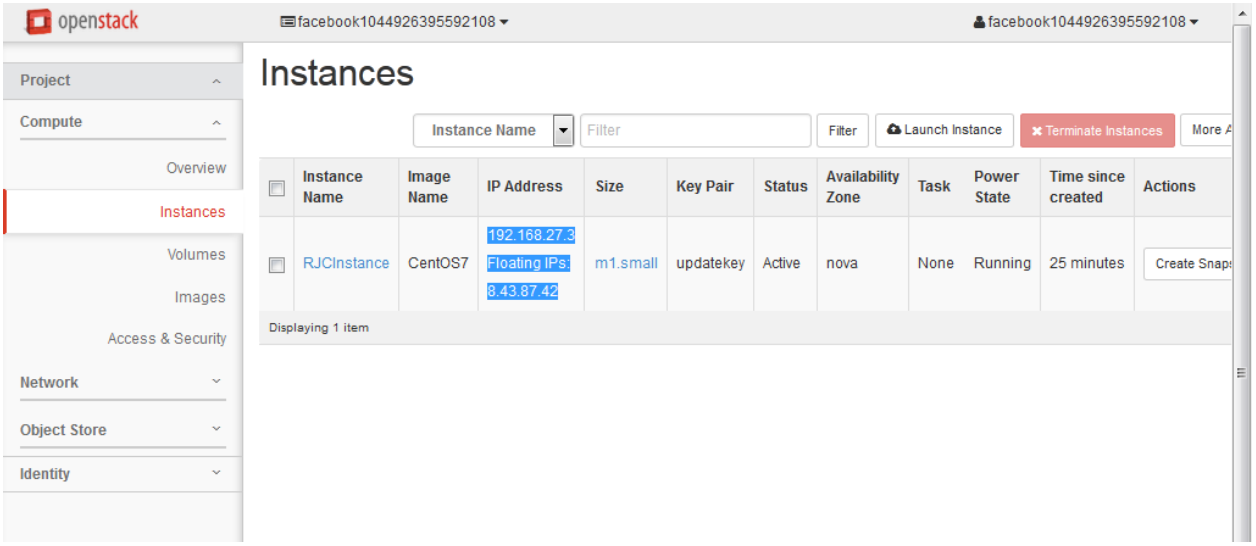
Click on " + " sign



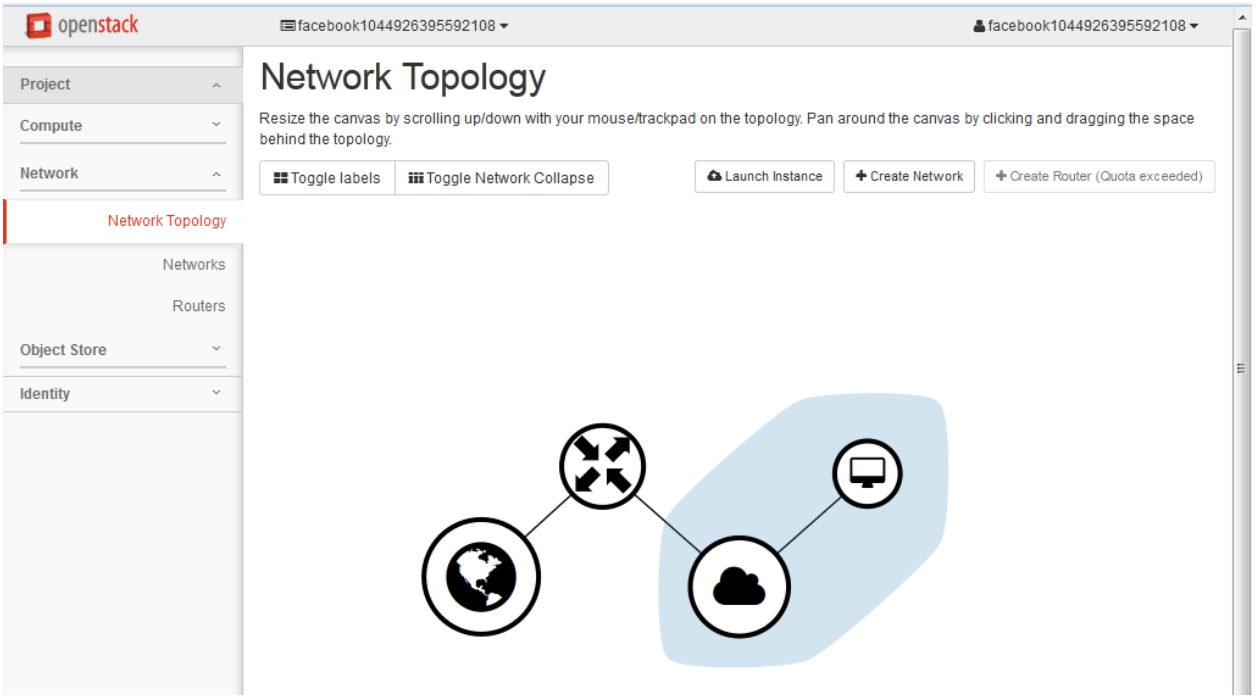
Click on Allocate IP



Click on Associate



Go to Network -> Network Topology



Conclusion : Successfully evaluated a private cloud.

Practical No: 6

Aim: Implement FOSS-Cloud Functionality - VDI (Virtual Desktop Infrastructure)

FOSS-Cloud (FOSS-Cloud software and hardware) is an integrated and redundant server infrastructure to provide virtualization- and cloud-services, Windows or Linux based SaaS-, Terminal Server-, Virtual Desktop Infrastructure (VDI) or virtual server environments.
FOSS-Cloud covers all aspects of an virtualized IT environment. FOSS-Cloud is a pure Open Source solution, is licensed under EUPL and is available on the sourceforge.net. FOSS-Cloud is the most advanced Open Source Cloud.

FOSS-Cloud is a cost-effective alternative to Citrix and VMware. We can Build our own private- or public-Cloud!

Functionality

- VDI (Virtual Desktop Infrastructure)
- VSI (Virtual Server Infrastructure)
- Infrastructure as a Service (IaaS)
- Platform as a Service (PaaS)
- Software as a Service (SaaS)
- Storage Cloud

Features

- Web-based management console
- Multi-tenancy

Minimal Requirements

Your server should fulfill the following minimal requirements on each node:

- Dedicated hardware
- 64-Bit Intel with VT-Technology
- 4 Gigabyte memory
- 320 Gigabyte disk space
- Network interface

The **Demo System** is really just a playing environment and not for professional use. It is very easy to install. The Demo System is made to get a feeling of the possibilities of the FOSS-Cloud.

Limitation of Demo System

There is no access to the VMs from outside of FOSS-Cloud. That means, for example, pinging a VM is not possible.

Steps to implement FOSS Cloud Demo System

1. Create FOSS-Cloud Boot Stick or DVD

Linux

First download the appropriate iso image from sourceforge:

root # wget
http://sourceforge.net/projects/foss-cloud/files/X.X.X/fo
ss-cloud- installer-X.X.X.iso

Most modern LiveCD's, like Gentoo are already in hybrid mode. If your LiveUSB doesn't boot then you have to convert it to hybrid mode. Hybrid mode means image that can be booted from either DVD or USB drive.

root # isohybrid

foss-cloud-installer-X.X.X.iso

Write the LiveCD image to the

USB drive

root # dd if=foss-cloud-installer-X.X.X.iso

of=/dev/sdb bs=4M Check the USB device. In

our case it

To create a live DVD on a Windows-PC, is /dev/sdb.

Windows

1. Do a right-click on the ISO-image and burn the DVD. To create a liveUSB-stick, download a tool like "liveUSB creator"

2. In the BIOS setup, enable VT technology for the CPU Options

3. Insert the FOSS-Cloud CD and boot the server.

Answer the questions as follows (the bold values are examples can be set through the administrator and are variable, according to the local setup):

Warning: ALL DATA WILL BE LOST!

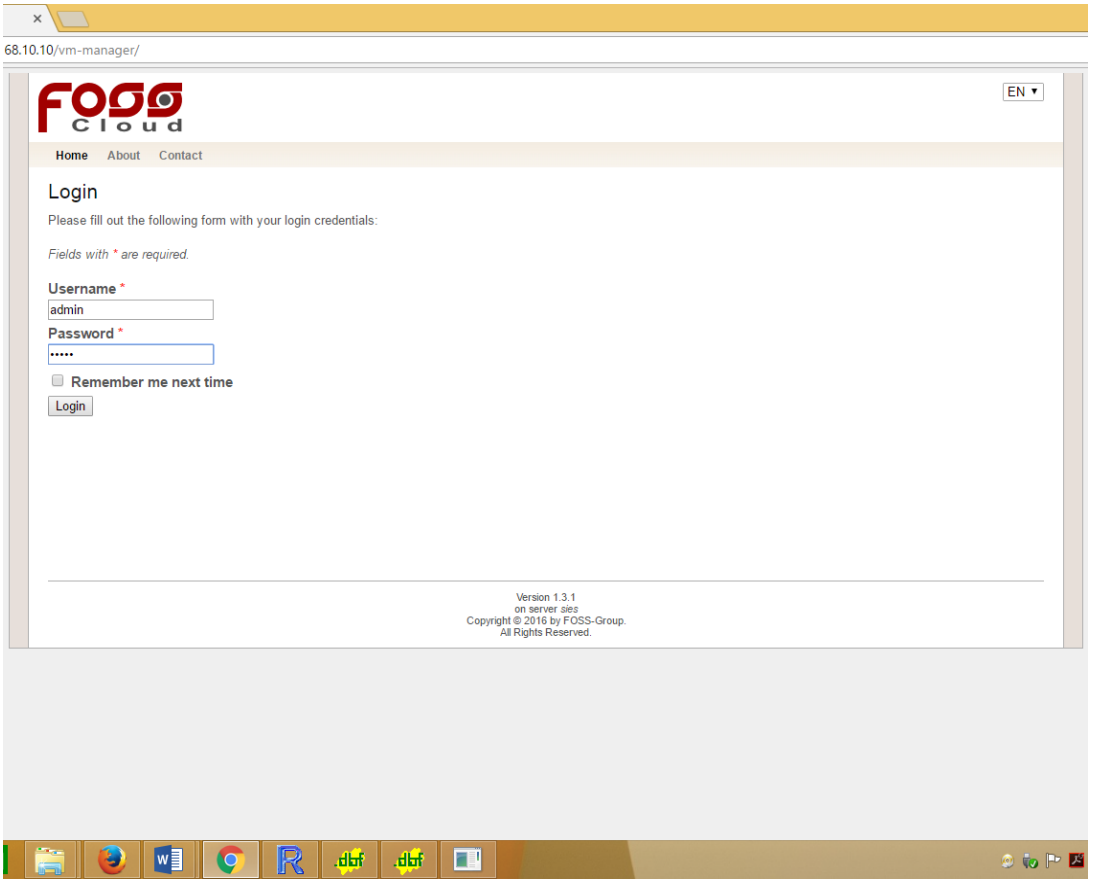
1. Choose your keymap

1 azerty	2 be	3 bg
4 br-a	5 br-l	6 by
7 cf	8 croat	9 cz
10 de	11 dk	12 dvorak
13 es	14 et	15 fi
16 fr	17 gr	18 hu
19 il	20 is	21 it
22 jp	23 la	24 lt
25 mk	26 nl	27 no
28 pl	29 pt	30 ro
31 ru	32 se	33 sg

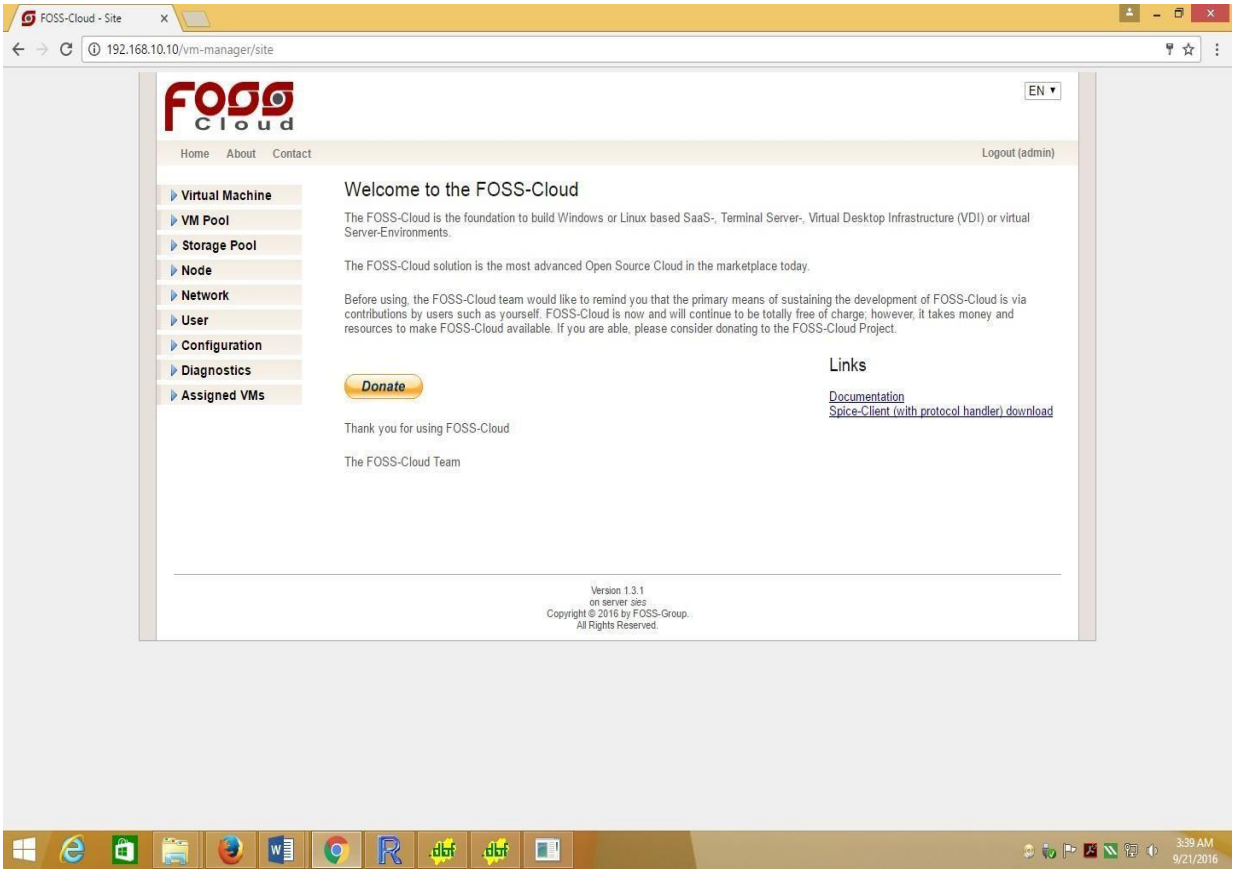
34 sk-y 35 sk-z 36 slovene
37 trf 39 ua 40 uk
41 us 42 wangbe 43 fr_CH
44 speakup 45 cs_CZ 46 de_CH
47 sg-lat1 48 fr-bepo 49 colemak

2. Confirm that you want to start: yes
3. Choose Single-Server: 1
4. Choose a Block-Device: sda
5. Confirm that you want to continue: yes
6. Confirm that you want to continue: yes
7. Choose the network interface: eth0
8. Choose if you want to use automatic network configuration: no
9. Choose hostname: e.g. foss-cloud
10. Choose domain name: e.g. foss-cloud.org
11. Choose a free IP address in your home network: 192.168.10.10
12. Choose the network mask in CDIR format: 24
13. Choose the broadcast IP address: 192.168.1.255
14. Confirm your entries: yes
15. Enter the IP address of your gateway: 192.168.1.1
16. Enter the IP address of your DNS resolver: 192.168.1.1
17. Do you like another DNS resolver: no
18. Confirm your entries: yes
19. Reboot your system: yes
20. Install Patch: Missing_roles_of_VM-User

Access your IP 192.168.10.10 with a web browser and login with *admin* and the password *admin*



Home page of FOSS Cloud at Client



Uploading ISO files to foss cloud

ISO files need to be uploaded manually.

Windows

- Local Windows computer with WinSCP installed (Download WinSCP)
- You've downloaded a ISO-File to your computer. This example is done with: ubuntu-11.04- desktop-i386.iso
- The IP-address of the FOSS-Cloud Node. This example is done with: 192.168.1.216

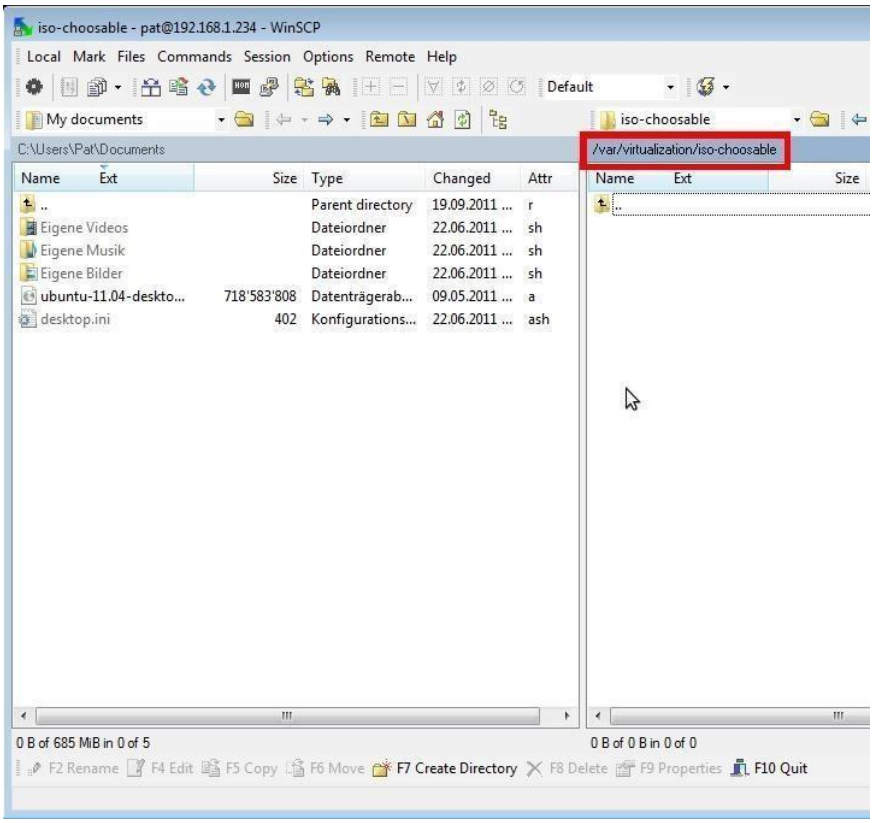
Run WinSCP and establish a connection to the

FOSS-Cloud Node: username: root password:

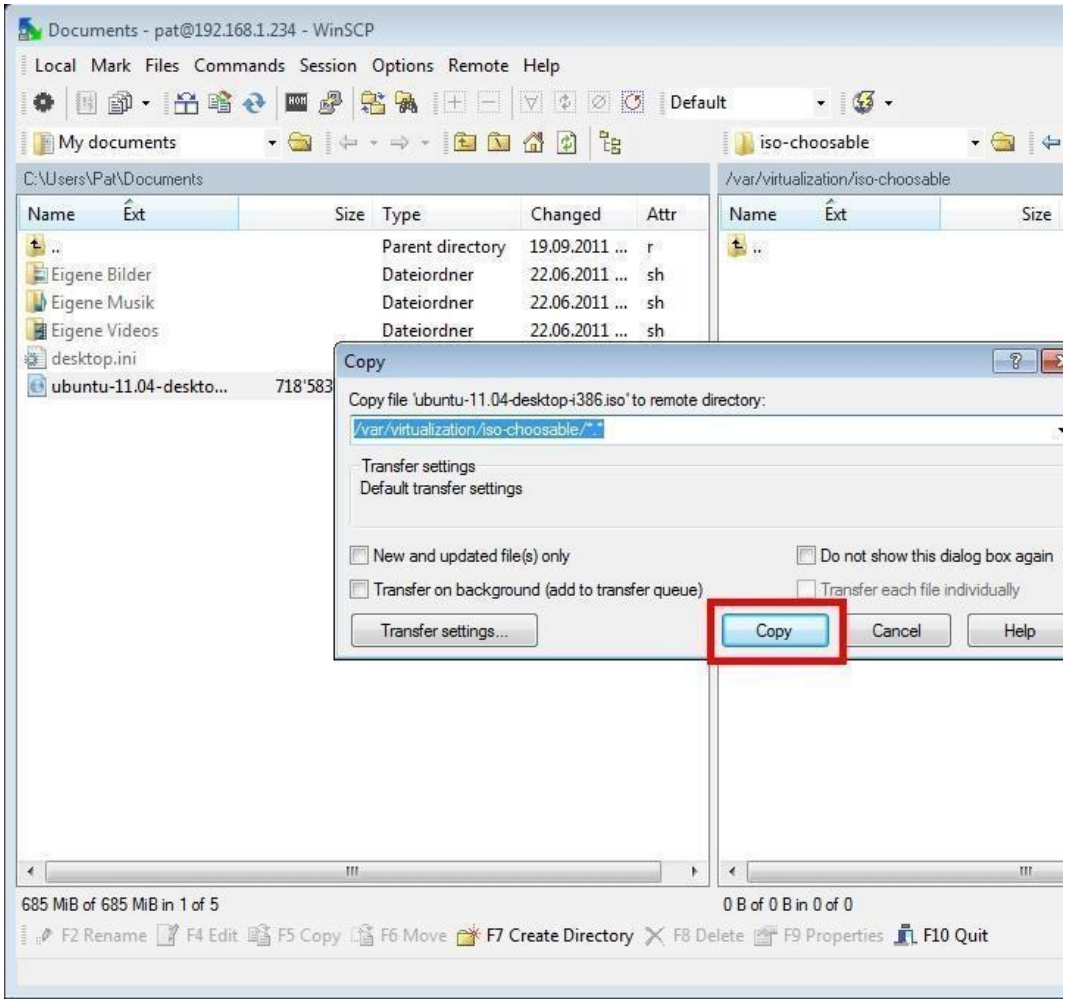
admin:



On the guest machine change to the '/var/virtualization/iso-choosable' direcotry. On the host, change to the directory the ISO-file is located:\\

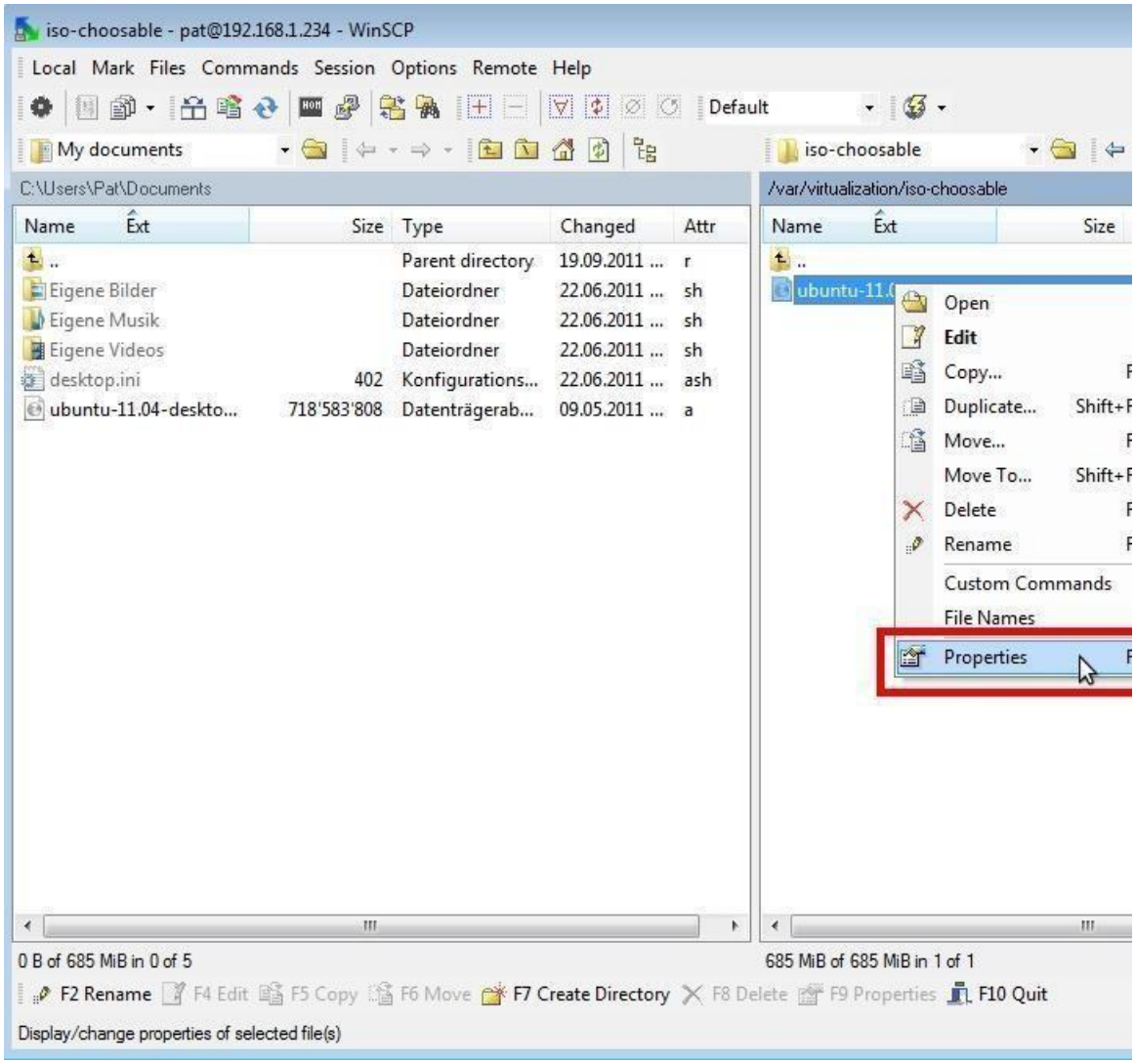


Drag (Drag & Drop) the ISO-file in the '/var/virtualization/iso-choosable' directory and commit the copy:



After the copy, right-click on the ubuntu-11.04-desktop-i386.iso file in the '/var/virtualization/iso-choosable' directory and select 'properties'.

Or simply select the ISO-file and press 'F9':



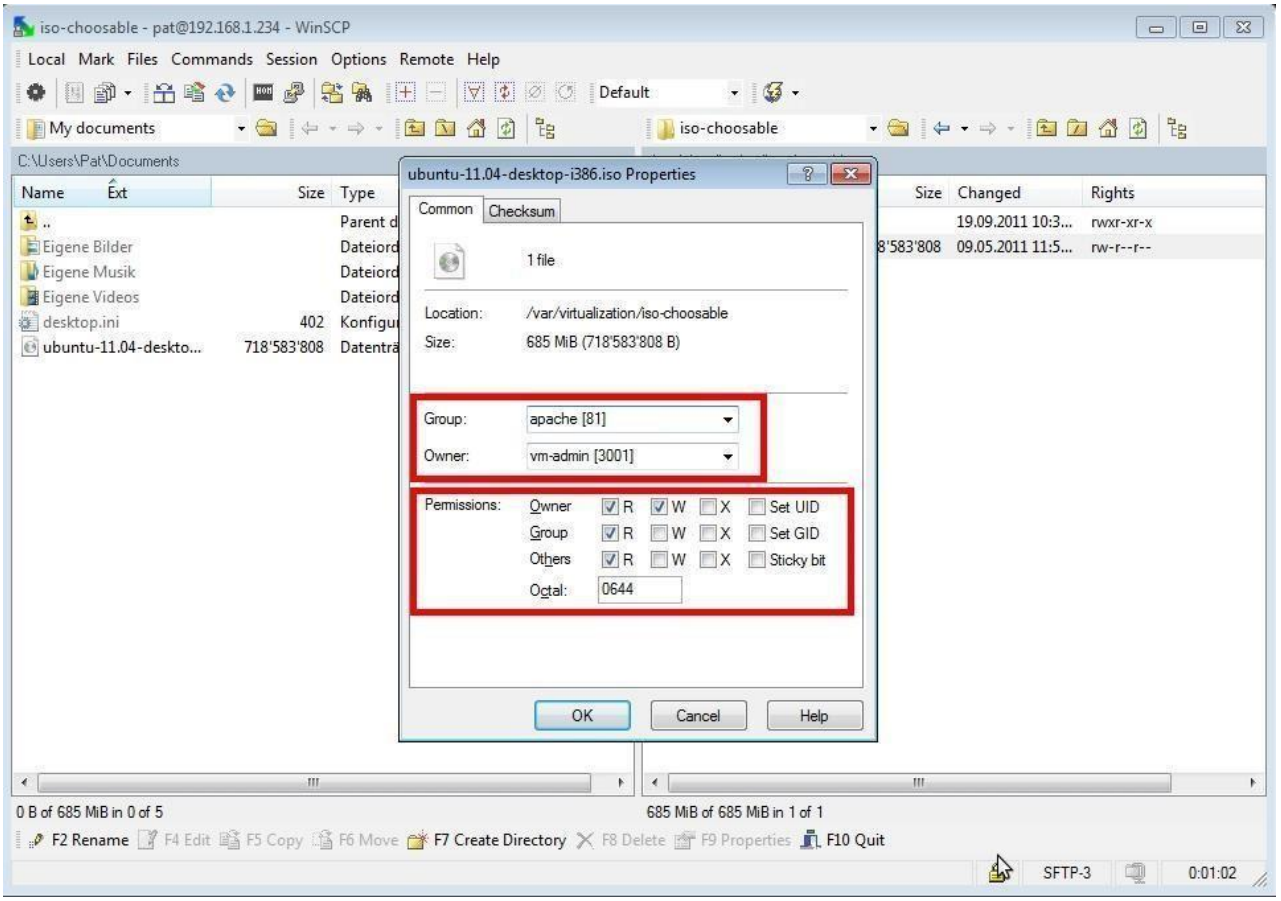
Set the files ownership as follows:

- Group: root
- Owner vm-storage Caution the UID may be different to 3000

And the rights:

- Owner: R and W
- Group: R
- Other: R Or octal: 0644

When you create a new Virtual Machine Profile, the uploaded ISO-file will be listed.



Linux

Upload ISO with scp

- Local Linux computer with shell access.
- You've downloaded a ISO-File to your computer. This example is done with openSUSE-11.3- KDE4-LiveCD-i686.iso
- The IP-address of the FOSS-Cloud Node is 192.168.1.216.

Upload the ISO-file with Secure Copy (scp). The root password is admin:

```
root # scp openSUSE-11.3-KDE4-LiveCD-i686.iso
root@192.168.1.216:/var/virtualization/iso-choosable/.
```

Make sure, that you have the correct permissions on the newly uploaded file:

```
root # cd /var/virtualization/iso-choosable
```

ls -al

```
-rw-r--r-- 1 root vm-storage 728760320 Mär 24 17:54
openSUSE-11.4-KDE- LiveCD-x86_64.iso.iso
```

If not, you can set them as follows:

```
root # chown root:vm-storage openSUSE-11.4-KDE-LiveCD-x86_64.iso.iso
root # chmod 644 openSUSE-11.4-KDE-LiveCD-x86_64.iso.iso
```

When you create a new Virtual Machine Profile, the uploaded ISO-file will be listed. Upload ISO with wget

Login to a VM node as root, then execute the following command

```
root # cd /var/virtualization/iso-choosable/
root # wget http://<your_download_Link/your_image.iso>
root # chown apache:vm-storage <your_image.iso>
root # chmod 644 <your_image.iso>
```

Examples:

```
root # wget
http://download.fedoraproject.org/pub/fedora/linux/releases/19/Live/x86_64/Fe dora-Live-Desktop-x86_64-19-1.iso
root # wget http://mirror.switch.ch/ftp/mirror/centos/6.5/isos/x86_64/CentOS-6.5-x86_64-netinstall.iso
```

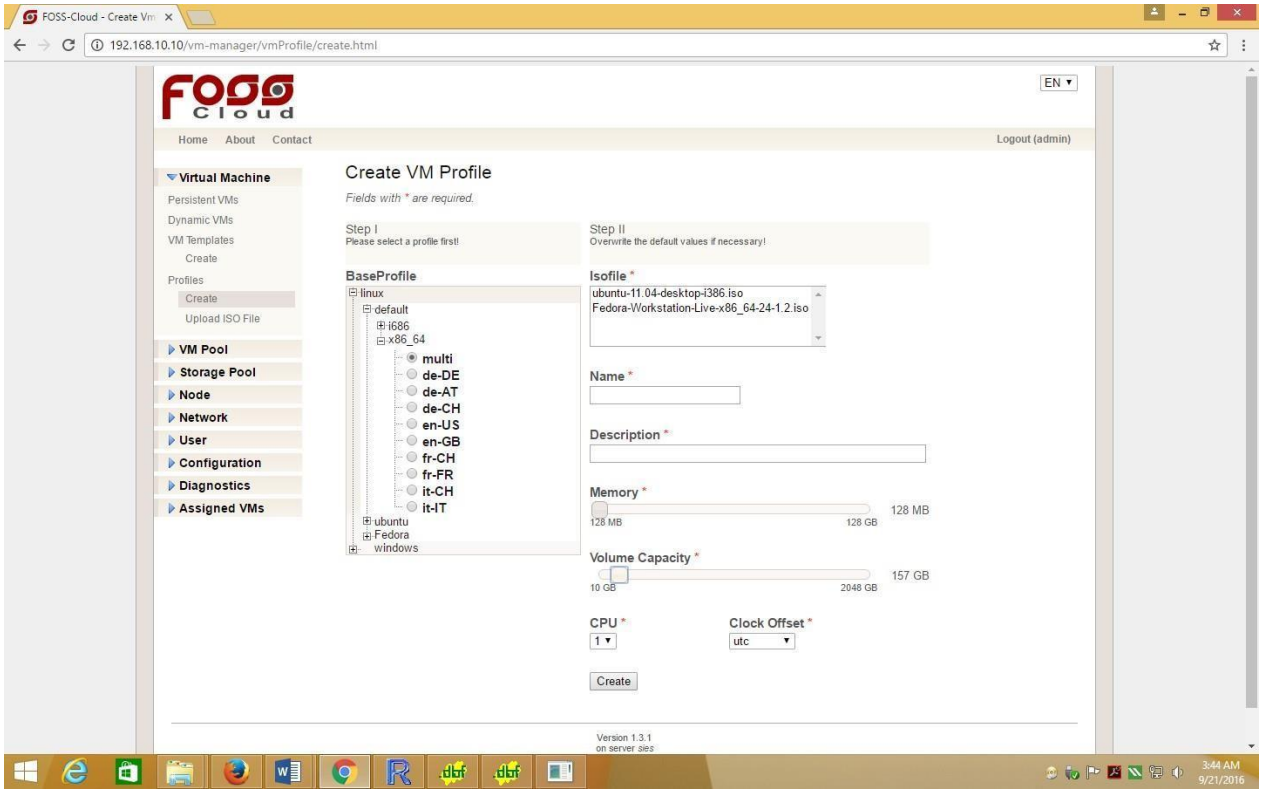
Practical No: 7

Aim: Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure)

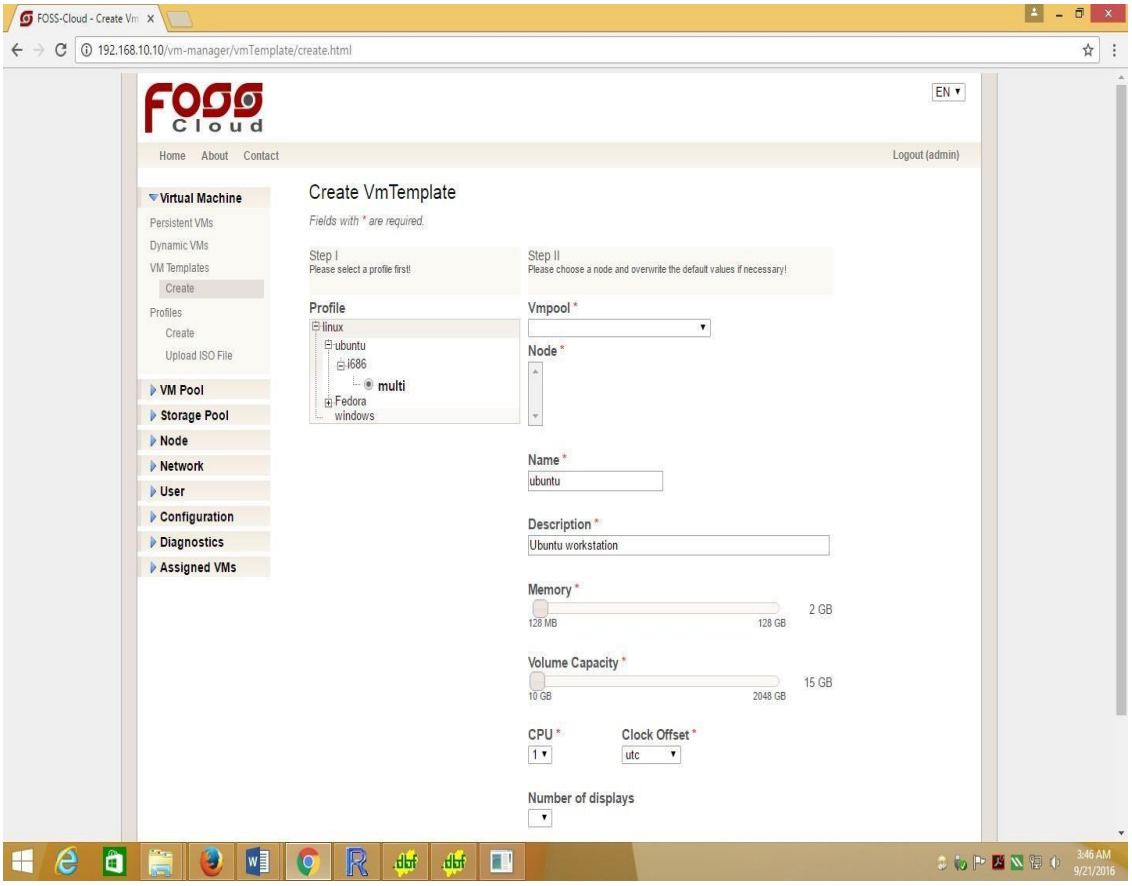
Infrastructure as a Service (IaaS)

Creating Virtual Machines

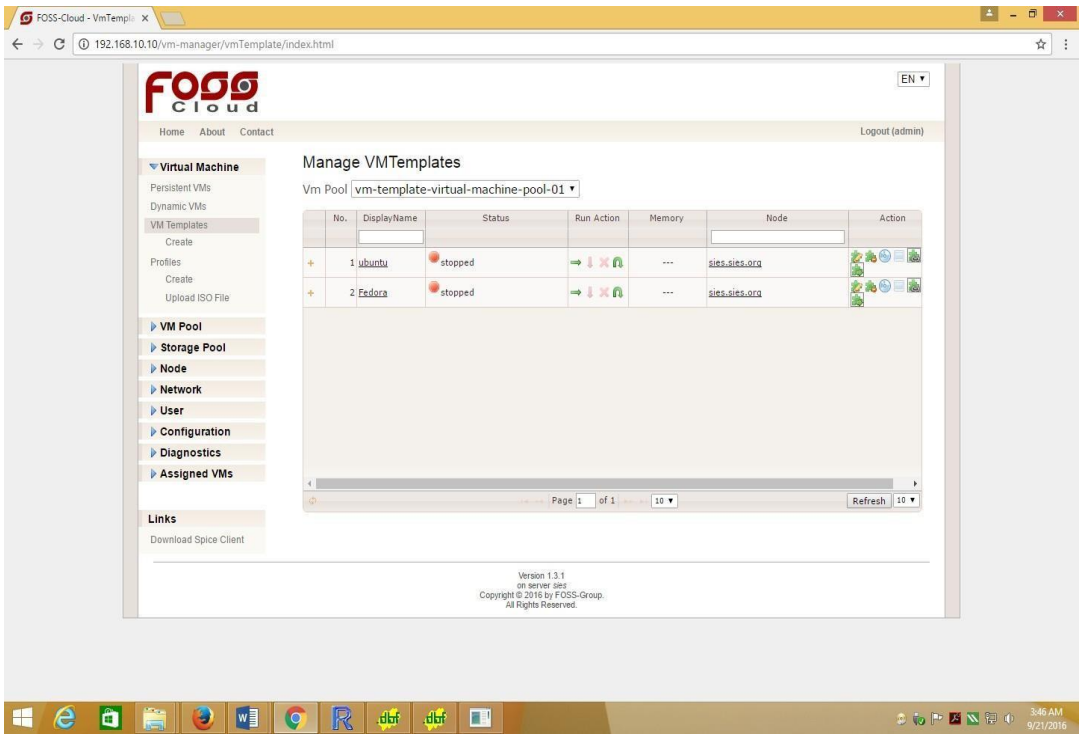
Step 1.Create a Profile



2. Create VM Template

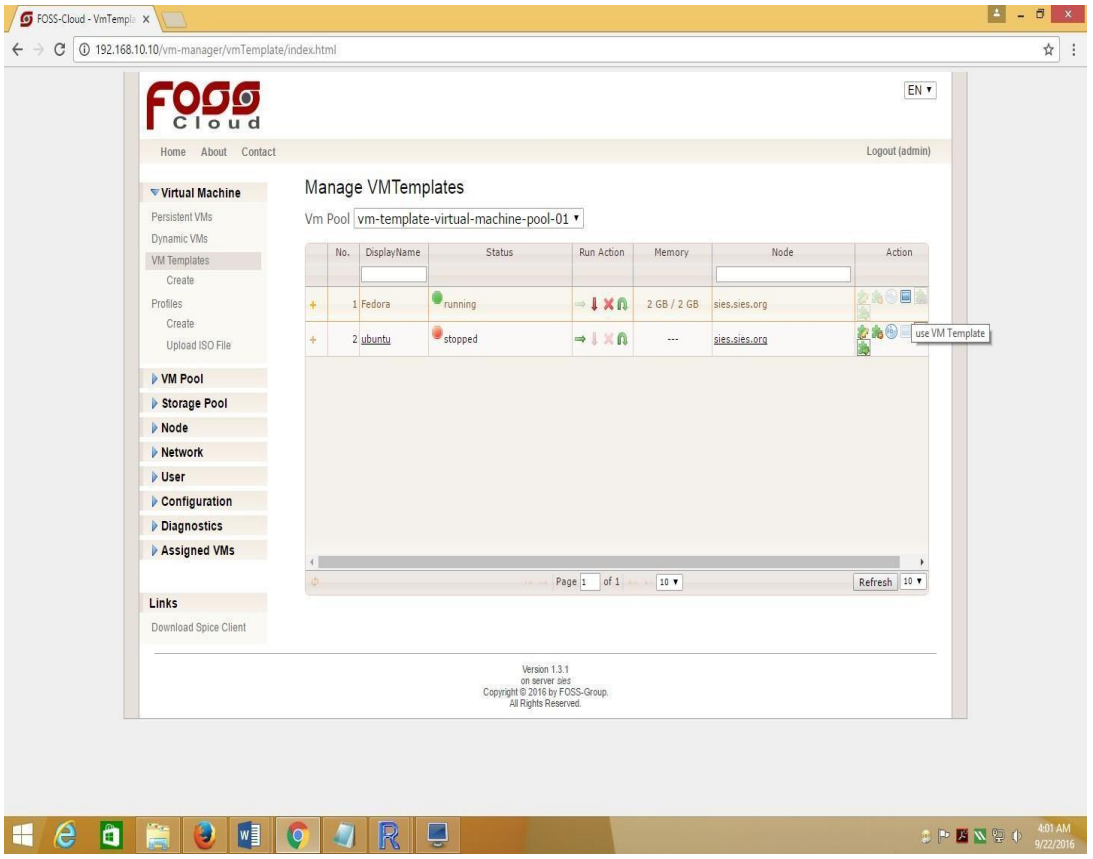


3. Manage VM Template



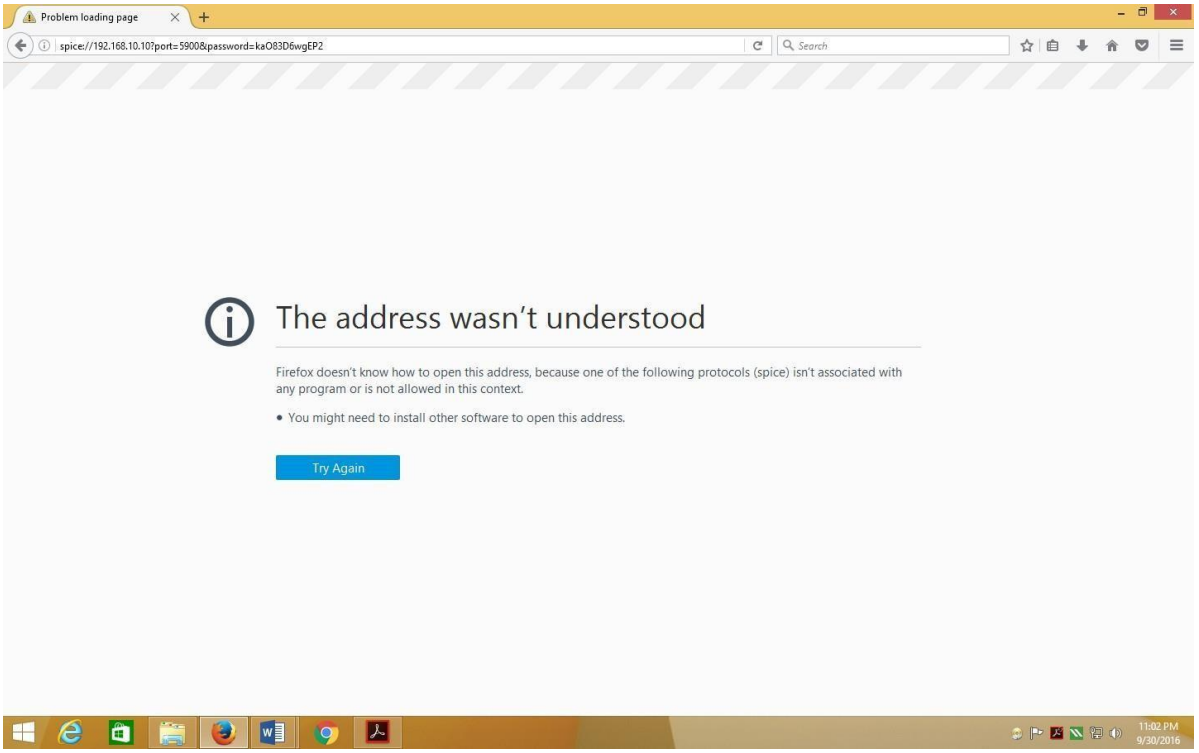
Implement FOSS-Cloud Functionality – VDI (Virtua Desktop Infrastructure)

Today, Linux distributions like Ubuntu, Suse, Fedora etc. contains the spice protocol in their distribution. The client is named virt- or remote-viewer. FOSS-Cloud needs the remote viewer which is part of the virt-viewer package. In general the client is working out of the box.



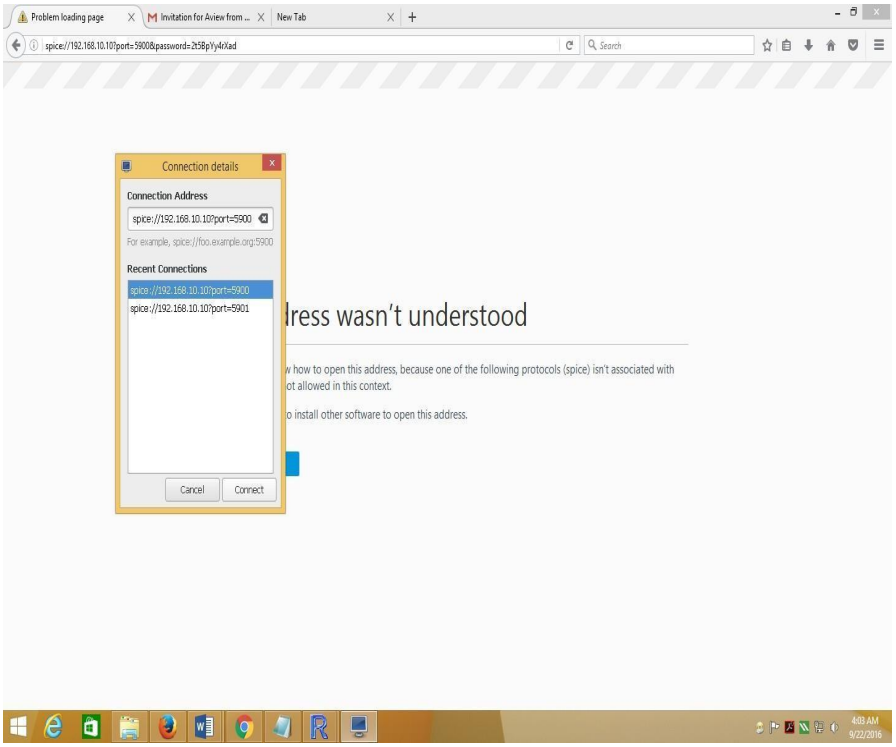
The Browser window opens up with following content in the address bar

spice://192.168.10.10?port=5900&password=kaO83D6wgEP2



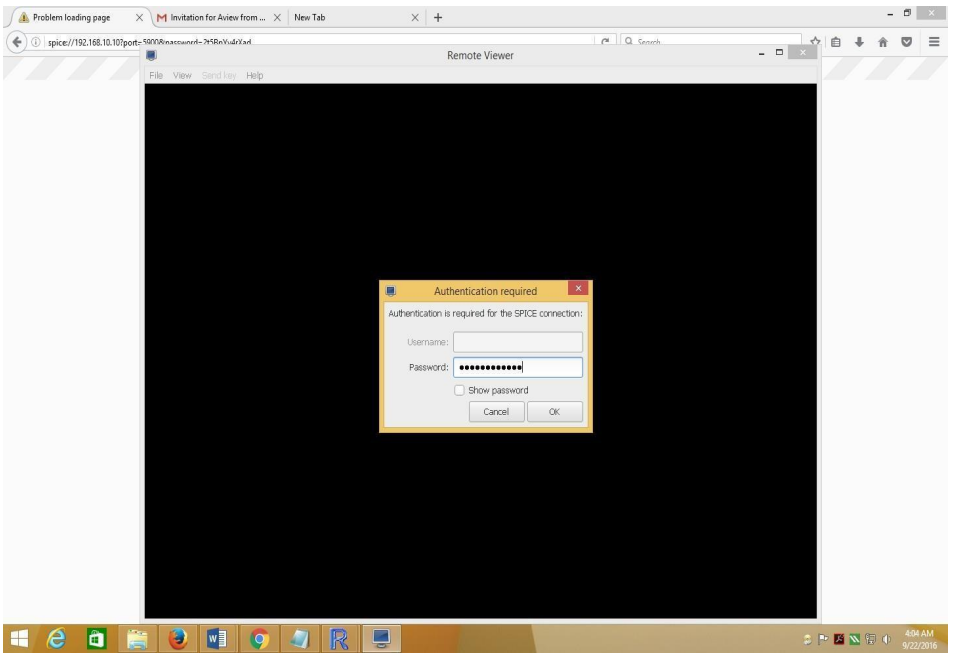
Open the Remote Viewer put the spice address in Connection Address as

spice://192.168.10.10?port=5900

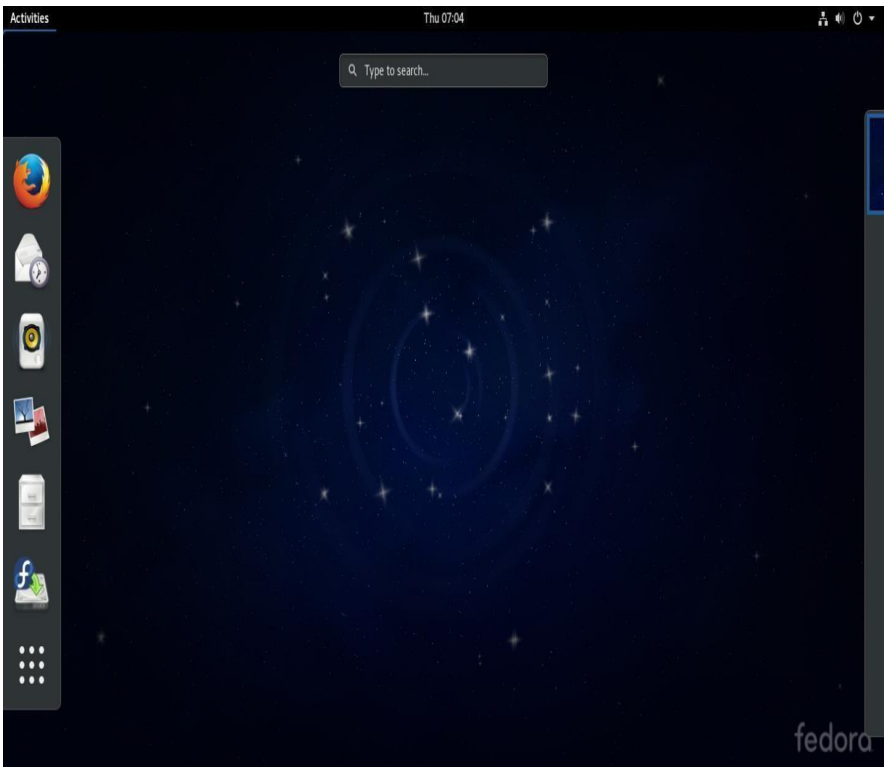


Enter the password

password=kaO83D6wgEP2



The Virtual Desktop Interface opens up for that Virtual Machine as follows:



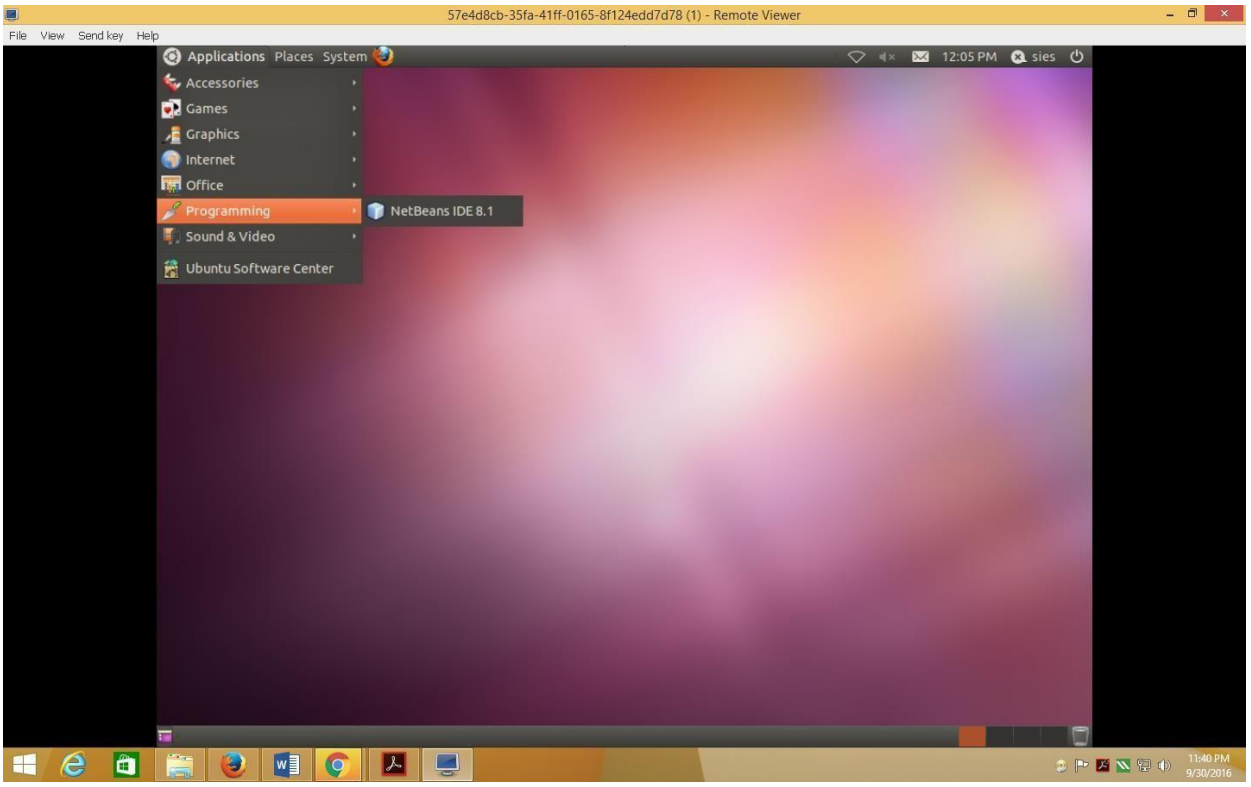
Practical No: 8

Aim: Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure)

Platform as a Service (PaaS)

Software development Kits can be made available in the Virtual Machines that can be implemented as Platform as a Service.

Installation of Netbeans, Eclipse, Visual Studio and DBMS can be done in the appropriate Virtual Machines.

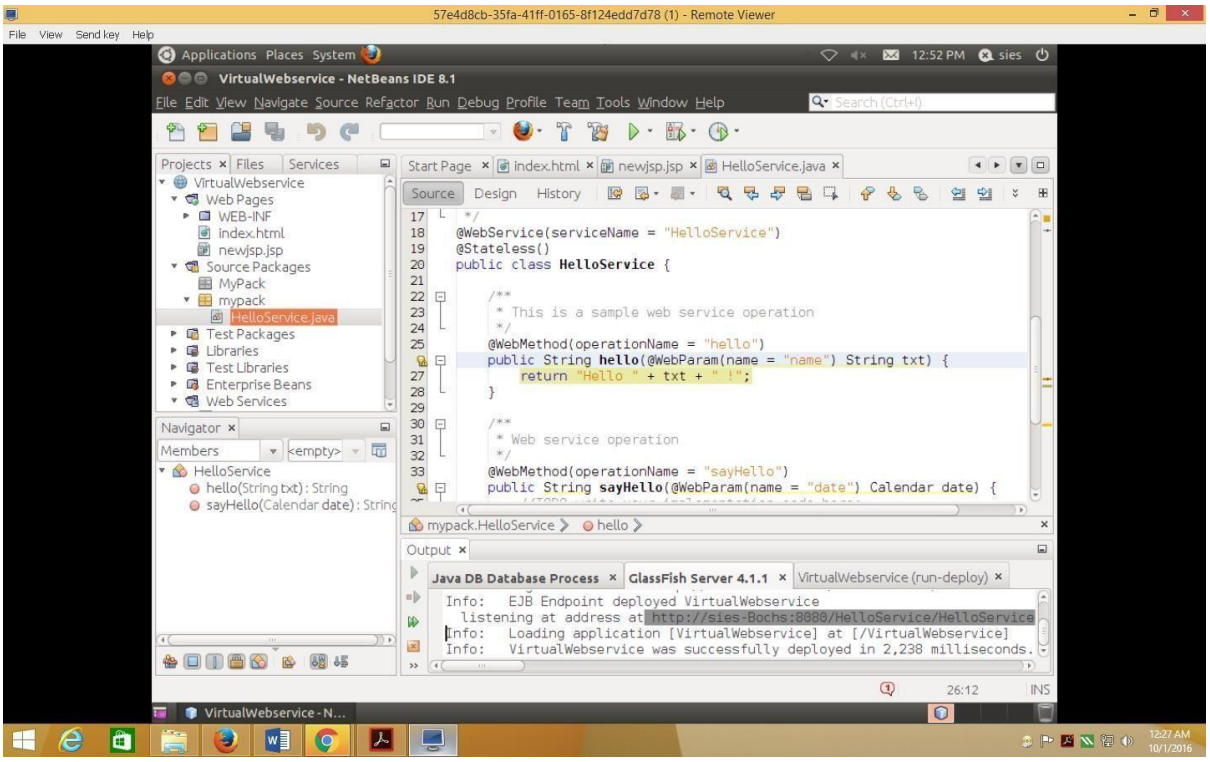


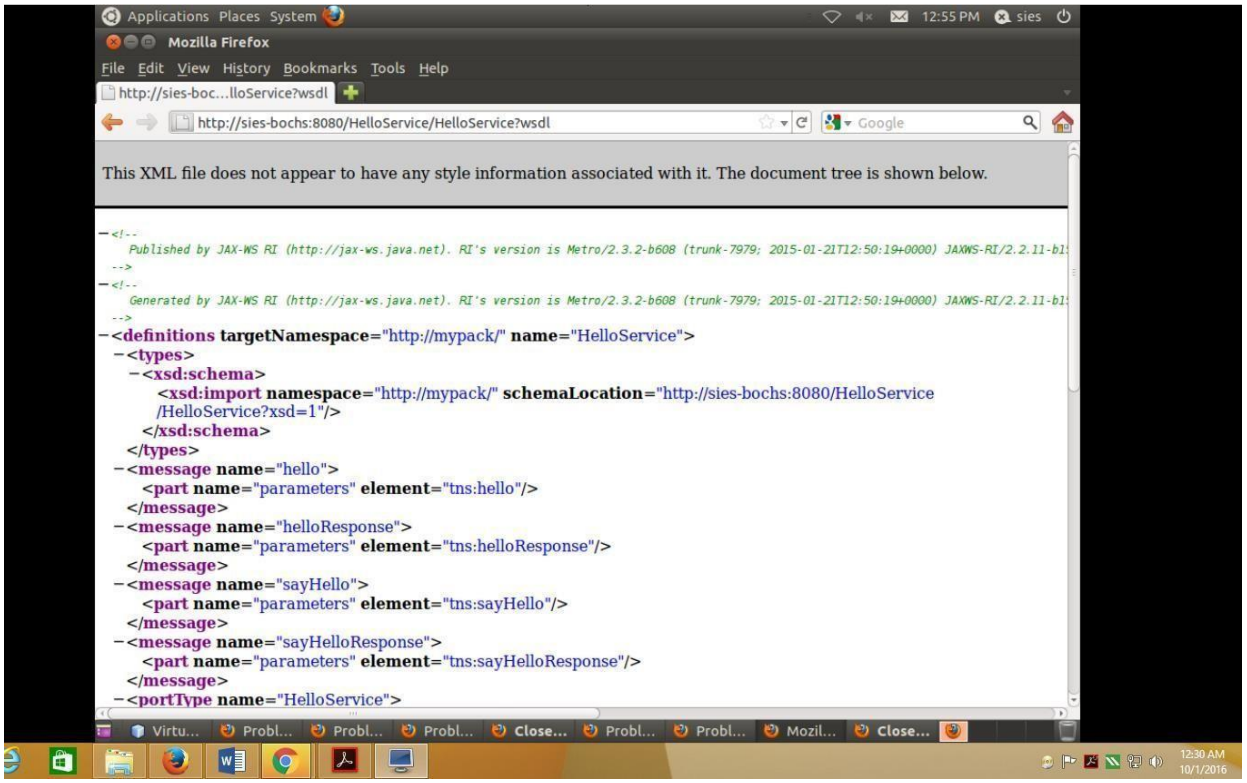
Practical No: 9

Aim:Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure)

Software as a Service (SaaS)

Applications created and deployed in the virtual machines can be accessed by outside world. This can be implemented as Software as Service.




Applications Places System
12:55 PM sies
Mozilla Firefox
File Edit View History Bookmarks Tools Help
http://sies-boc...lloService?wsdl
http://sies-bochs:8080/HelloService/HelloService?wsdl
Google
This XML file does not appear to have any style information associated with it. The document tree is shown below.
<!--
Published by JAX-WS RI (http://jax-ws.java.net). RI's version is Metro/2.3.2-b608 (trunk-7979; 2015-01-21T12:50:19+0000) JAXWS-RI/2.2.11-b1
-->
<!--
Generated by JAX-WS RI (http://jax-ws.java.net). RI's version is Metro/2.3.2-b608 (trunk-7979; 2015-01-21T12:50:19+0000) JAXWS-RI/2.2.11-b1
-->
<definitions targetNamespace="http://mypack/" name="HelloService">
<types>
<xsd:schema>
<xsd:import namespace="http://mypack/" schemaLocation="http://sies-bochs:8080/HelloService
/HelloService?xsd=1"/>
</xsd:schema>
</types>
<message name="hello">
<part name="parameters" element="tns:hello"/>
</message>
<message name="helloResponse">
<part name="parameters" element="tns:helloResponse"/>
</message>
<message name="sayHello">
<part name="parameters" element="tns:sayHello"/>
</message>
<message name="sayHelloResponse">
<part name="parameters" element="tns:sayHelloResponse"/>
</message>
<portType name="HelloService">
Virtu... Probl... Probl... Probl... Close... Probl... Probl... Mozil... Close...
12:30 AM
10/1/2016

Practical No: 10

Aim: Explore GCP / AWS / Azure

Theory:

What is DynamoDB?

DynamoDB is a hosted NoSQL database offered by Amazon Web Services (AWS). It offers:
reliable performance even as it scales;
a managed experience, so you won't be SSH-ing into servers to upgrade the crypto libraries;
a small, simple API allowing for simple key-value access as well as more advanced query patterns.

DynamoDB is a particularly good fit for the following use cases:

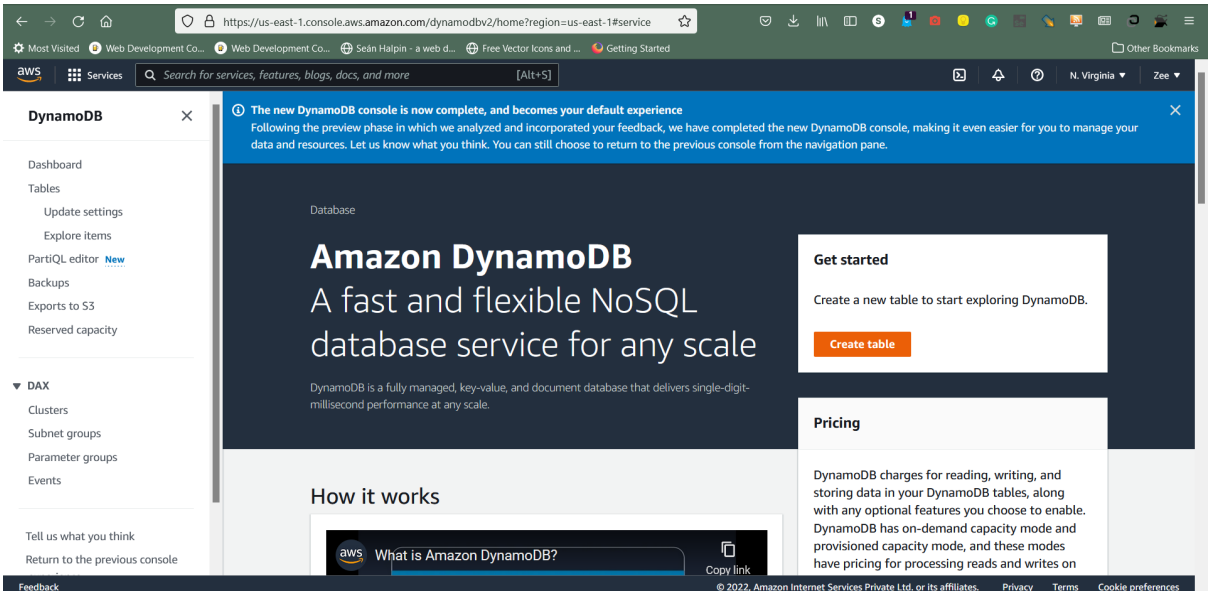
Applications with large amounts of data and strict latency requirements. As your amount of data scales, JOINs and advanced SQL operations can slow down your queries. With DynamoDB, your queries have predictable latency up to any size, including over 100 TBs!

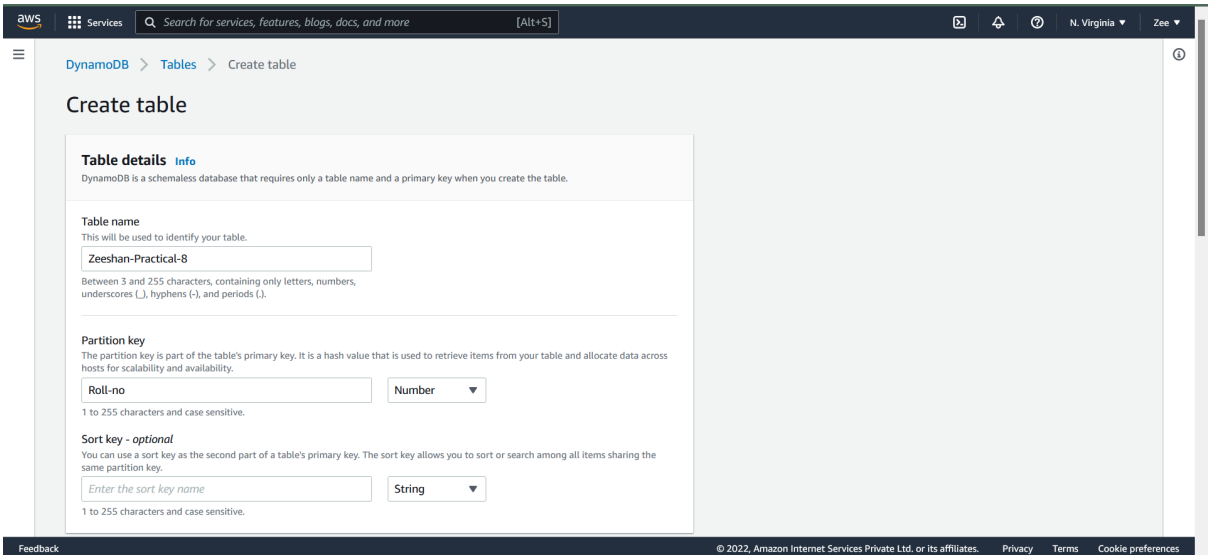
Serverless applications using AWS Lambda. AWS Lambda provides auto-scaling, stateless, ephemeral compute in response to event triggers. DynamoDB is accessible via an HTTP API and performs authentication & authorization via IAM roles, making it a perfect fit for building Serverless applications.

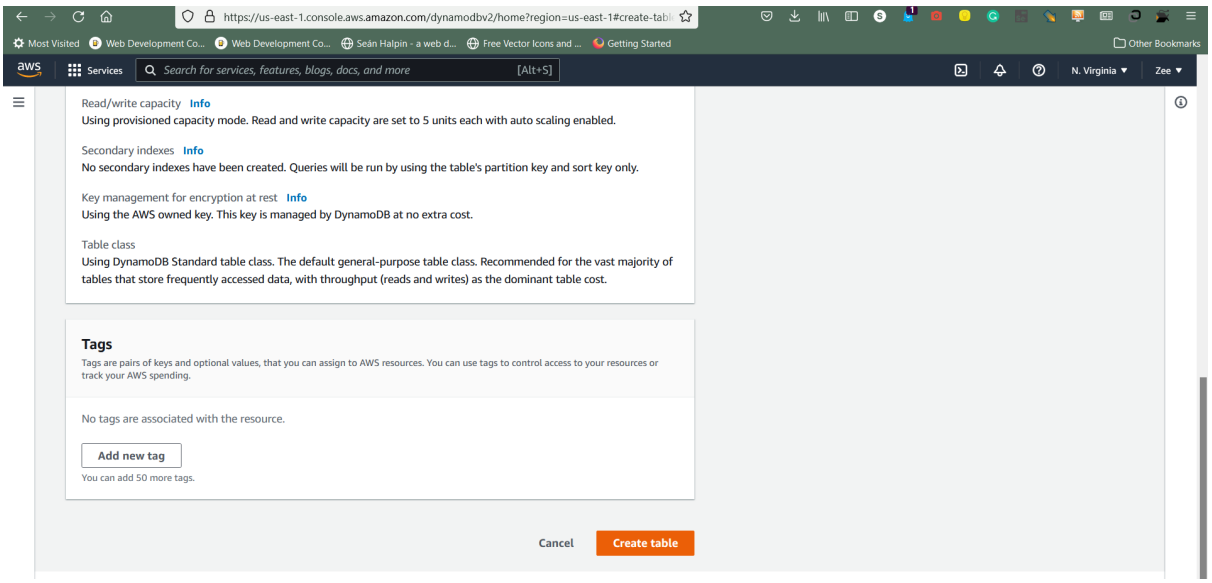
Data sets with simple, known access patterns. If you're generating recommendations and serving them to users, DynamoDB's simple key-value access patterns make it a fast, reliable choice.

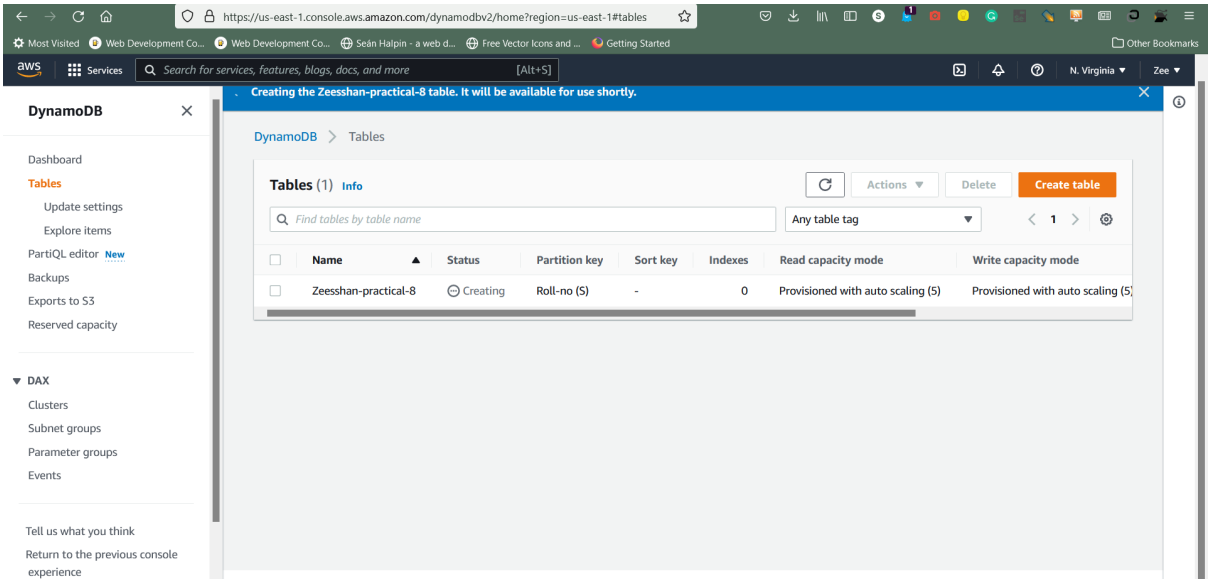
Steps:

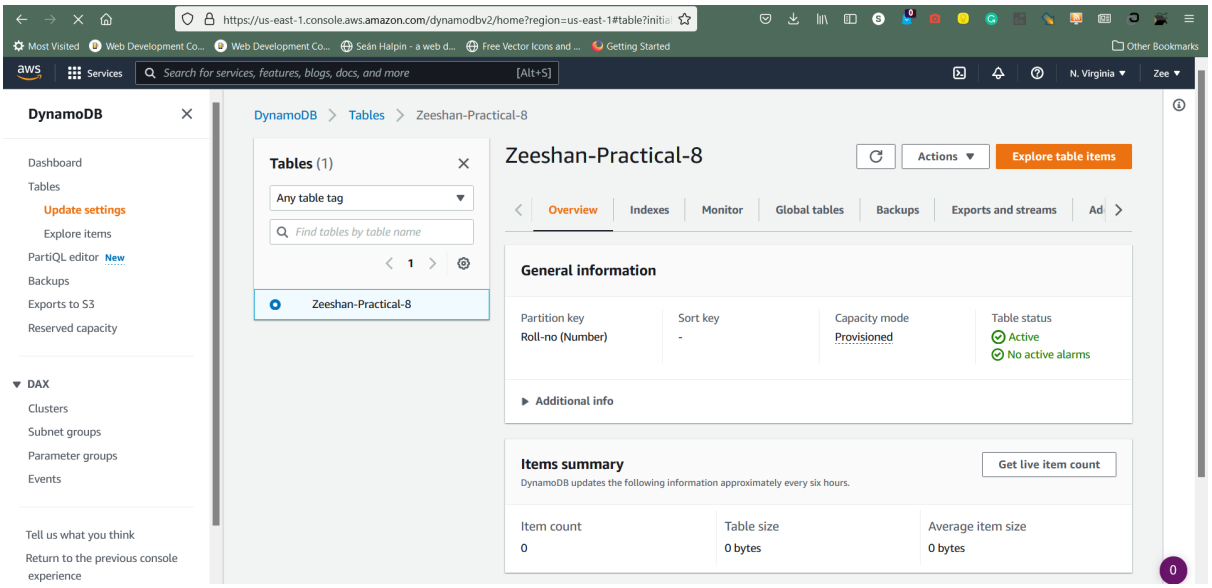
1) Create table

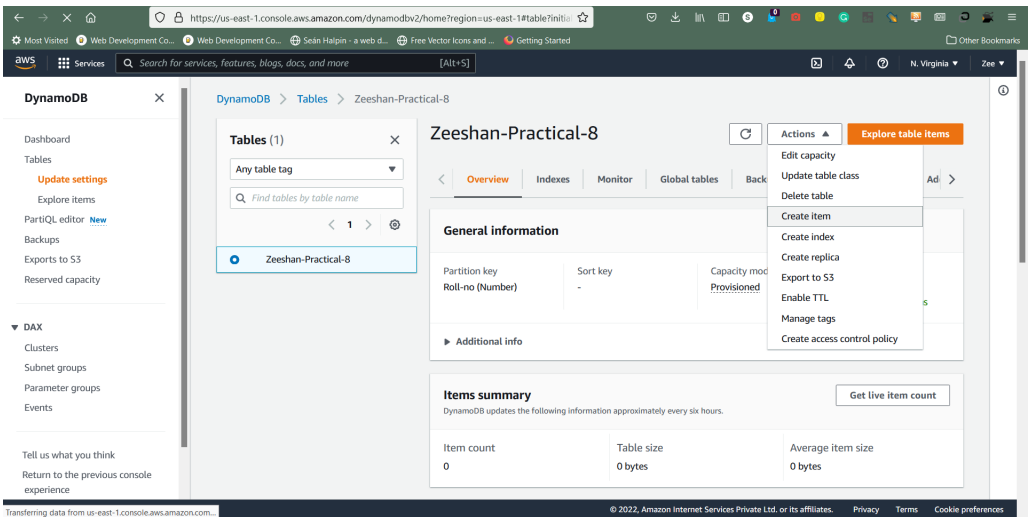


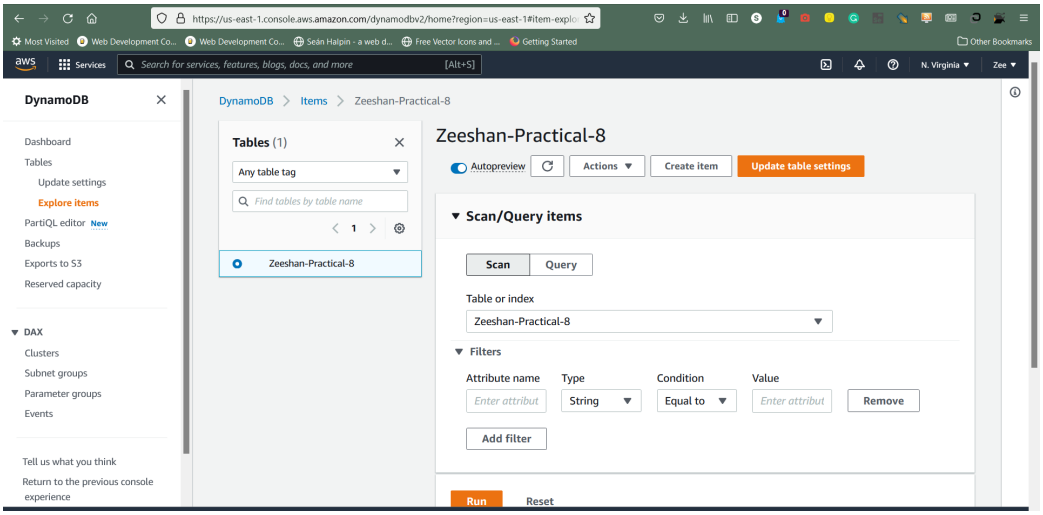




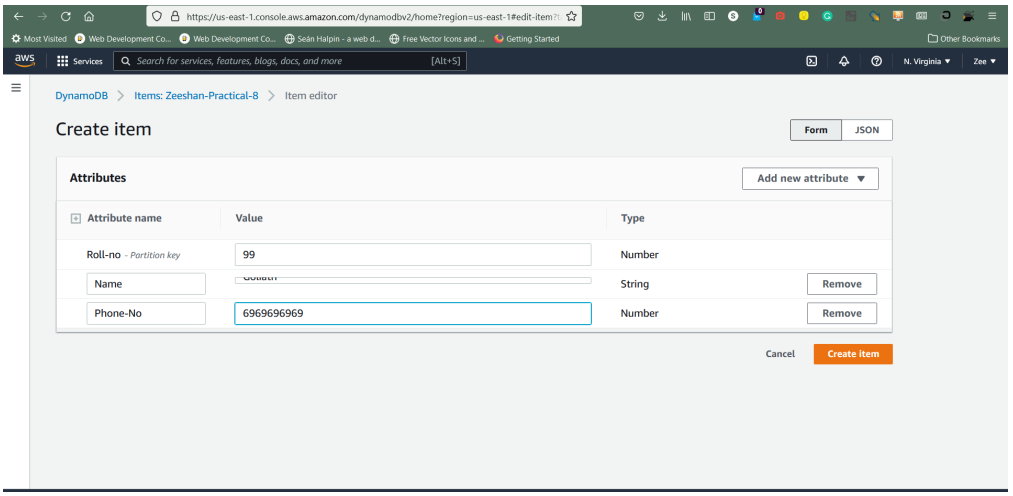

https://us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#tables
Creating the Zeesshan-practical-8 table. It will be available for use shortly.
DynamoDB
Tables
Tables (1) Info
Actions
Delete
Create table
Find tables by table name
Any table tag
Name
Status
Partition key
Sort key
Indexes
Read capacity mode
Write capacity mode
Zeesshan-practical-8
Creating
Roll-no (S)
-
0
Provisioned with auto scaling (5)
Provisioned with auto scaling (5)
Dashboard
Tables
Update settings
Explore items
PartiQL editor New
Backups
Exports to S3
Reserved capacity
DAX
Clusters
Subnet groups
Parameter groups
Events
Tell us what you think
Return to the previous console experience


https://us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#table?initia
DynamoDB
Tables
Zeeshan-Practical-8
Tables (1)
Any table tag
Find tables by table name
Zeeshan-Practical-8
Actions
Explore table items
Overview
Indexes
Monitor
Global tables
Backups
Exports and streams
General information
Partition key
Roll-no (Number)
Sort key
-
Capacity mode
Provisioned
Table status
Active
No active alarms
Additional info
Items summary
DynamoDB updates the following information approximately every six hours.
Get live item count
Item count
0
Table size
0 bytes
Average item size
0 bytes

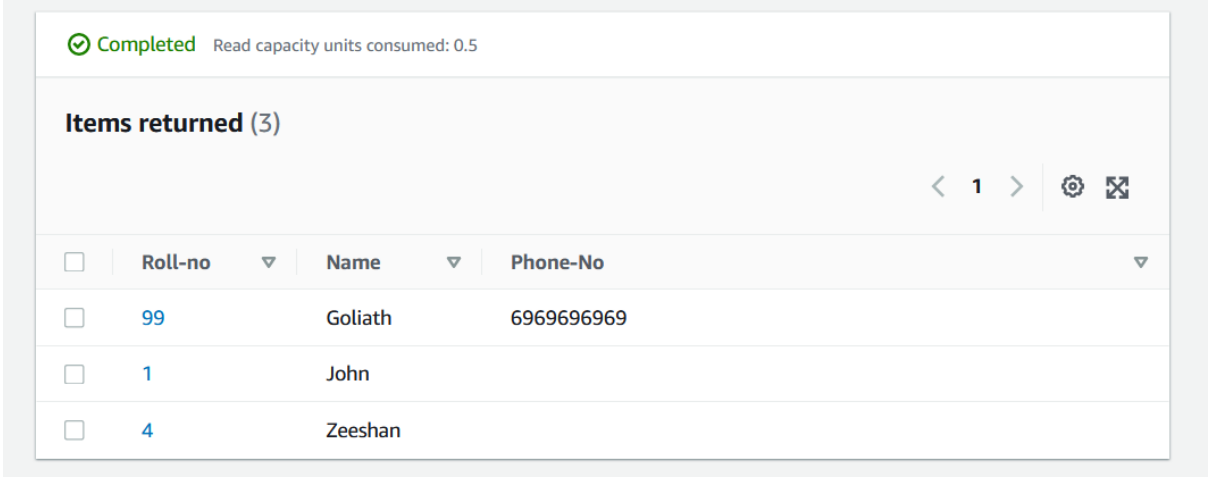

Edit capacity
Update table class
Delete table
Create item
Create index
Create replica
Export to S3
Enable TTL
Manage tags
Create access control policy
© 2022, Amazon Internet Services Private Ltd. or its affiliates.
Privacy
Terms
Cookie preferences

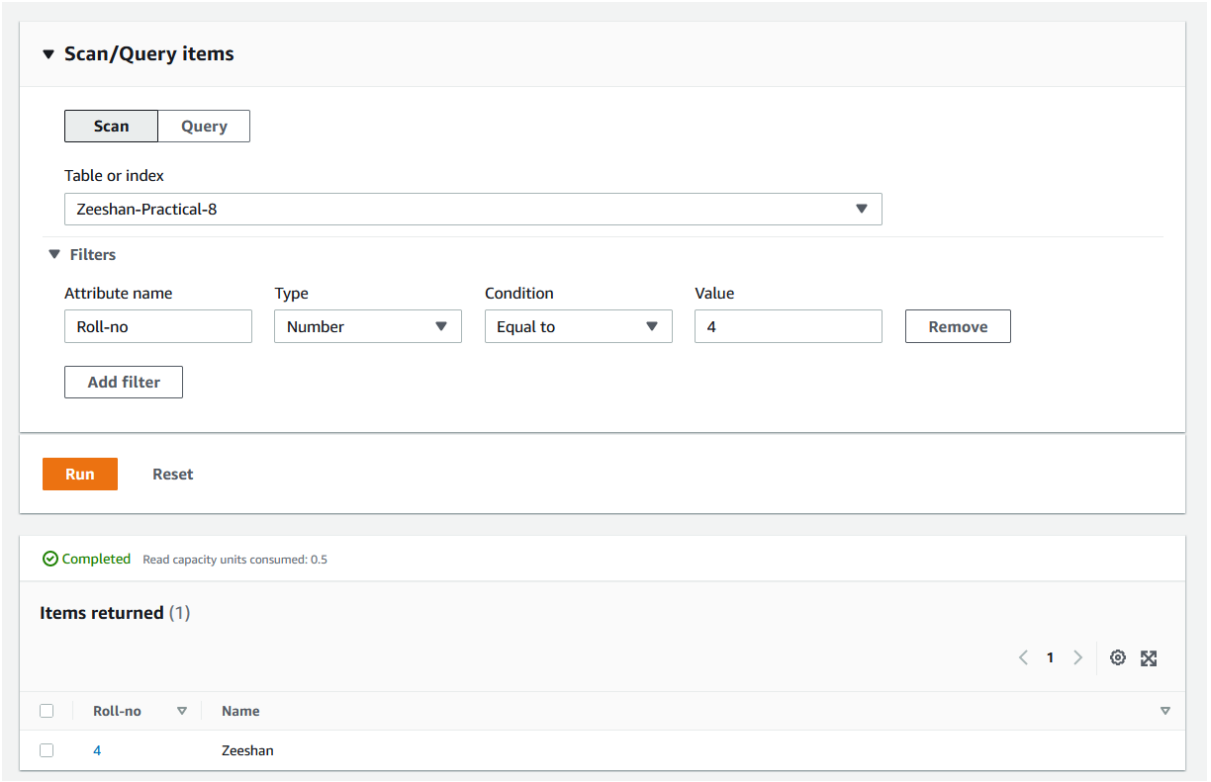


2) Create Item

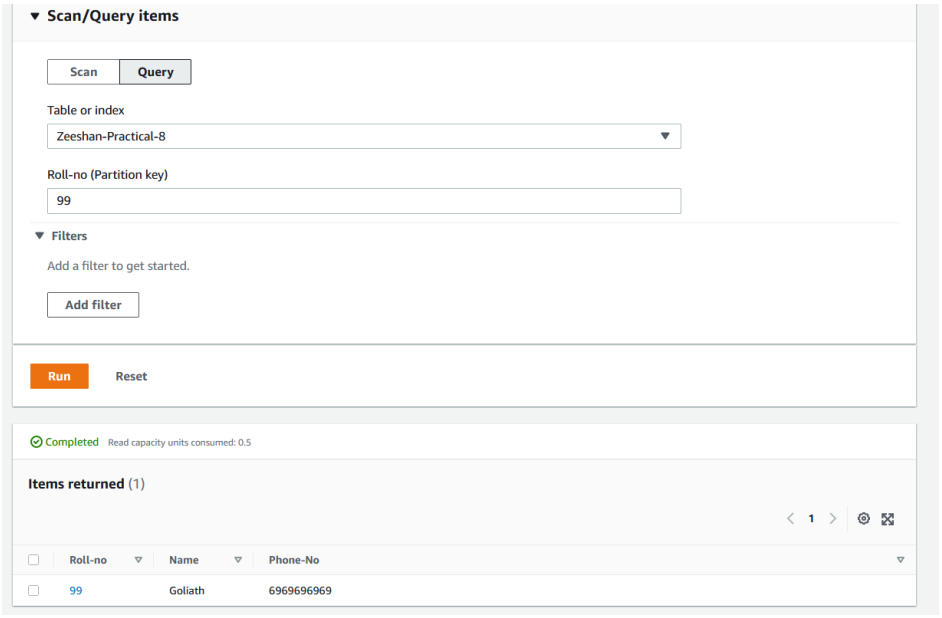


3) Scan Items.

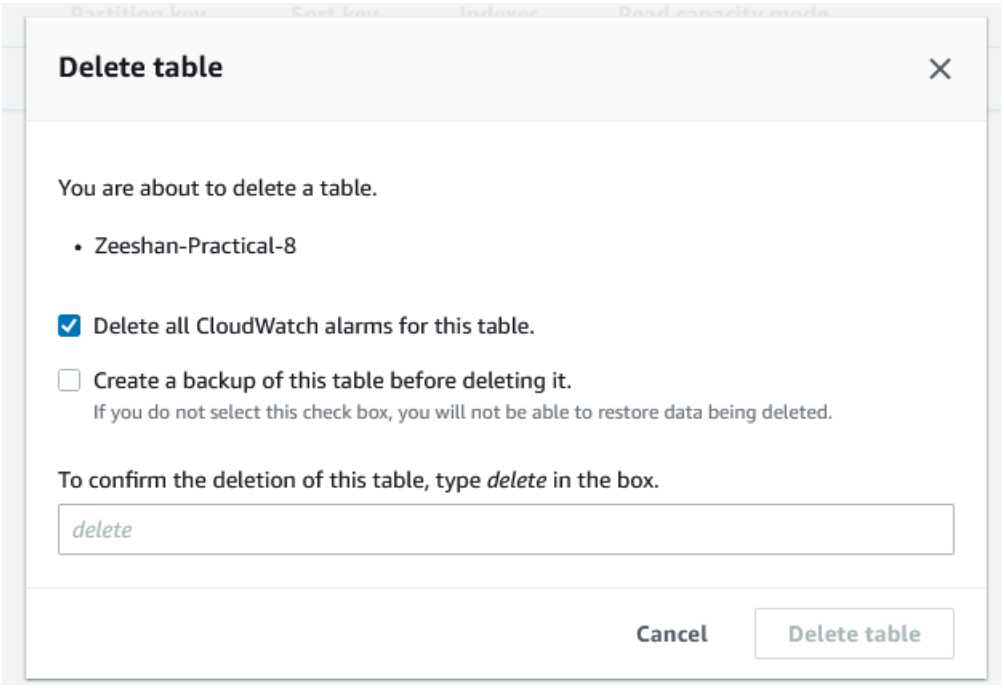


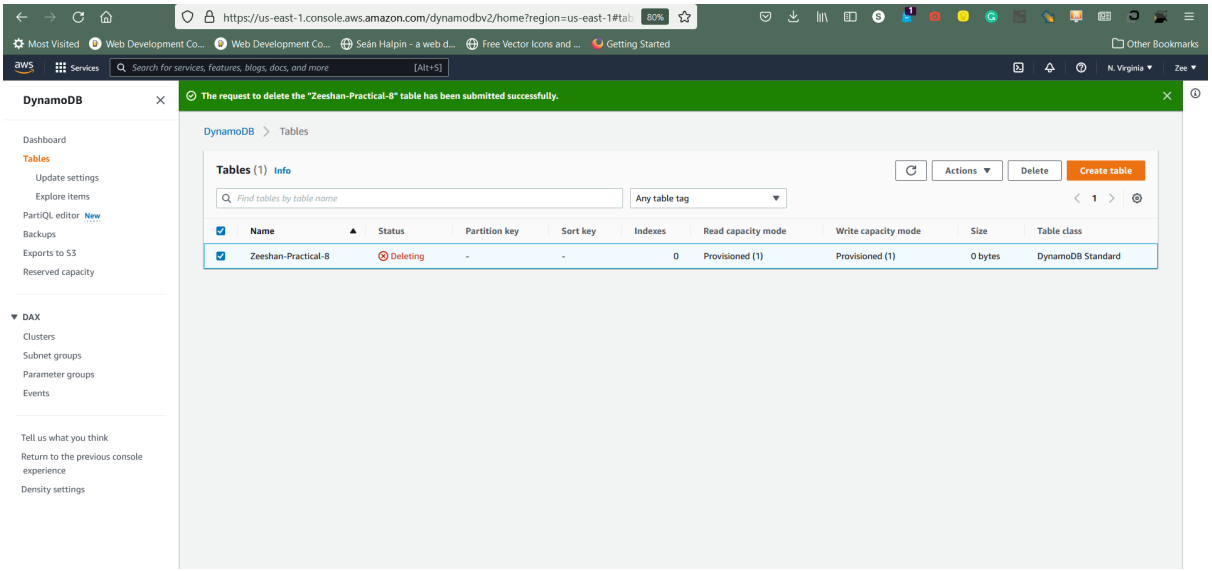


4) Query Items.



5) Delete table





Conclusion: Add, Update, Scan, Query and Delete table operation successfully implemented on DynamoDB.